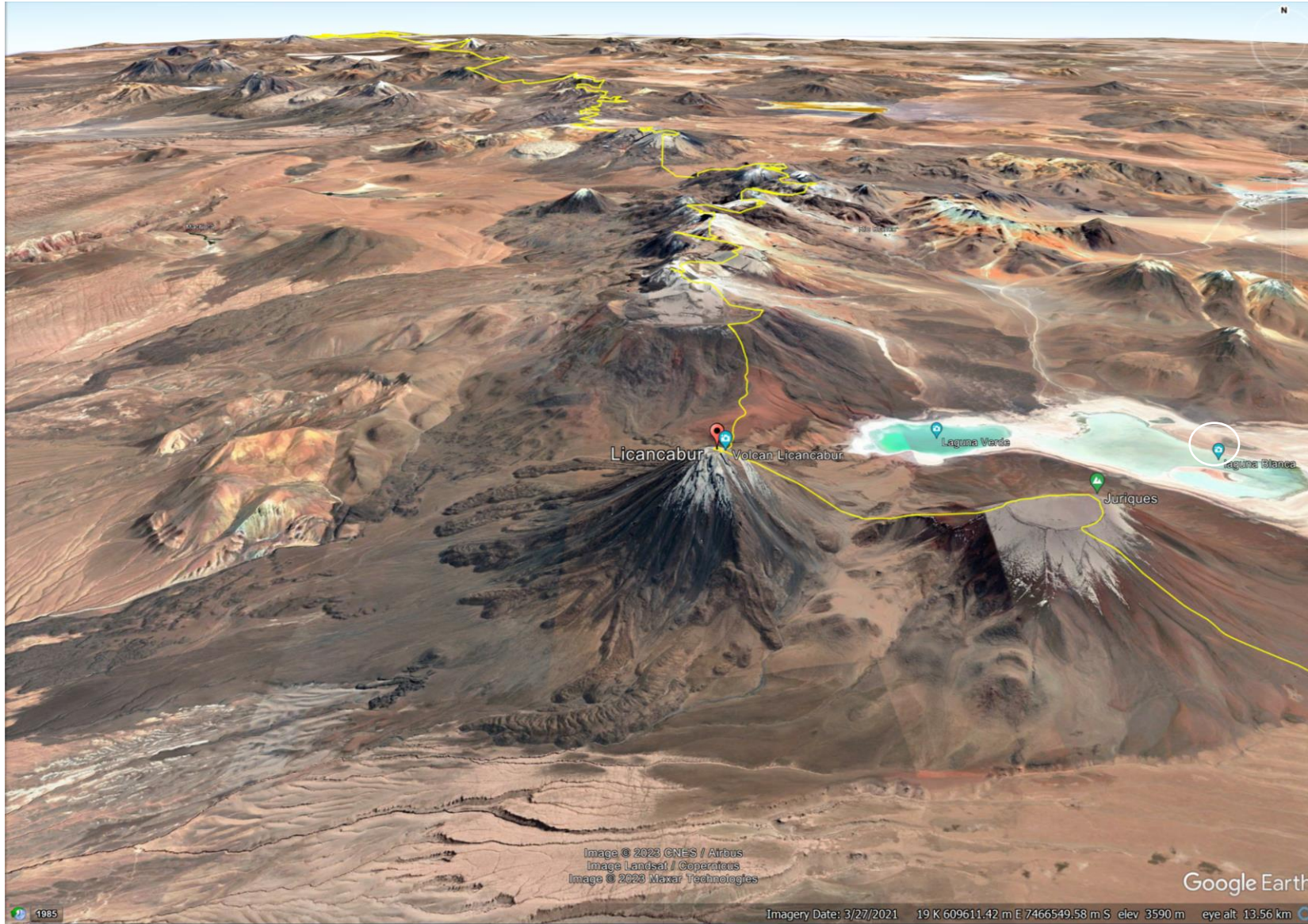
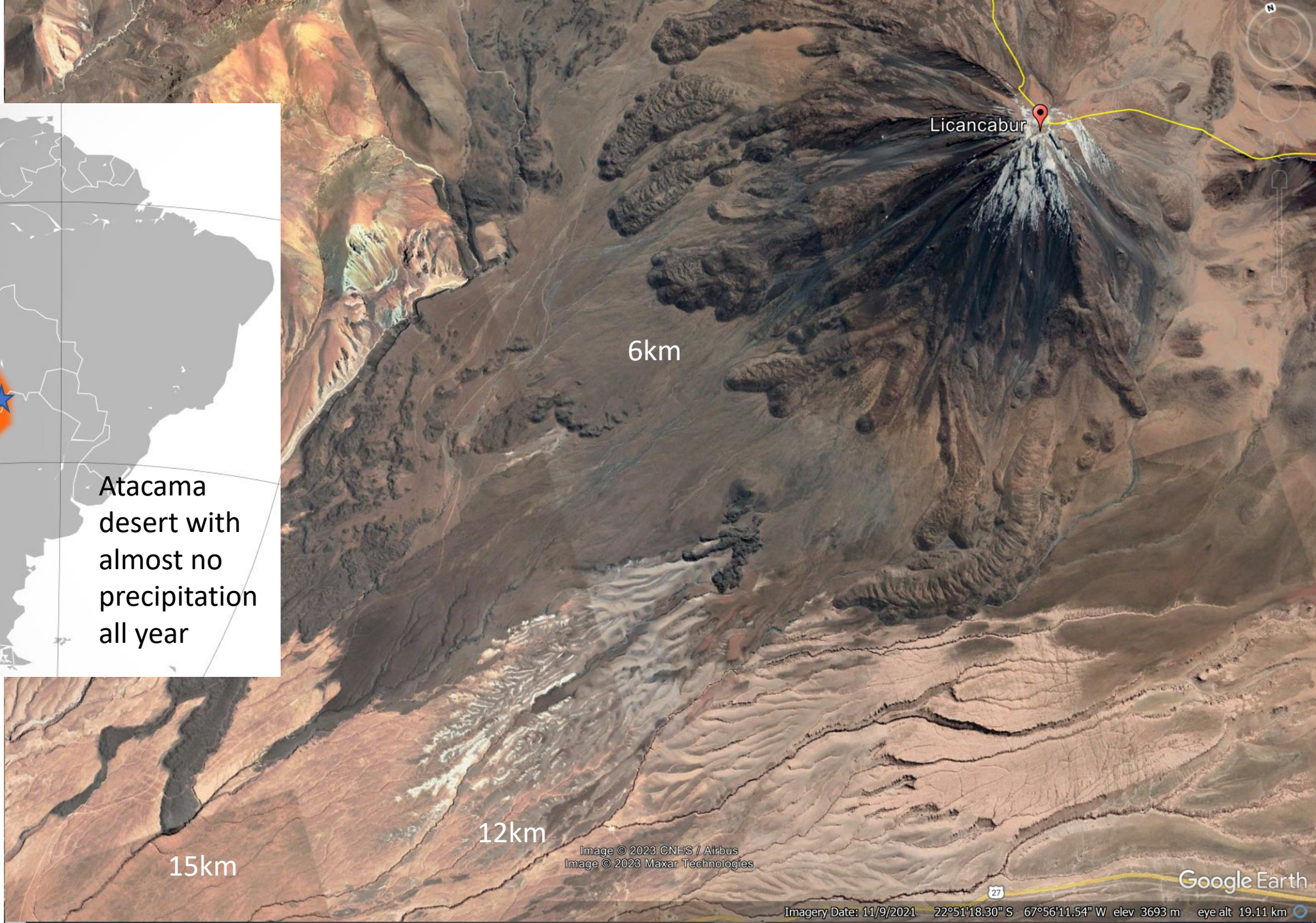


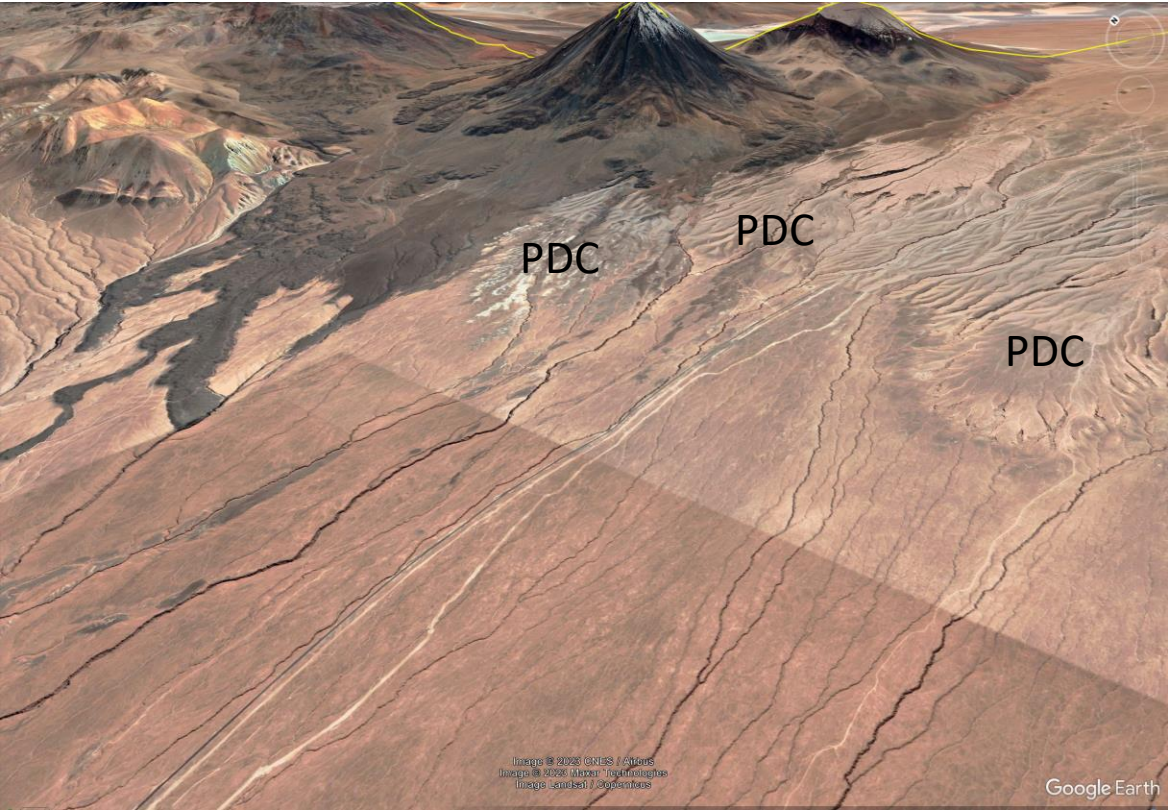


LICANCABUR LAHARZ Exercise



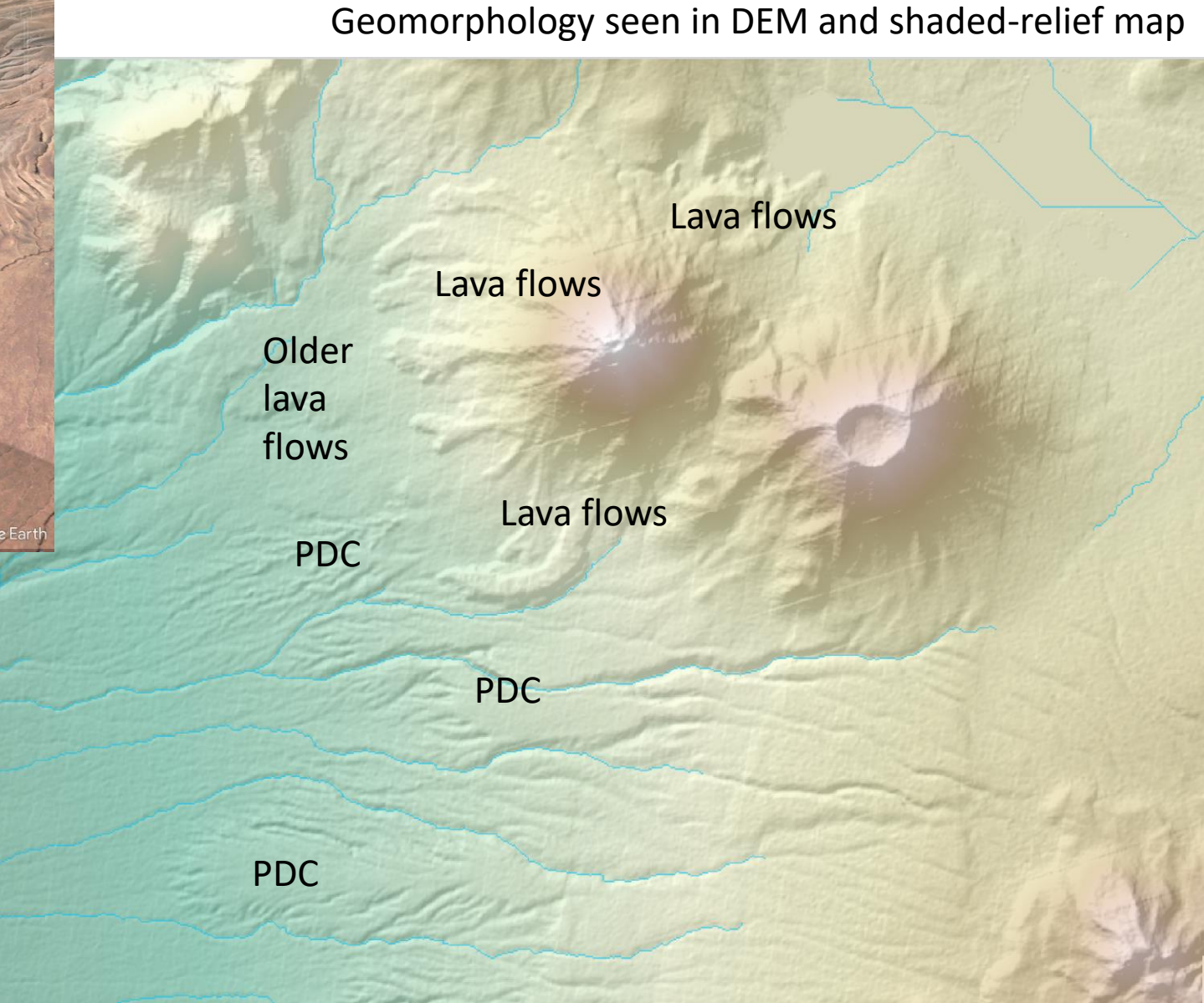






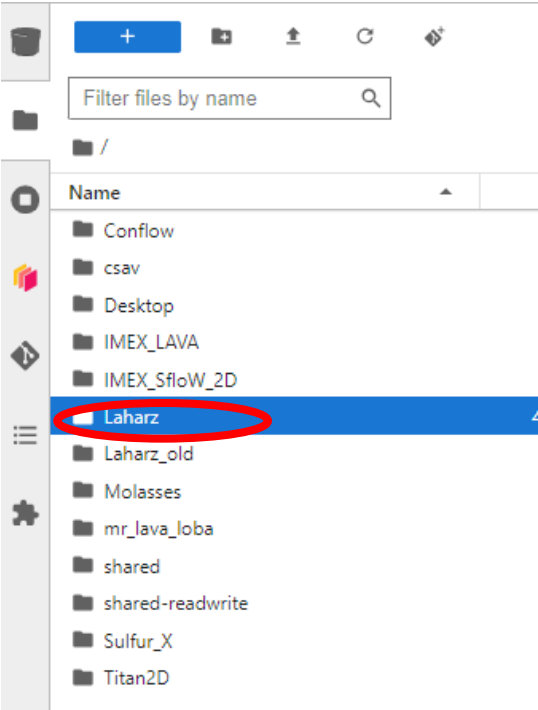
Google Earth and satellite imagery

PDCs = Pyroclastic density currents

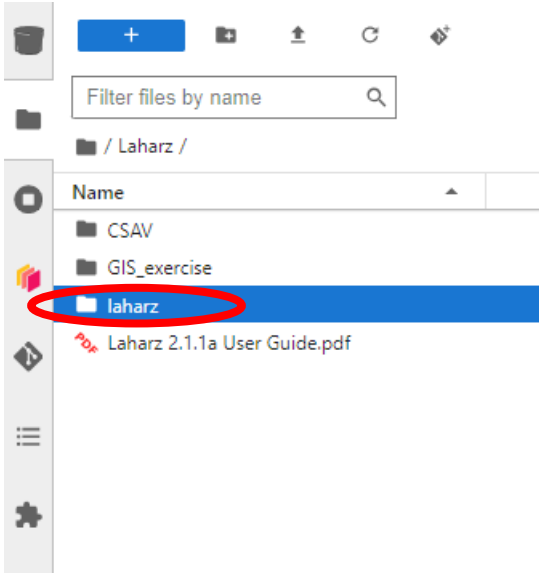


Geomorphology seen in DEM and shaded-relief map

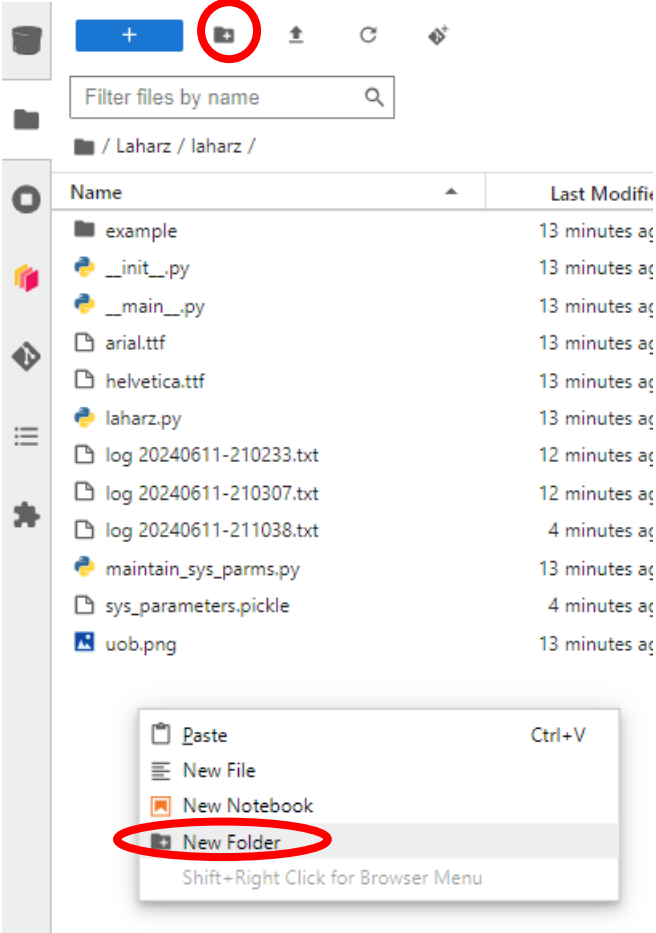
Let's make a folder to work in!



Double click on “Laharz”

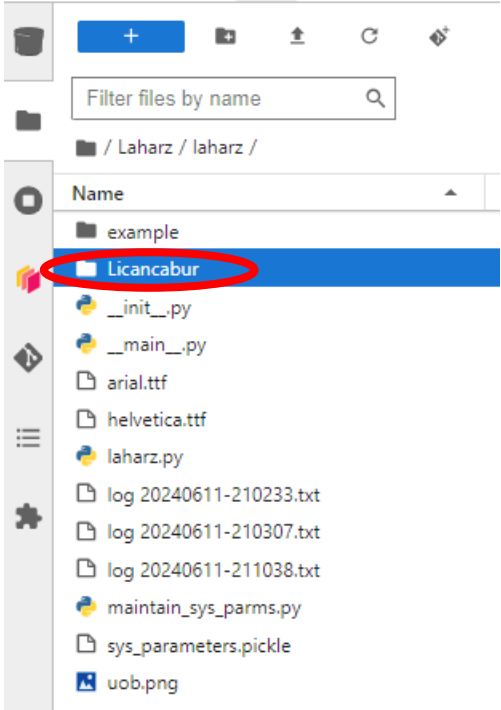


Double click on “laharz”

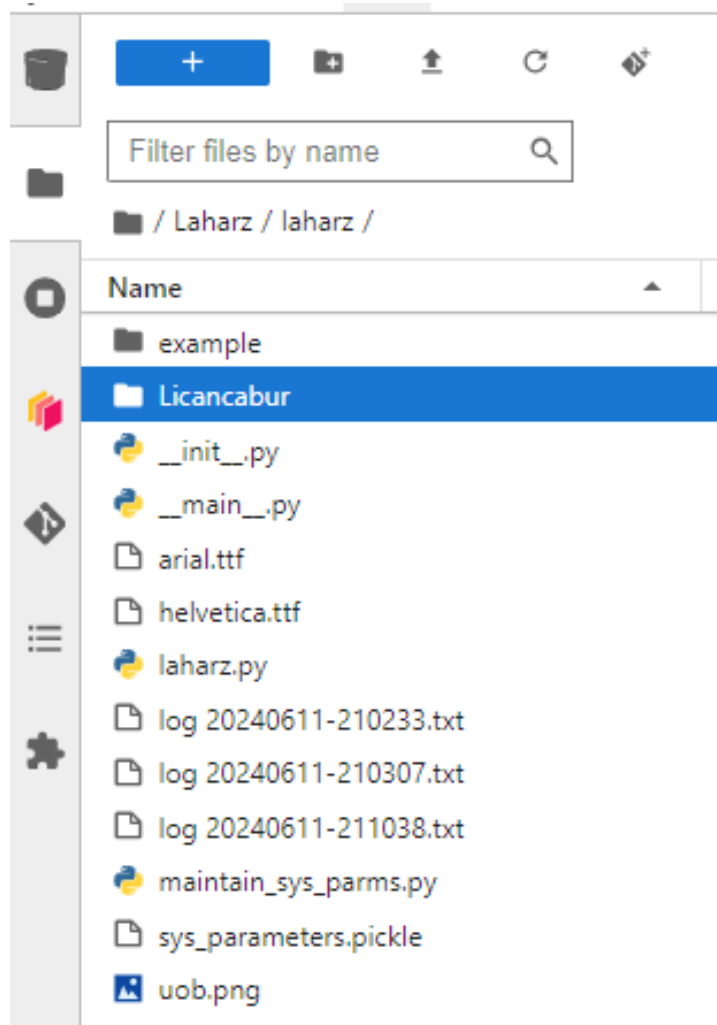


Right click in a blank space and choose “New Folder”

Or click the “New folder” icon at the top

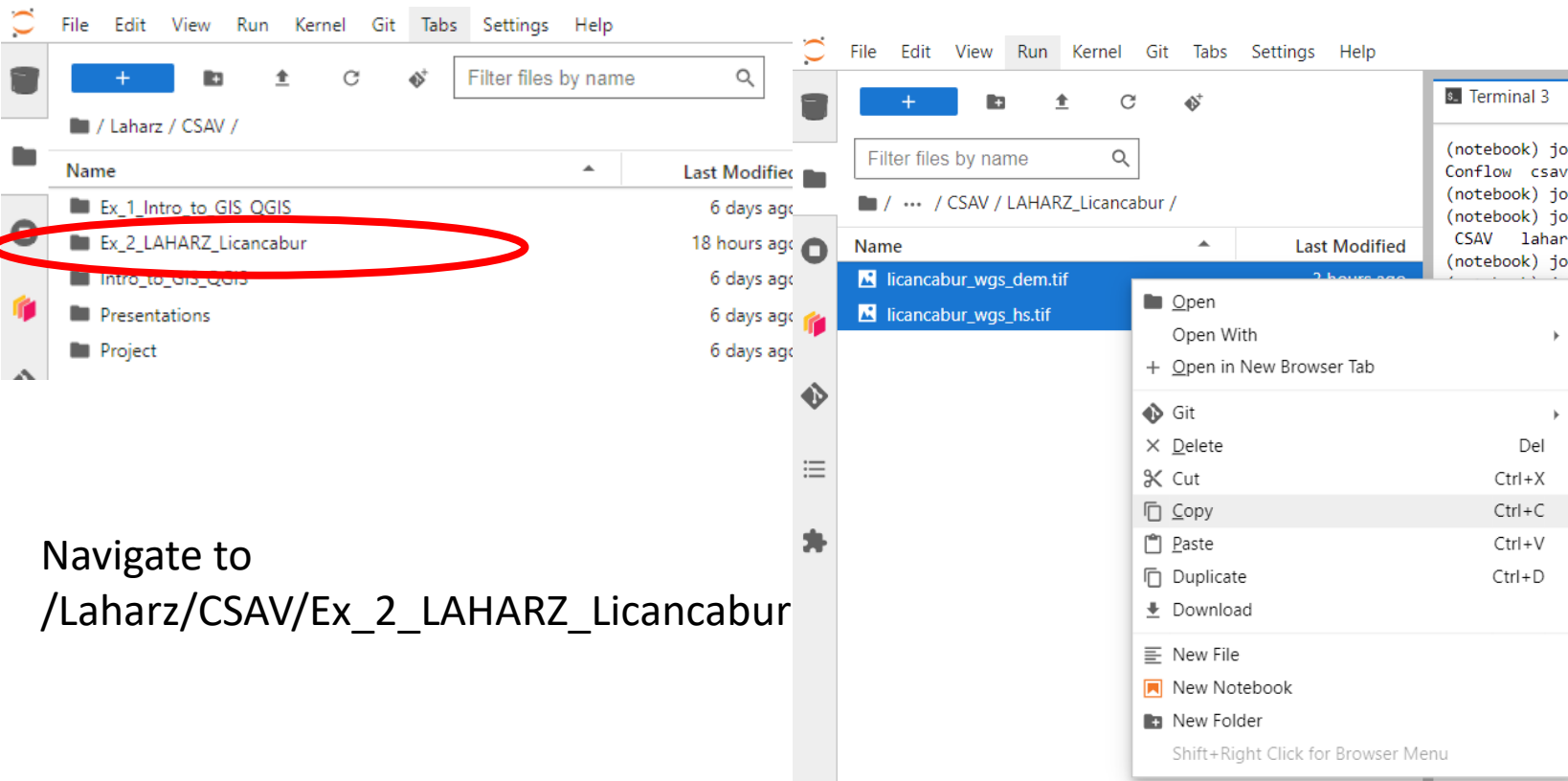


Name the new folder “Licancabur”



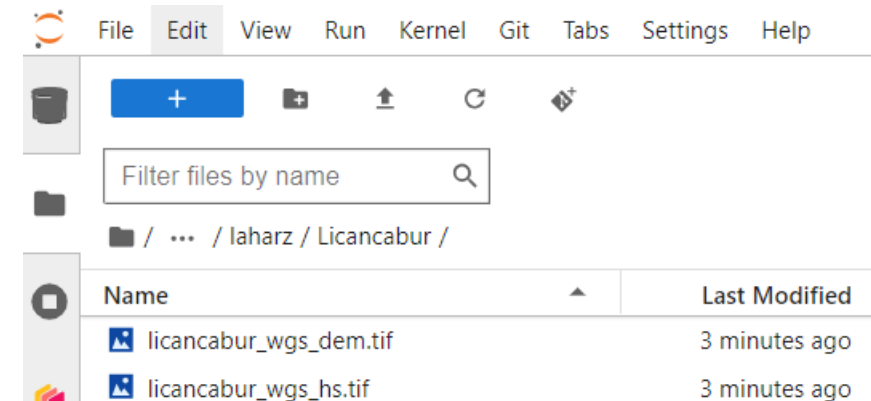
When you work on your projects, you'll want to make a new folder here too
/Laharz/laharz/new_volcano_name

Let's add the Licancabur DEM to the new folder

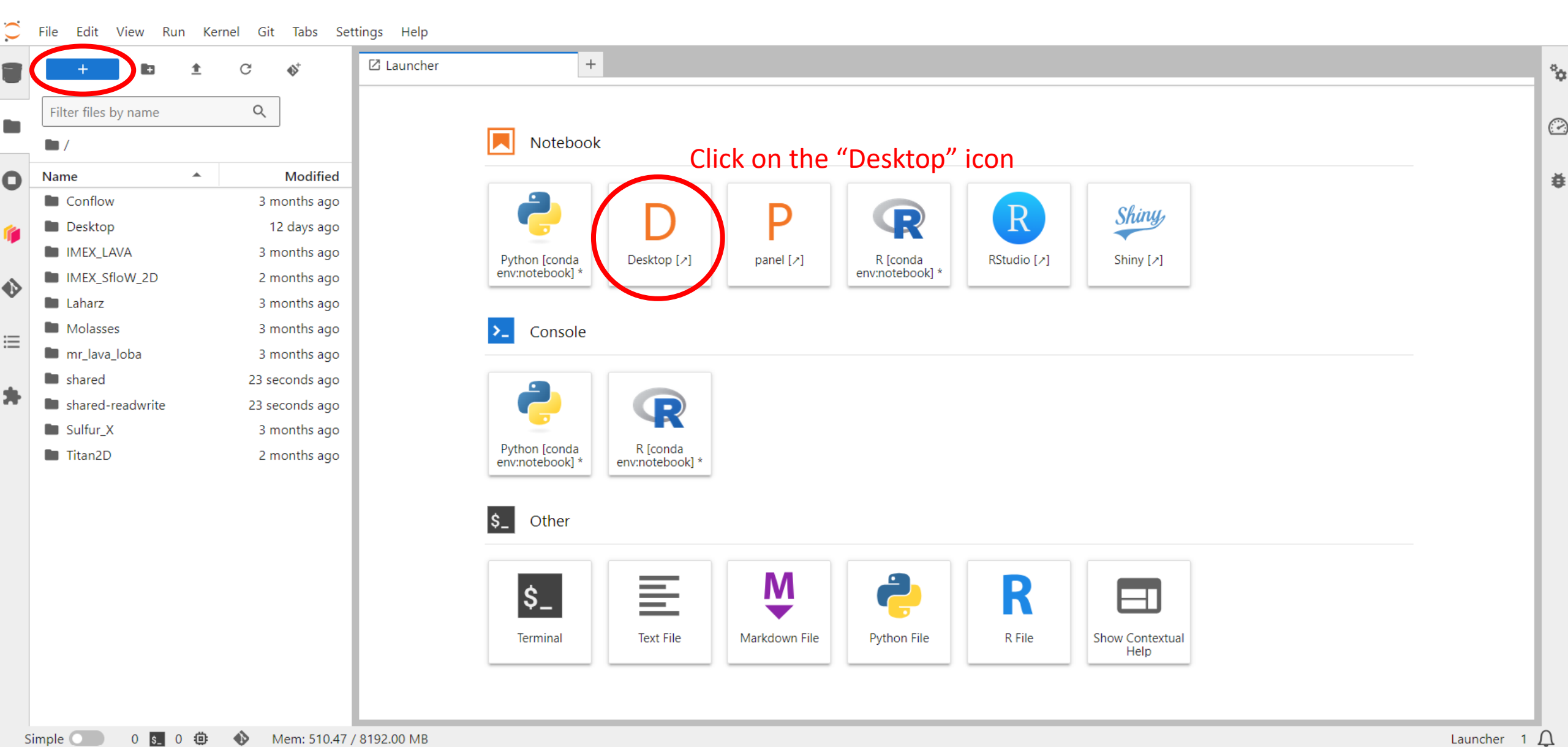


Navigate to
/Laharz/CSAV/Ex_2_LAHARZ_Licancabur

- Highlight the two Licancabur files (DEM and hillshade)
- Right click
- Click “copy”



- Navigate to
/Laharz/laharz/Licancabur/
- Right click
- Click “paste”



Click on the “Desktop” icon

Tip: If you lose your “Launcher” screen, click the “plus” icon at the top left for “New Launcher”

Applications

Wed 5 Jun, 16:20 Default user



File System



Home



QGIS

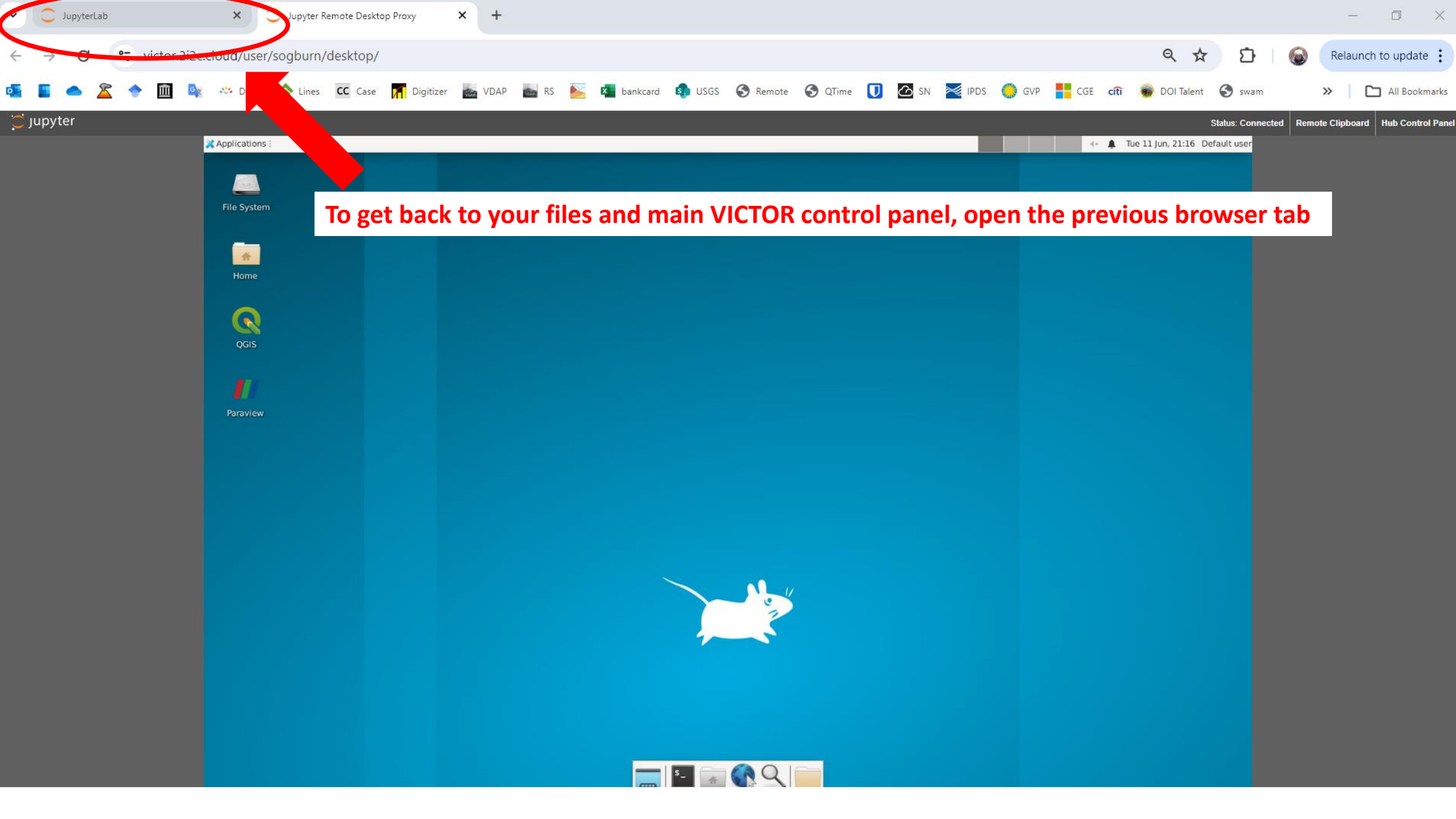


Paraview

QGIS

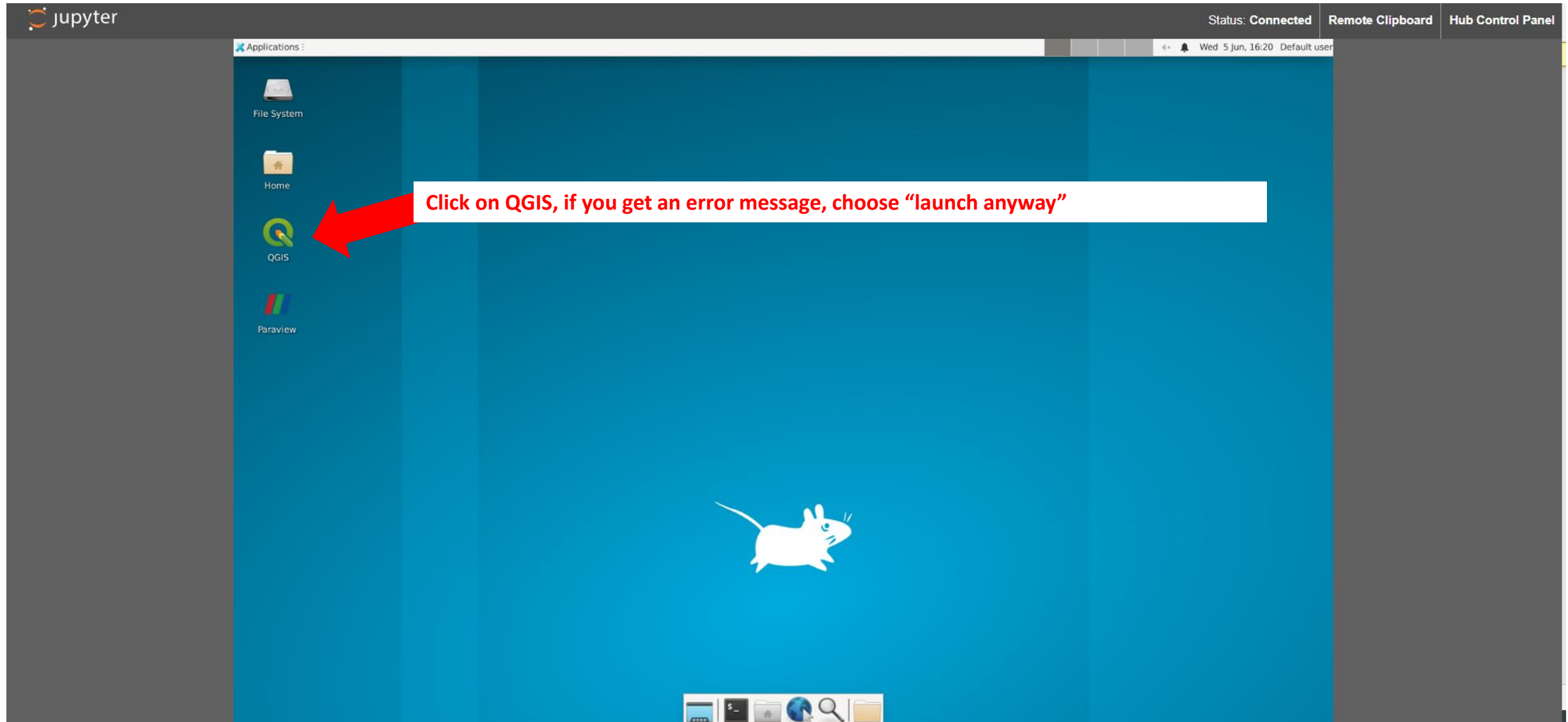
Command line

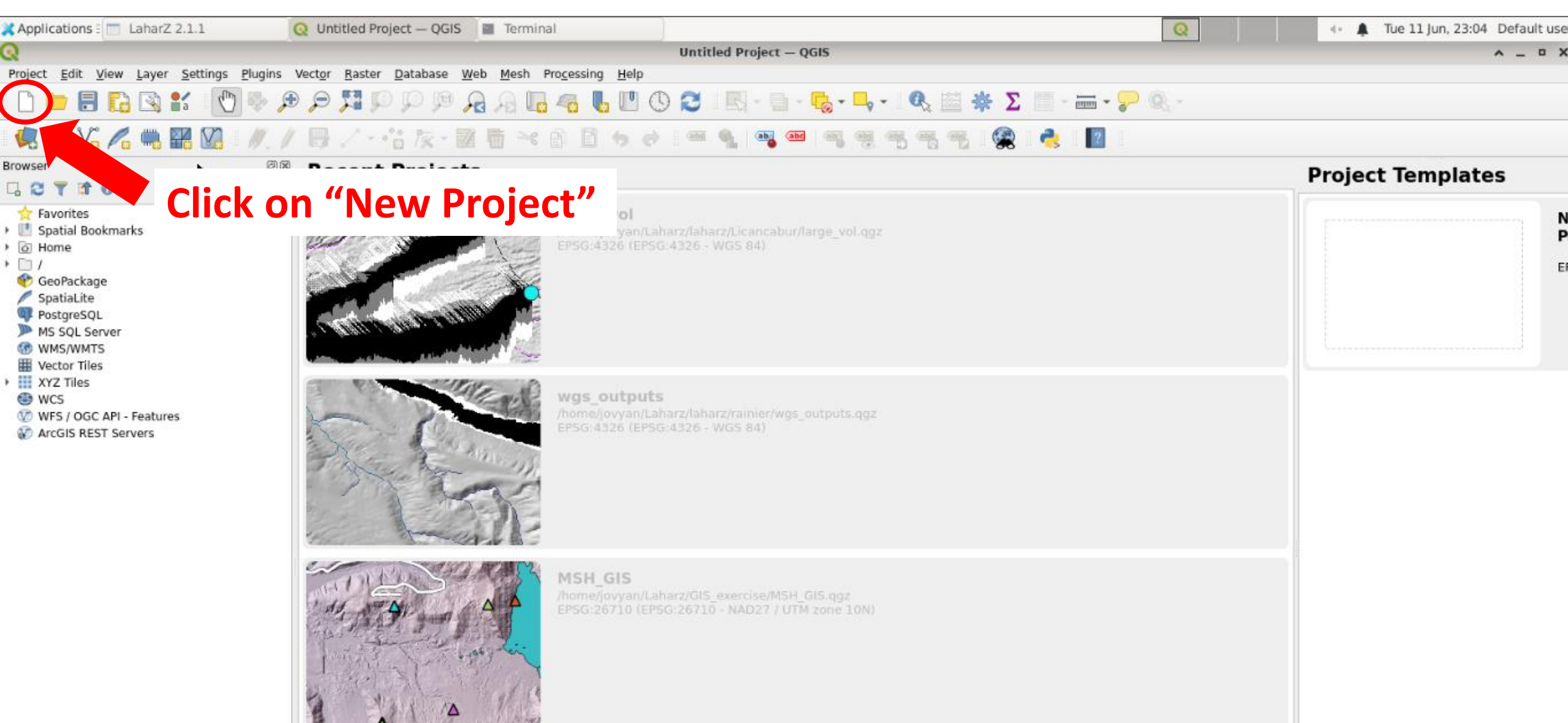




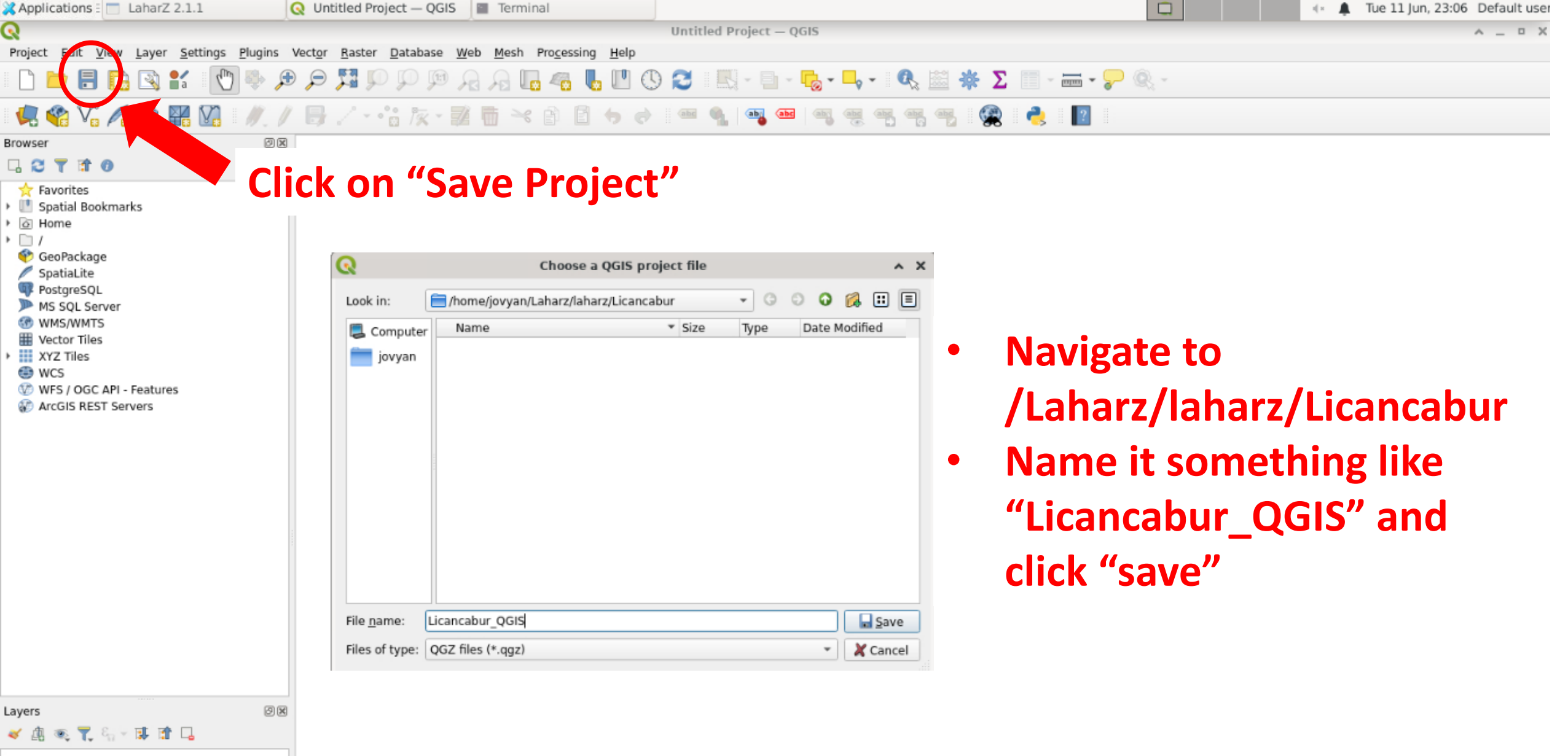
To get back to your files and main VICTOR control panel, open the previous browser tab

Let's make a filled DEM.





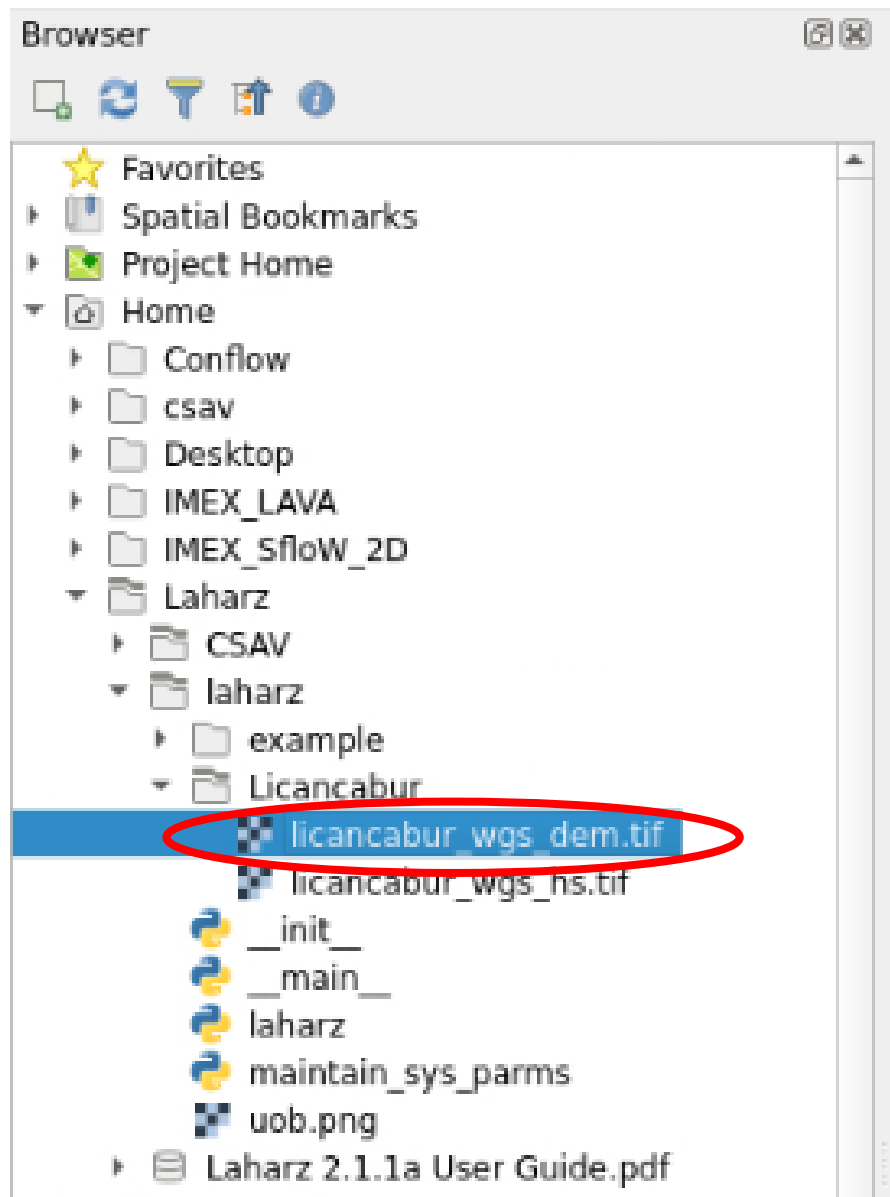
Click on "New Project"



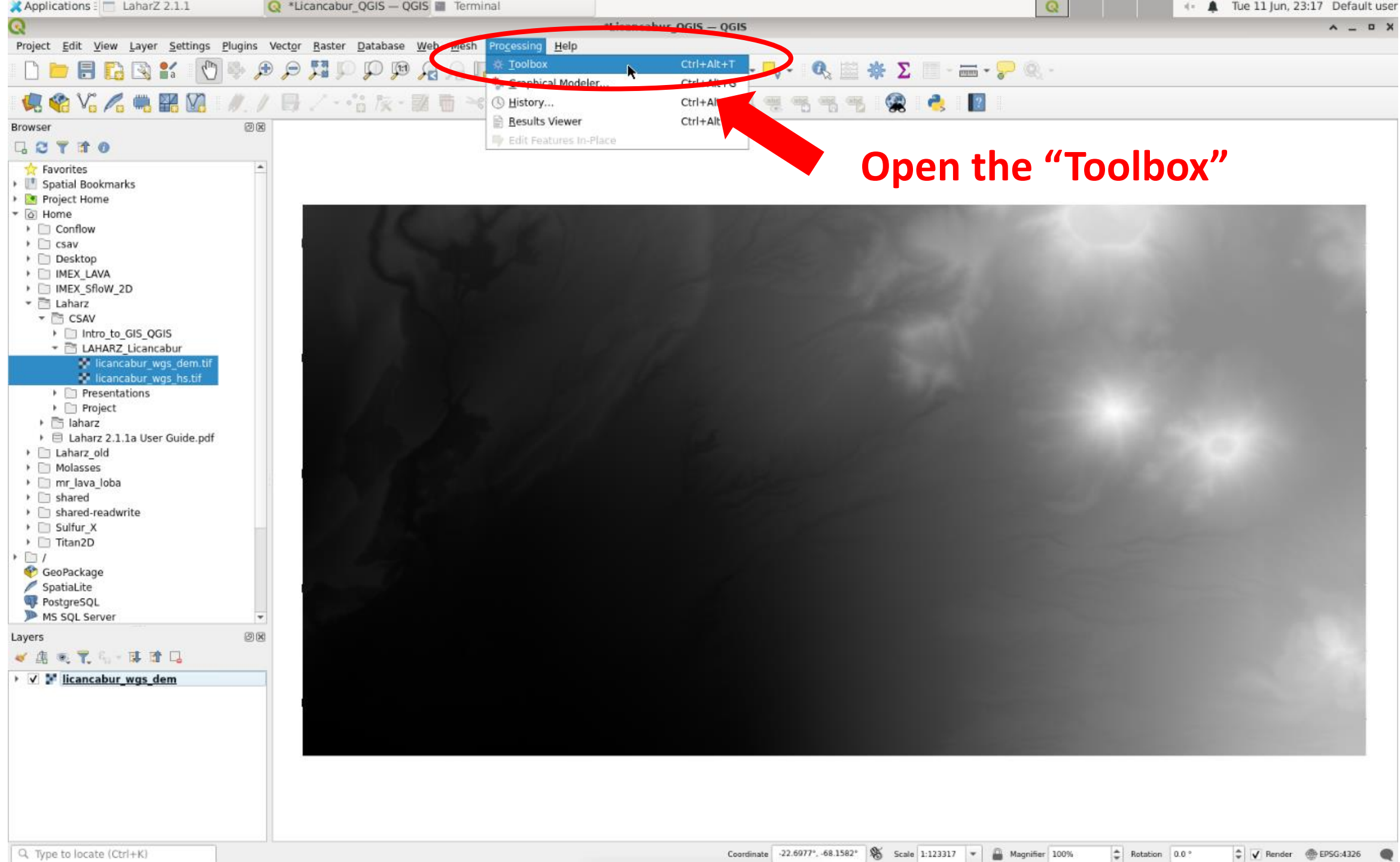
The image shows the QGIS desktop application interface. At the top, the menu bar includes Project, Edit, View, Layer, Settings, Plugins, Vector, Raster, Database, Web, Mesh, Processing, and Help. Below the menu bar is a toolbar with various icons. A red circle highlights the 'Save Project' icon (a floppy disk), and a red arrow points to it from the text 'Click on "Save Project"'. Below the toolbar is the 'Browser' panel on the left, showing a tree view of file locations. At the bottom is the 'Layers' panel. In the center, a 'Choose a QGIS project file' dialog box is open. The 'Look in:' dropdown shows the path '/home/jovyan/Laharz/laharz/Licancabur'. The file list is empty. The 'File name:' field contains 'Licancabur_QGIS'. The 'Files of type:' dropdown is set to 'QGZ files (*.qgz)'. The 'Save' button is highlighted.

Click on "Save Project"

- Navigate to /Laharz/laharz/Licancabur
- Name it something like "Licancabur_QGIS" and click "save"



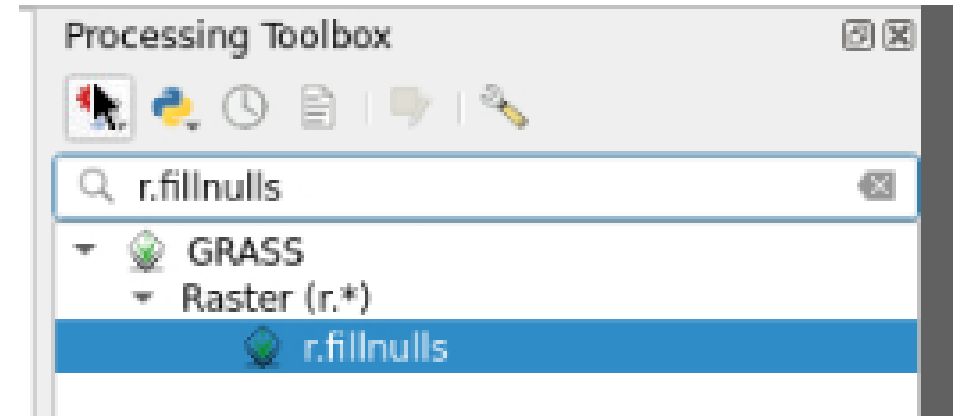
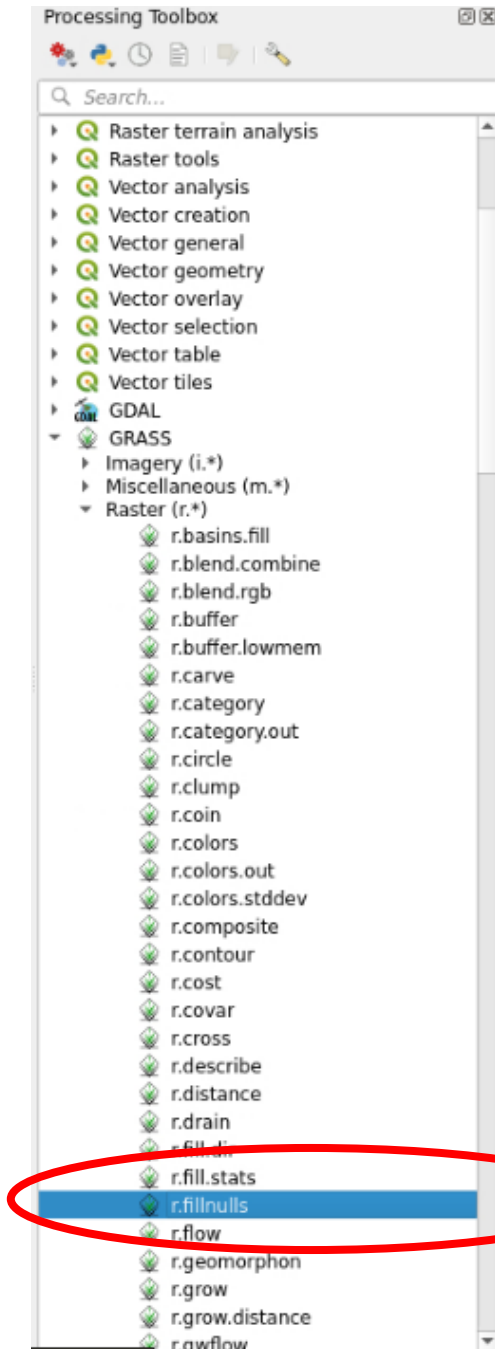
- **Navigate to the Laharz/laharz/Licancabur folder**
- **Drag and drop the “licancabur_wgs_dem.tif” DEM file onto your map**



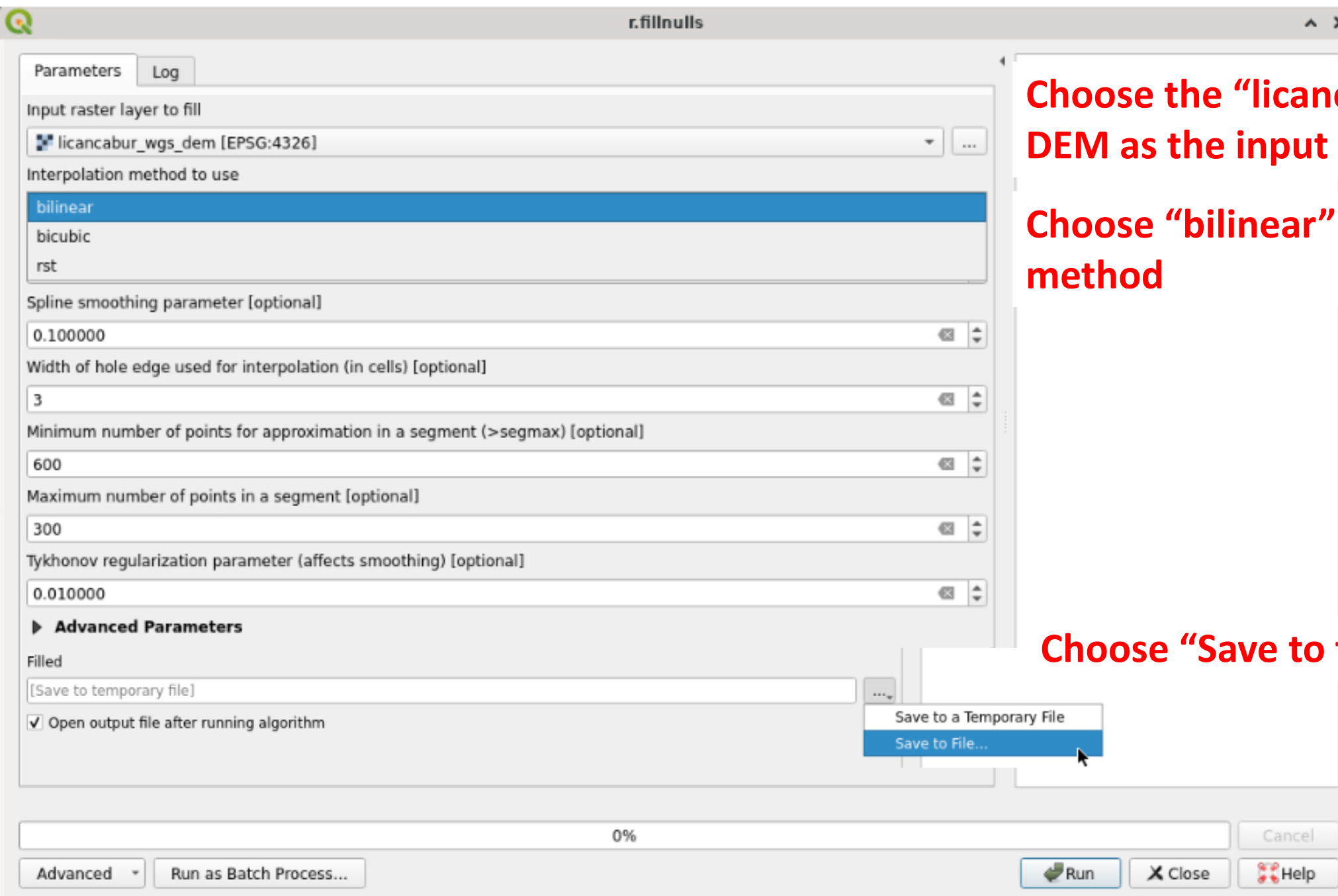
Open the "Toolbox"

Let's make a filled DEM.

Navigate to
GRASS/Raster/r.fillnulls
and double click on it



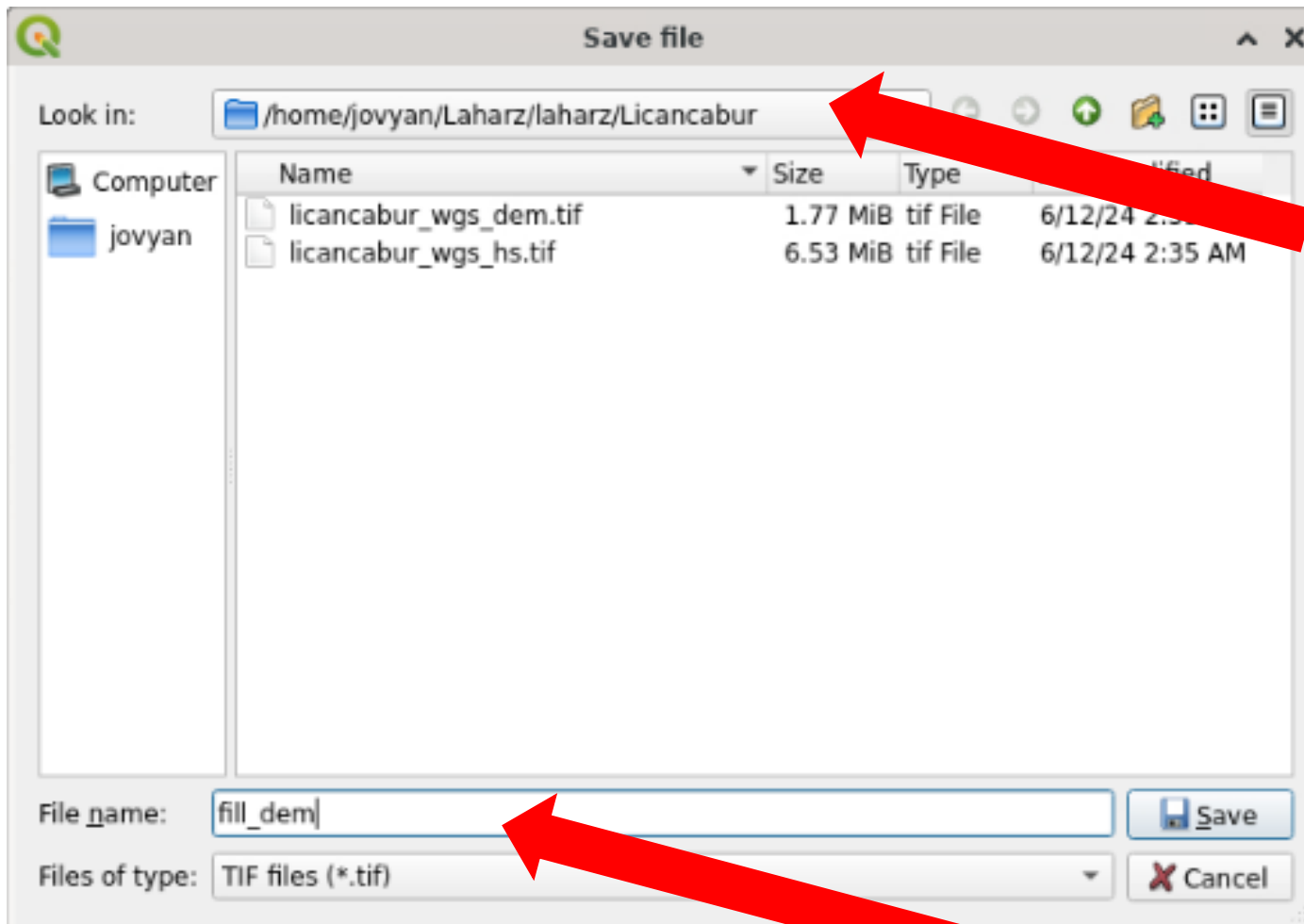
OR type
“r.fillnulls” into the
search box and double
click on it



Choose the "licancabur_wgs_dem.tif" DEM as the input raster

Choose "bilinear" as the interpolation method

Choose "Save to file"



Navigate to the
/Laharz/laharz/Licancabur folder

Name the file "fill_dem"
Click "save"



r.fillnulls



Parameters

Log

Input raster layer to fill

licancabur_wgs_dem [EPSG:4326]

Interpolation method to use

bilinear

Spline tension parameter [optional]

40.000000

Spline smoothing parameter [optional]

0.100000

Width of hole edge used for interpolation (in cells) [optional]

3

Minimum number of points for approximation in a segment (>segmax) [optional]

600

Maximum number of points in a segment [optional]

300

Tykhonov regularization parameter (affects smoothing) [optional]

0.010000

► Advanced Parameters

Filled

/home/jovyan/Laharz/laharz/Licancabur/fill_dem.tif

☒ Open output file after running algorithm**r.fillnulls**

Fills no-data areas in raster maps using spline interpolation.

0%

Advanced ▾

Run as Batch Process...

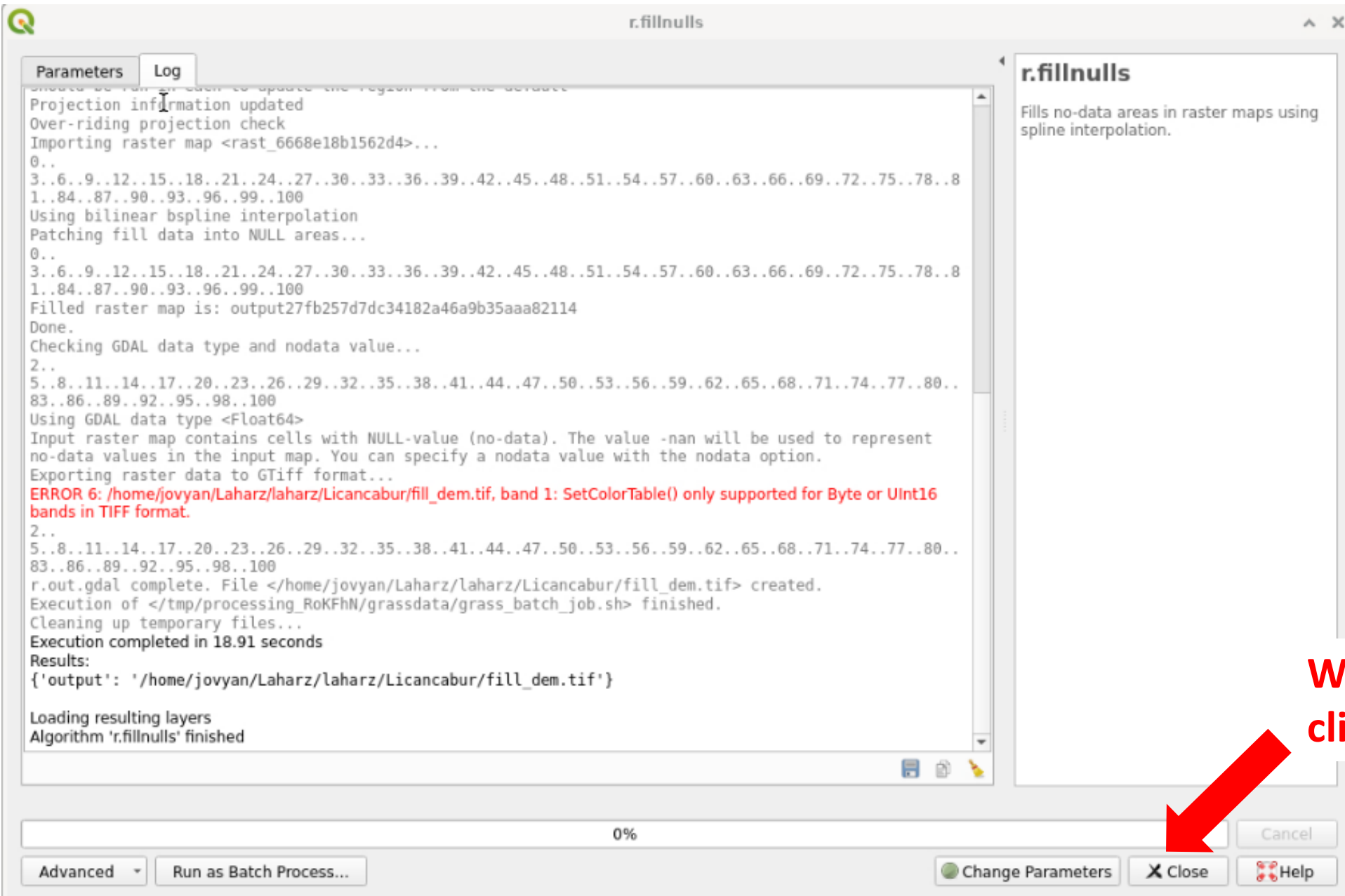


Run

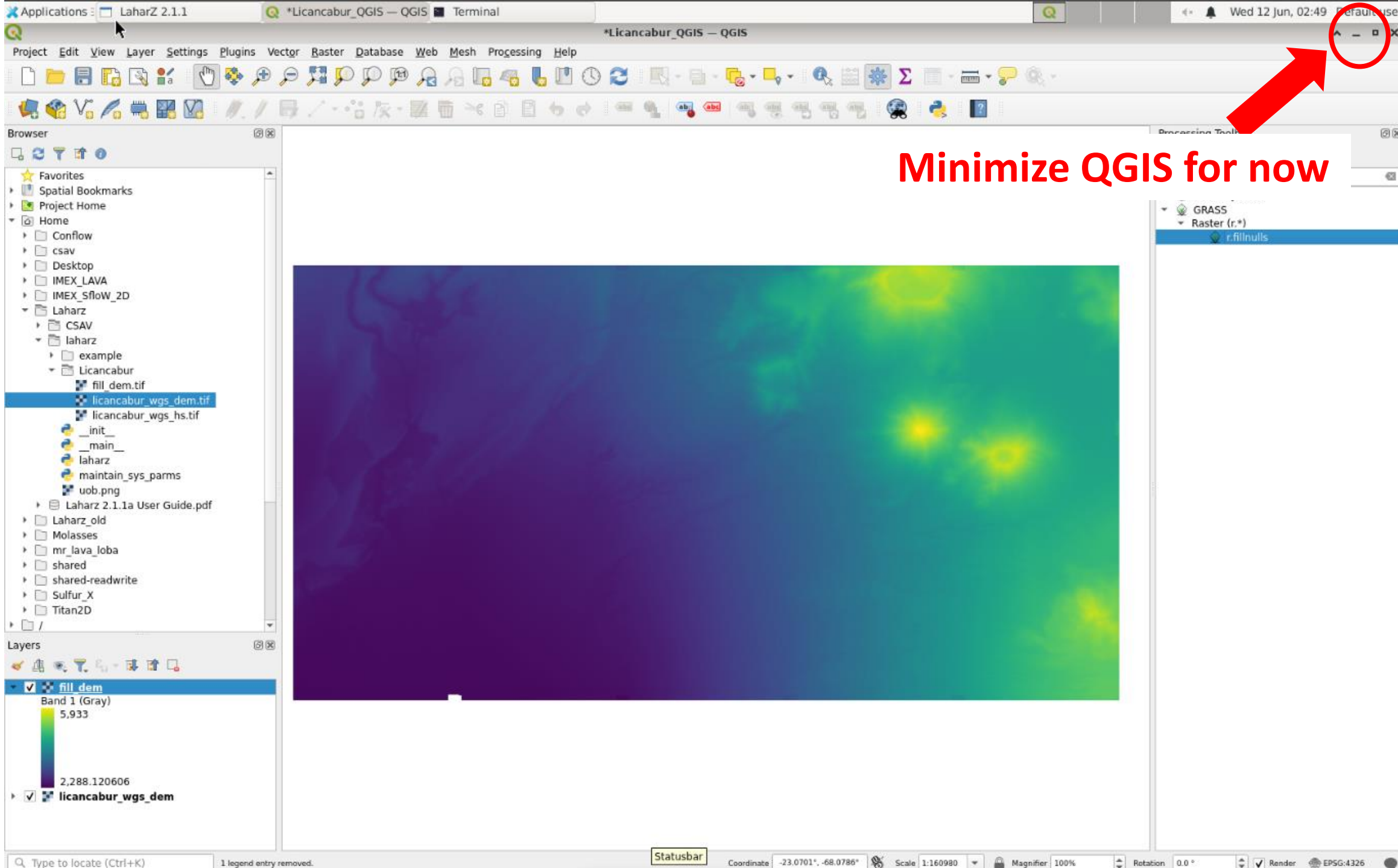
Close

Help

Click "Run"

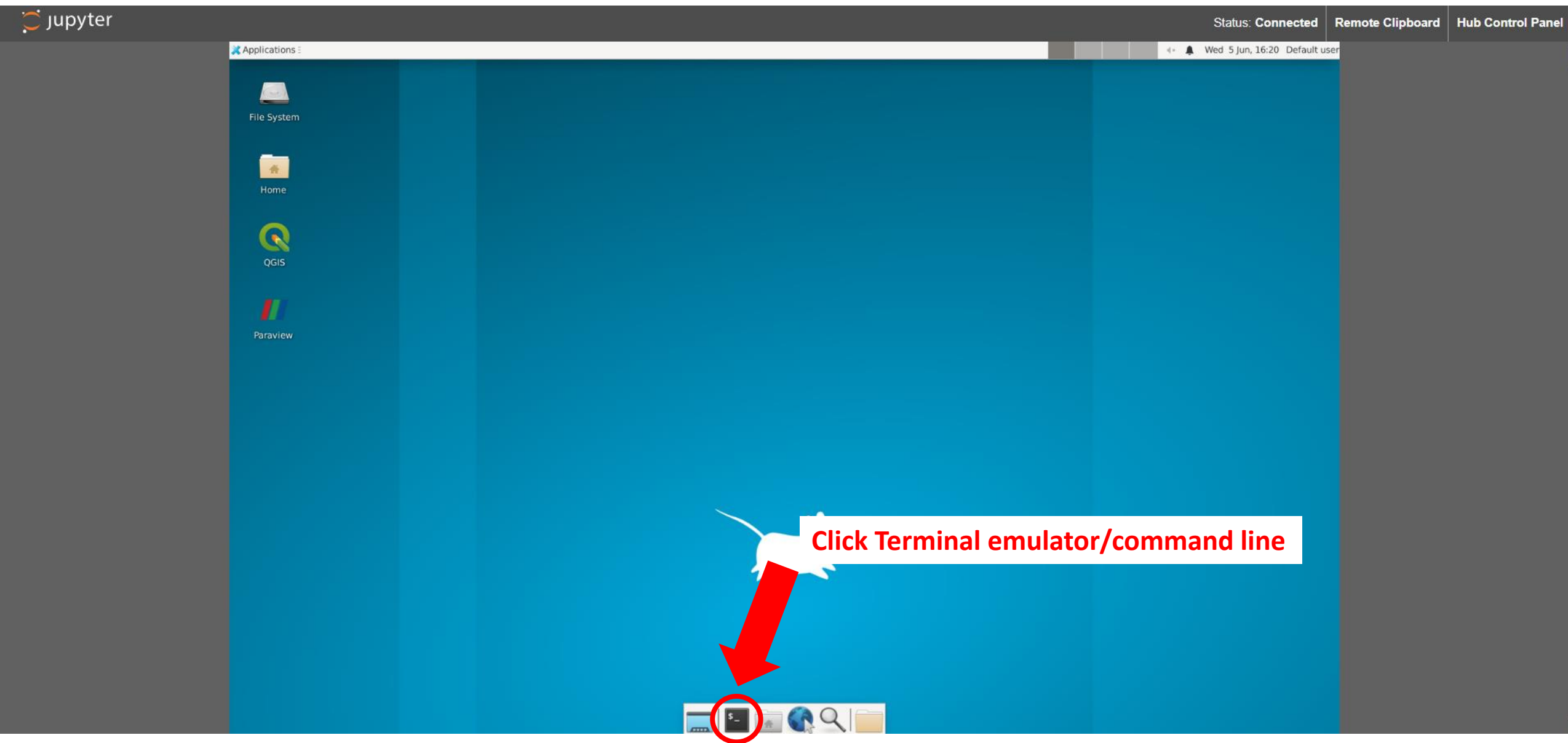


When it's done,
click "Close"



Minimize QGIS for now

Let's run LAHARZ



Using the command line

- Type a command and hit “enter”
- To list the files in the folder you’re in: **ls**
- To move to a folder: **cd name_of_folder**
 - If you type part of a folder name and then click the “tab” key, it will autocomplete
 - Move to Laharz folder: **cd Laharz/**
- List what’s in the “Laharz” folder: **ls**
 - You’ll see a “laharz” folder, the “CSAV” folder you copied there, and the user manual
- Move to “laharz” folder: **cd laharz/**
- List what’s in the “laharz” folder: **ls**
 - You should have the “Licancabur” folder that you created earlier and a bunch of other files

```
Terminal
File Edit View Search Terminal Help
jovyan@jupyter-sogburn:~$ ls
Conflow  IMEX_LAVA    Laharz  mr_lava_loba  shared-readwrite  Titan2D
Desktop  IMEX_Sflow_2D  Molasses  shared        Sulfur_X
jovyan@jupyter-sogburn:~$
```

```
Terminal
File Edit View Search Terminal Help
jovyan@jupyter-sogburn:~$ ls
Conflow  IMEX_LAVA    Laharz  mr_lava_loba  shared-readwrite  Titan2D
Desktop  IMEX_Sflow_2D  Molasses  shared        Sulfur_X
jovyan@jupyter-sogburn:~$ cd Laharz/
jovyan@jupyter-sogburn:~/Laharz$
```

```
Terminal
File Edit View Search Terminal Help
jovyan@jupyter-sogburn:~$ ls
Conflow  IMEX_LAVA    Laharz  mr_lava_loba  shared-readwrite  Titan2D
Desktop  IMEX_Sflow_2D  Molasses  shared        Sulfur_X
jovyan@jupyter-sogburn:~$ cd Laharz/
jovyan@jupyter-sogburn:~/Laharz$ ls
CSAV  laharz  'Laharz 2.1.1a User Guide.pdf'
```

```
Terminal
File Edit View Search Terminal Help
jovyan@jupyter-sogburn:~$ ls
Conflow  IMEX_LAVA    Laharz_old  shared  Titan2D
csav     IMEX_Sflow_2D  Molasses    shared-readwrite
Desktop  Laharz        mr_lava_loba  Sulfur_X
jovyan@jupyter-sogburn:~$ cd Laharz
jovyan@jupyter-sogburn:~/Laharz$ ls
CSAV  laharz  'Laharz 2.1.1a User Guide.pdf'
jovyan@jupyter-sogburn:~/Laharz$ cd laharz/
jovyan@jupyter-sogburn:~/Laharz/laharz$ ls
arial.ttf      laharz.py      'log 20240611-211038.txt'  uob.png
example        'log 20240611-210233.txt'  main_.py
helvetica.ttf  'log 20240611-210307.txt'  maintain_sys_parms.py
__init__.py    sys_parameters.pickle
jovyan@jupyter-sogburn:~/Laharz/laharz$
```

Some command line tips

- If you type part of a folder name and then click the “tab” key, it will autocomplete
- Clicking the up arrow on the keyboard will cycle through commands you’ve typed before

Launch LAHARZ

- Launch LAHARZ: **python laharz.py**

```
Terminal
File Edit View Search Terminal Help
jovyan@jupyter-sogburn:~$ ls
Conflow  IMEX_LAVA      Laharz    mr_lava_loba  shared-readwrite  Titan2D
Desktop  IMEX_Sflow_2D  Molasses  shared        Sulfur_X
jovyan@jupyter-sogburn:~$ cd Laharz/
jovyan@jupyter-sogburn:~/Laharz$ ls
laharz  'Laharz 2.0.0c User Guide.pdf'
jovyan@jupyter-sogburn:~/Laharz$ cd laharz/
jovyan@jupyter-sogburn:~/Laharz/laharz$ ls
example                'log 20240523-182236.txt'  __main__.py
__init__.py            'log 20240523-182530.txt'  maintain_sys_parms.py
laharz.py              'log 20240523-214140.txt'  parameters.pickle
'log 20240306-212417.txt' 'log 20240523-214415.txt'  sys_parameters.pickle
'log 20240415-153916.txt' 'log 20240605-162531.txt'  uob.png
jovyan@jupyter-sogburn:~/Laharz/laharz$ python laharz.py
```

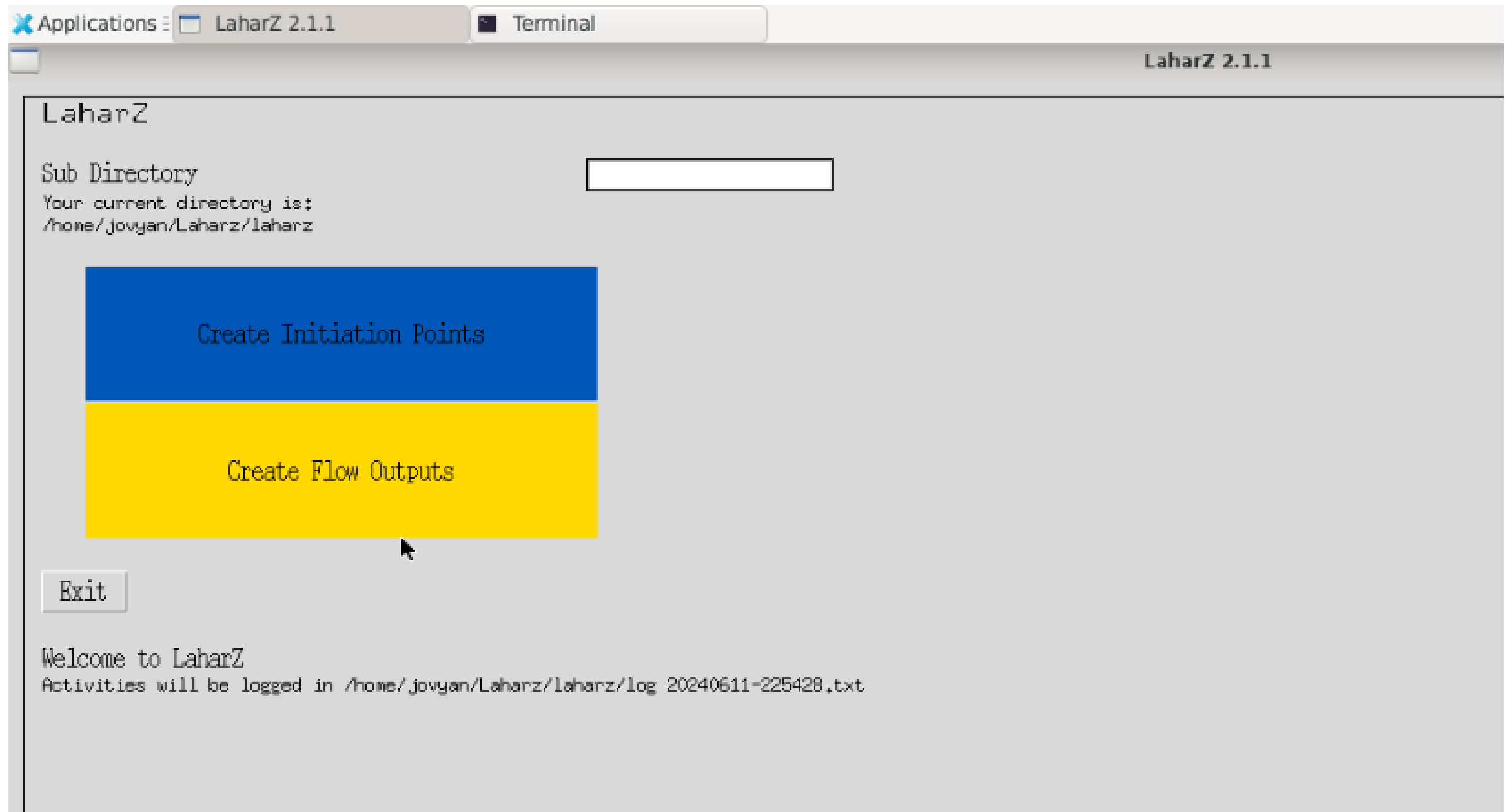
LaharZ

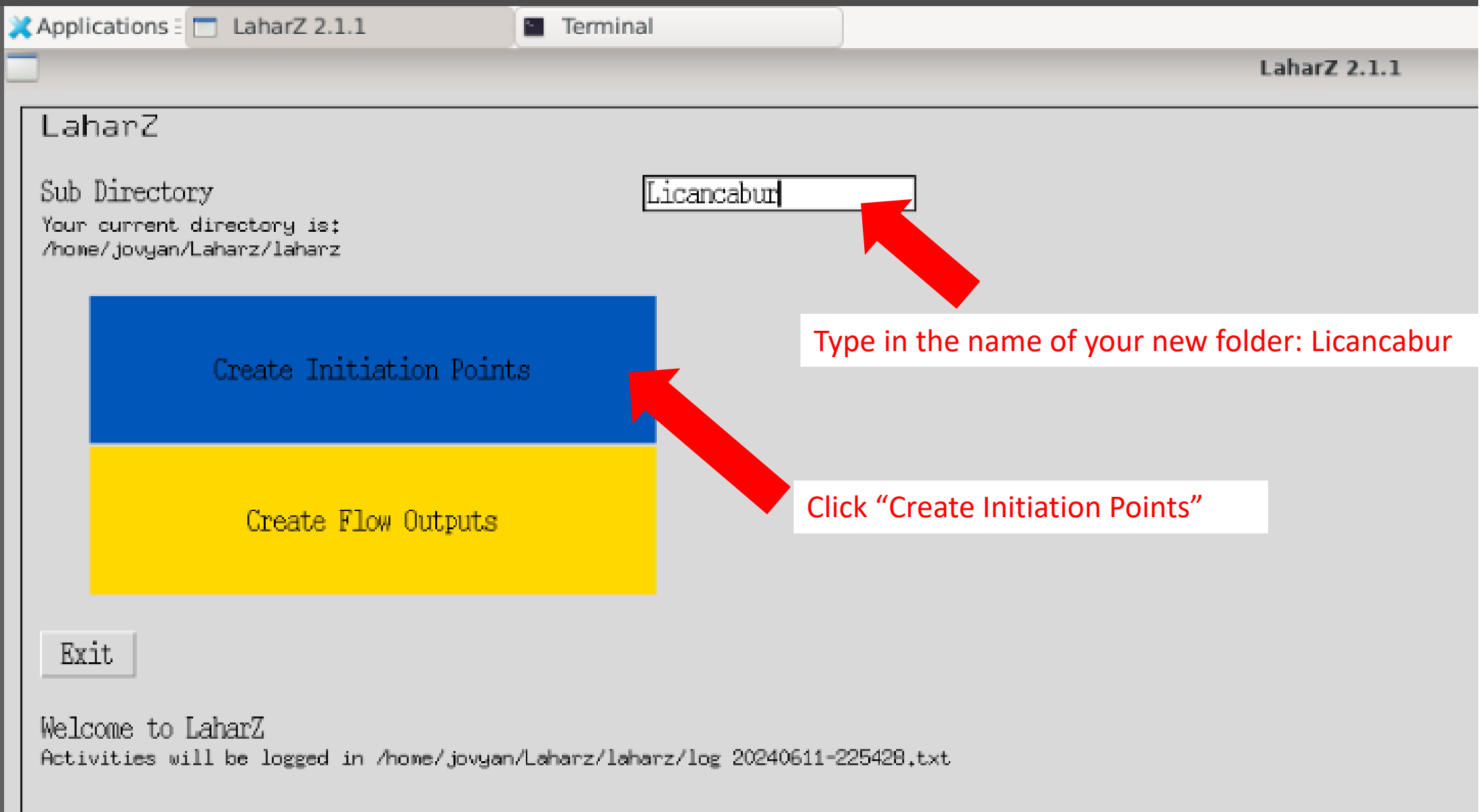


University of
BRISTOL

Version: 2.1.1 stand alone

Let's make a stream network, an H/L cone, and start points





We have some fields to fill in! There are some default values to change.

Applications: LaharZ 2.1.1 *Licancabur_QGIS — QGIS Terminal

LaharZ 2.1.1

Generate Initiation Points

DEM File Name of your DEM file in Licancabur

Stream and Flow Direction Files

- ☒ Generate files
- ☐ Use your own files

Streams file ☒ Name of your streams file to be created in Licancabur

Stream Threshold Threshold value for accumulation to form stream

Flow direction file ☒ Name of your flow direction file to be created in Licancabur

Determine Energy Cone Apex

- ☒ Use Search File
- ☐ Use Longitude/Latitude

Search File ☒ Name of your Search file in Licancabur

Search Option

- ☒ Use Highest Point in Search Area
- ☐ Use Centre of Search Area
- ☐ Use Point 'Centre'

Incremental Height Incremental height in metres

H/L Ratio H/L Ratio

Sea Level No initiation points will be created at sea level or below

☒ Plot Energy Cone Graphics

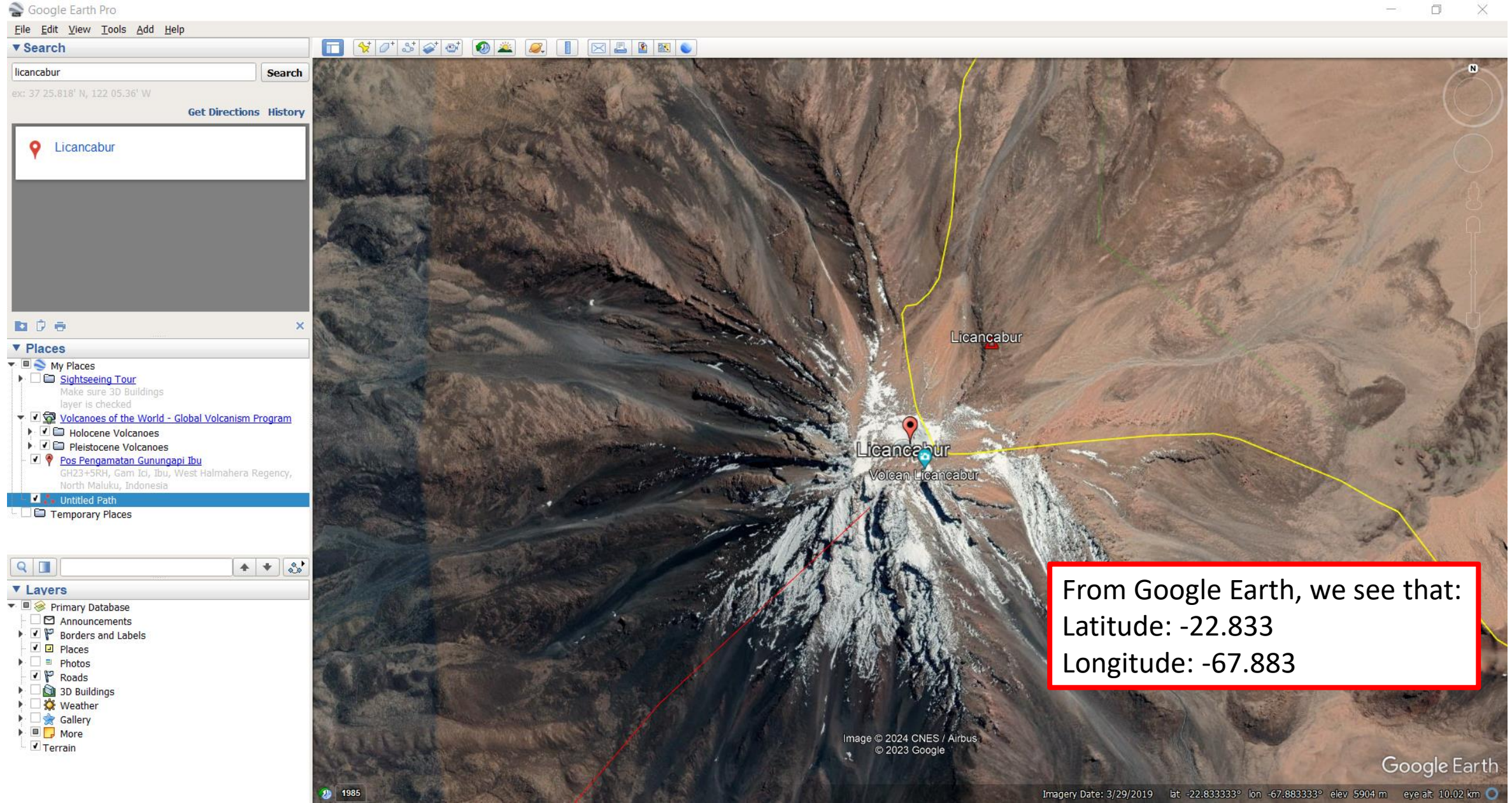
Energy Cone Graphics File ☒ Name of the file for the energy cone graphics in Licancabur

Extent Extent to plot the energy cone/surface (1.3 = 130% of L)

Energy Cone Line File ☒ Name of the file for the energy cone line in Licancabur

Initiation Points File ☒ Name of the file for the initiations points in Licancabur

For this exercise, we need coordinates of the summit



Applications ▾ LaharZ 2.1.1 *Licancabur_QGIS — QGIS Terminal

LaharZ 2.1.1

Generate Initiation Points

DEM File Name of your DEM file in Licancabur

Stream and Flow Direction Files

- ☒ Generate files
- ☐ Use your own files

Streams file ☒ Name of your streams file to be created in Licancabur

Stream Threshold Threshold value for accumulation to form stream

Flow direction file ☒ Name of your flow direction file to be created in Licancabur

Determine Energy Cone Apex

- ☐ Use Search File
- ☒ Use Longitude/Latitude

Apex Longitude, Latitude

Incremental Height Incremental height in metres

H/L Ratio H/L Ratio

Sea Level No initiation points will be created at sea level or below

☒ Plot Energy Cone Graphics

Energy Cone Graphics File ☒ Name of the file for the energy cone graphics in Licancabur

Extent Extent to plot the energy cone/surface (1.3 = 130% of L)

Energy Cone Line File ☒ Name of the file for the energy cone line in Licancabur

Initiation Points File ☒ Name of the file for the initiations points in Licancabur

For DEM File, type the name of your filled DEM:
fill_dem.tif

For this step, we're going to try a
stream threshold of 10000.

- Streams file: streams_10000.tif
- Stream threshold: 10000
- Flow direction file: flow_10000.tif

We know the coordinates of the summit, so choose
"Use Longitude/Latitude"

Longitude comes first! Type: -67.883,-22.833

- Licancabur is nice and conical, so don't add height
- Use an H/L ratio of 0.25

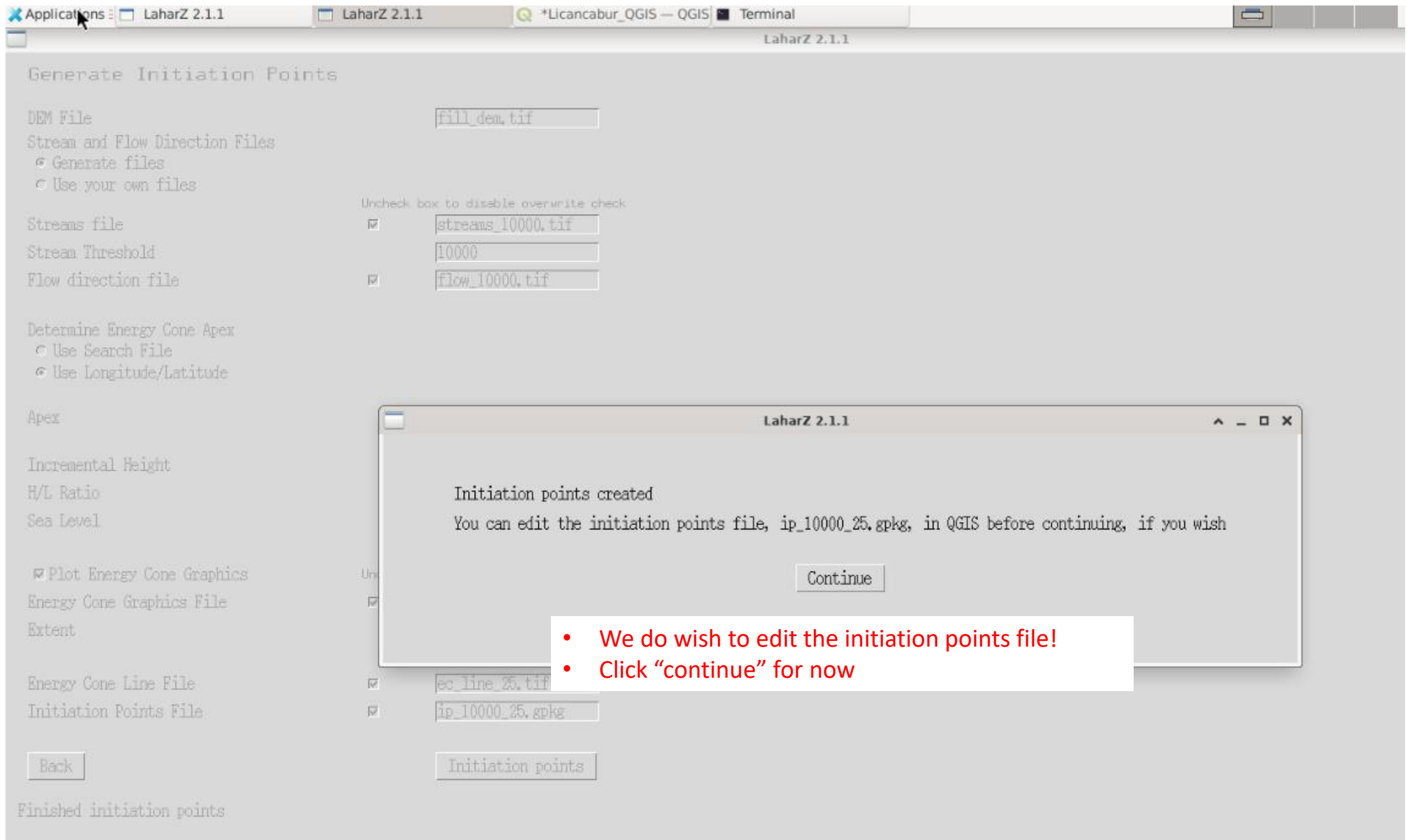
Name the energy cone files with the H/L
ratio used:

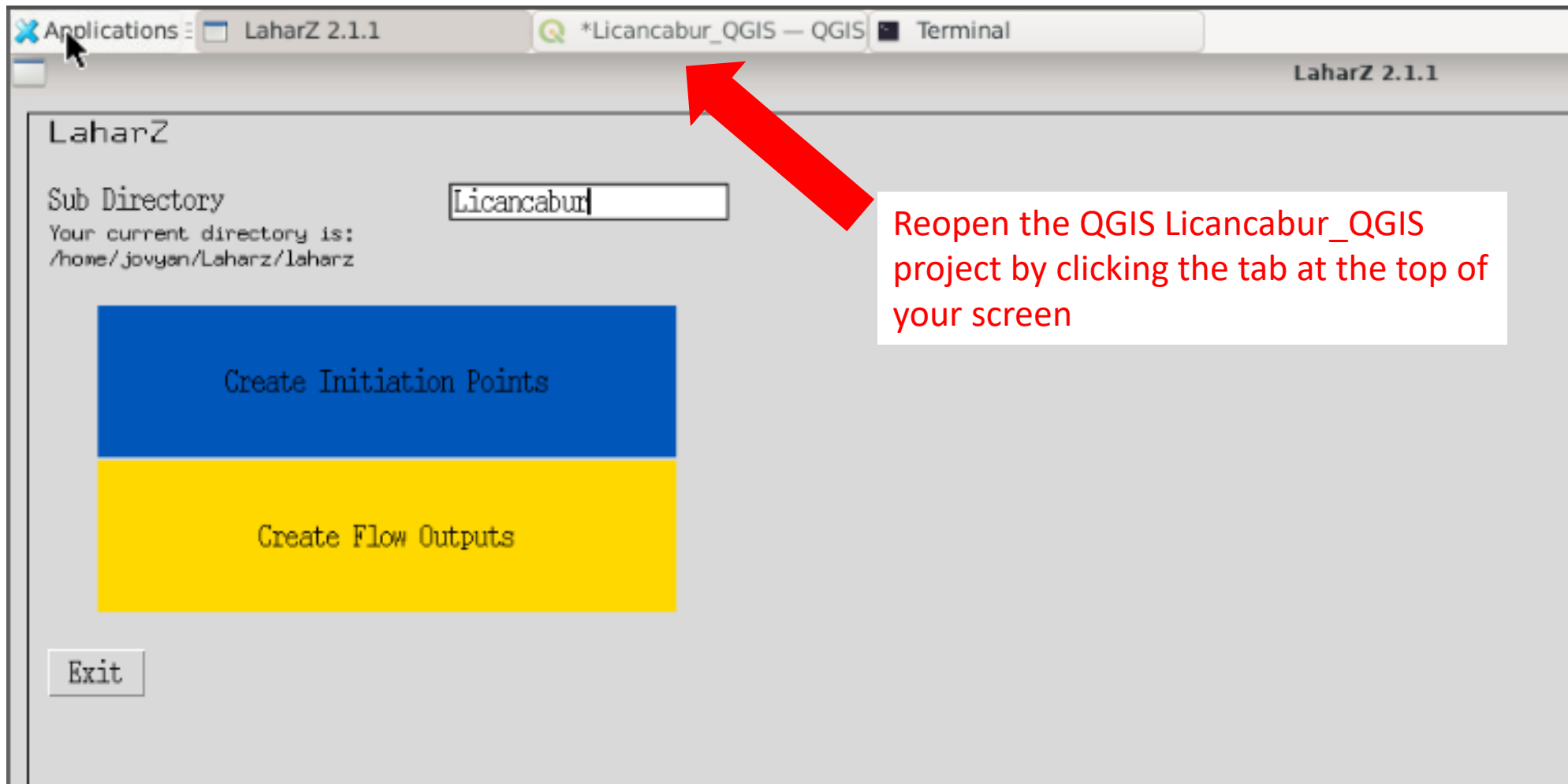
- Energy Cone Graphics: ec_cone_25.tif
- Energy Cone Line: ec_line_25.tif

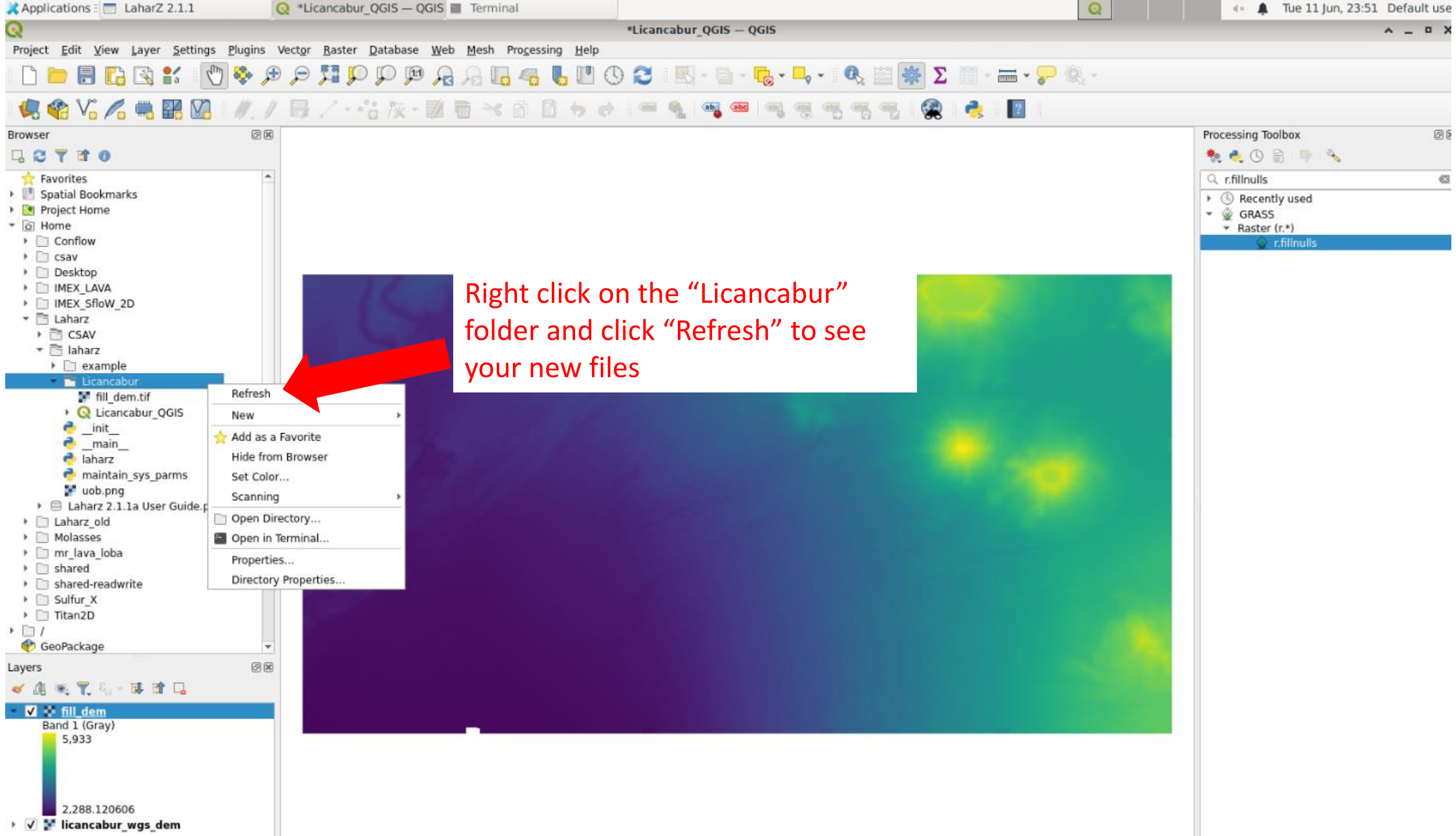
Name the initiation points with the
stream threshold and H/L ratio:

- ip_10000_25.gpkg

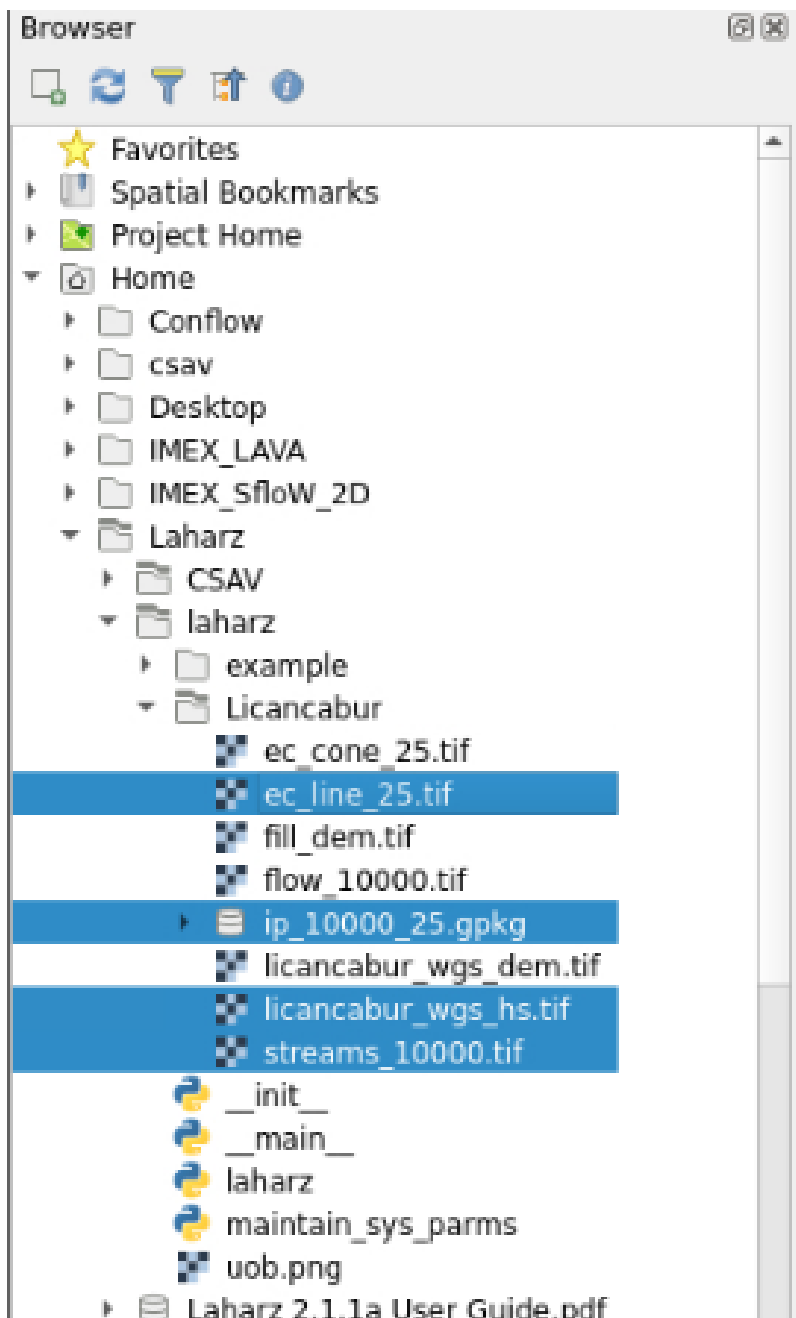
When you're done adding these choices, click
"Initiation points" button







Right click on the "Licancabur" folder and click "Refresh" to see your new files



Let's look at the stream network, H/L cone (ec_line), and the initiation points we created!

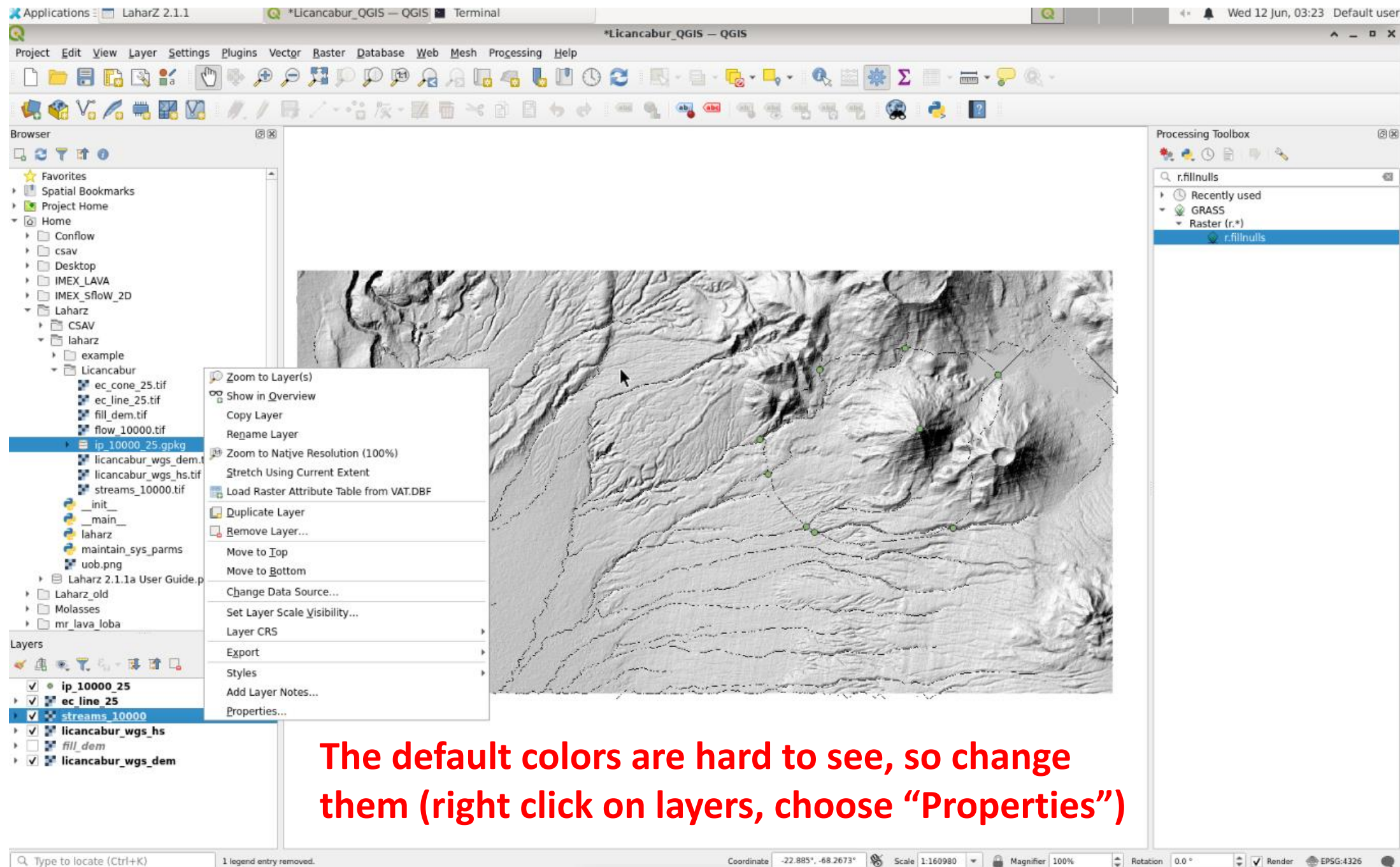
Add these files to your map:

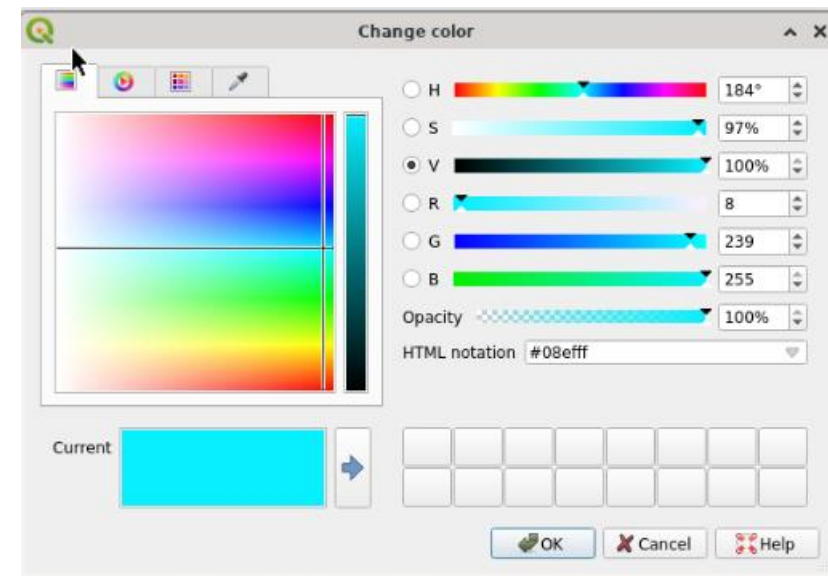
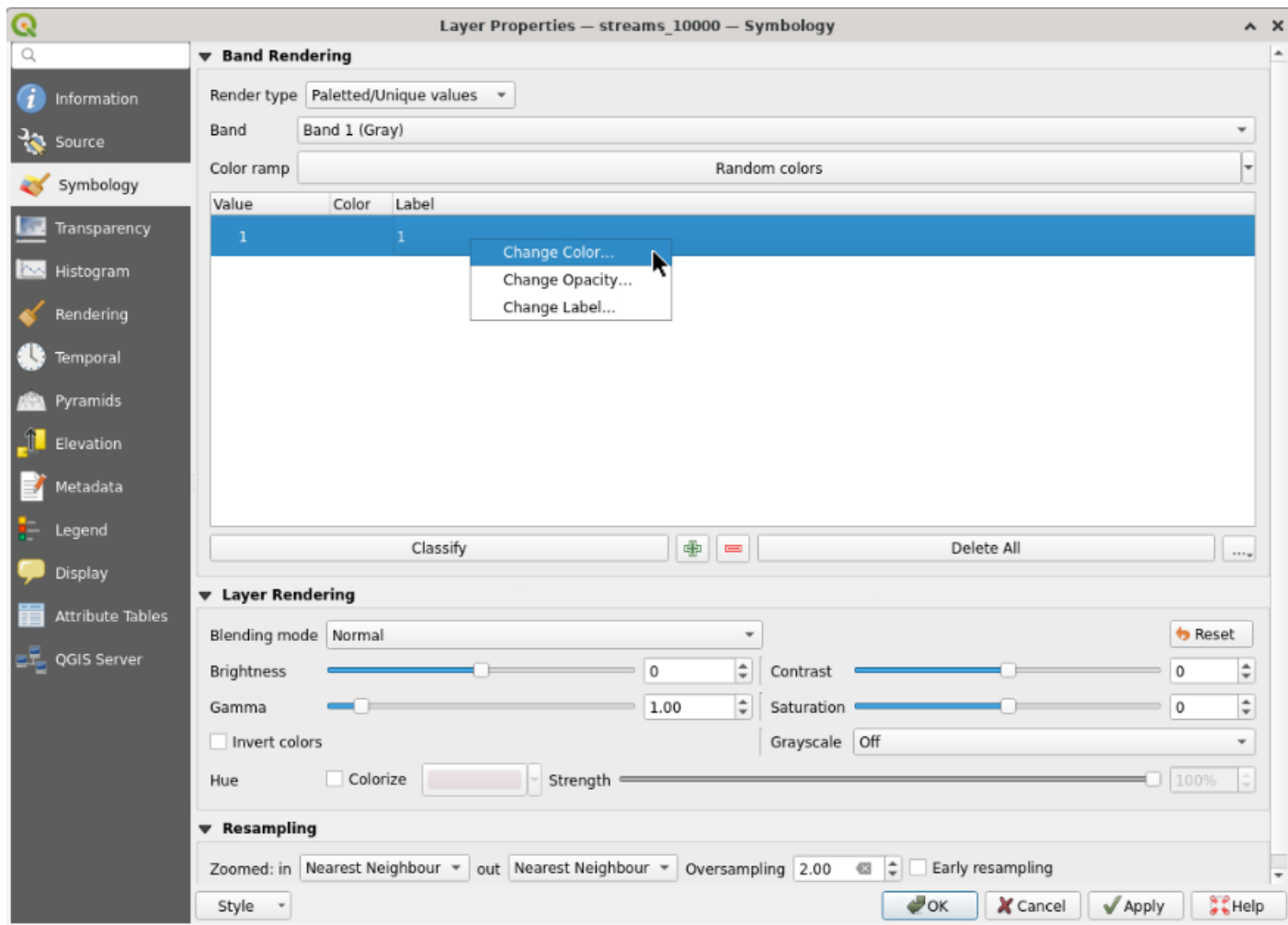
licancabur_wgs_hs.tif

streams_10000.tif

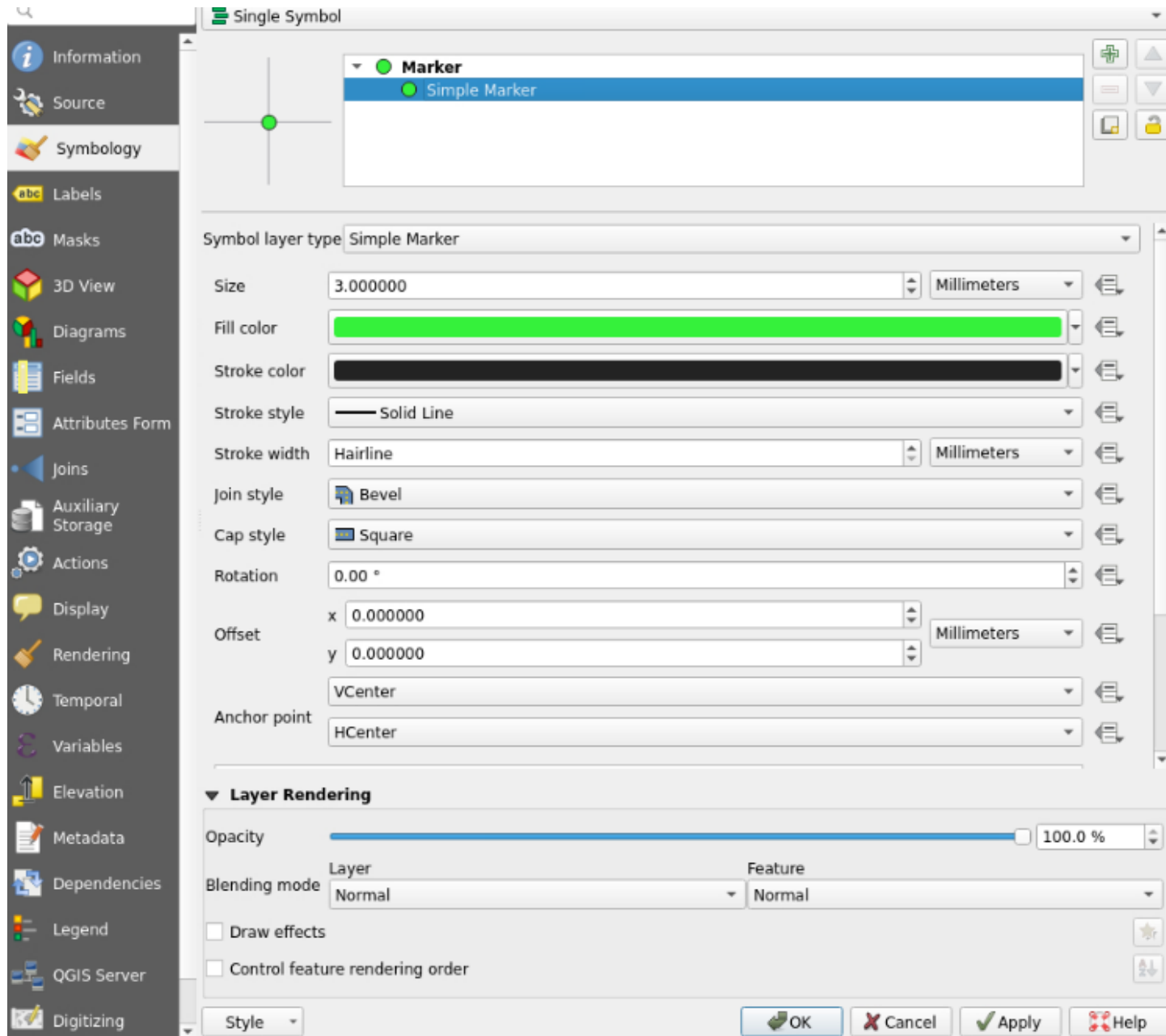
ec_line_25.tif

ip_10000_25.gpkg



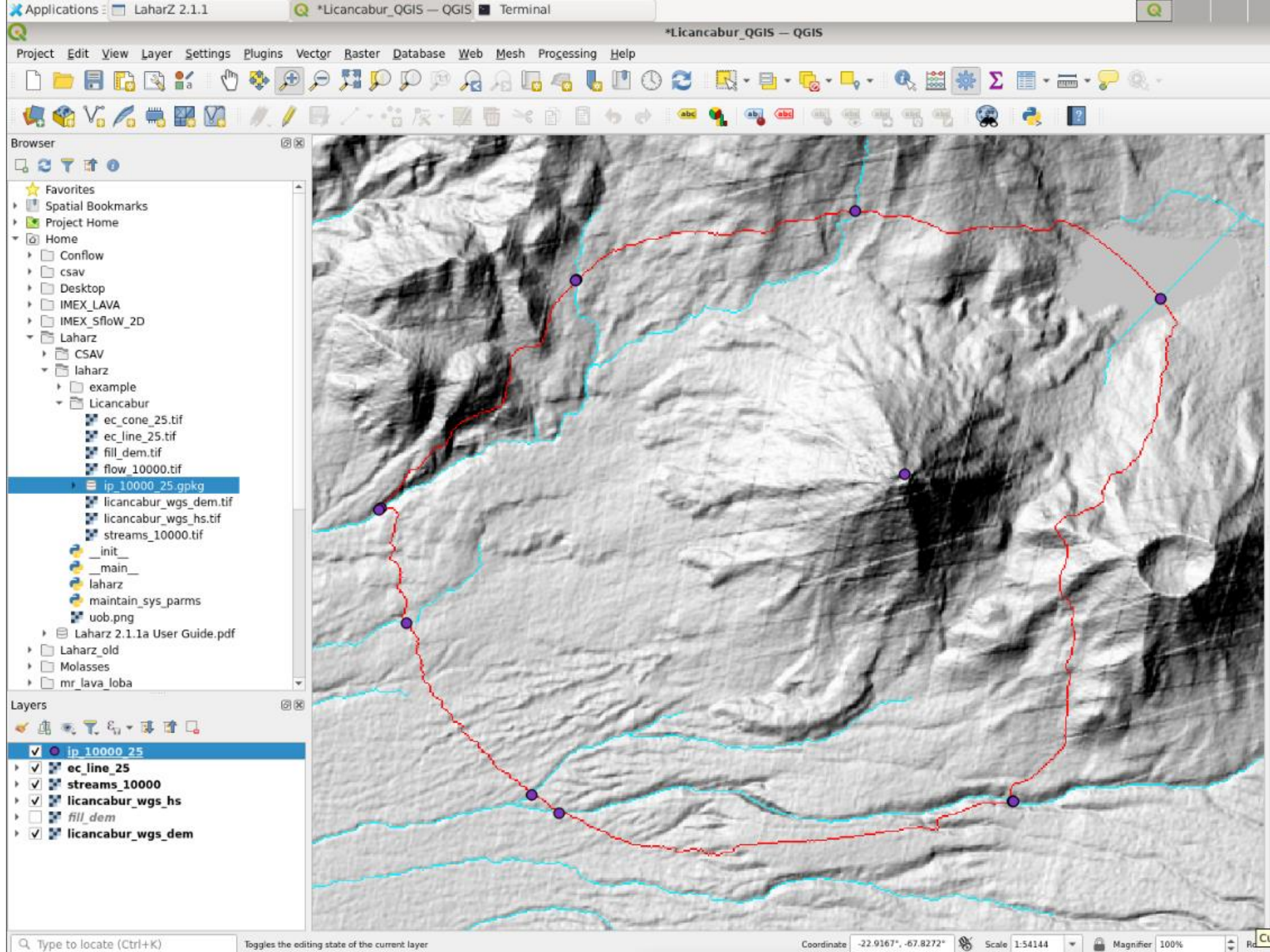


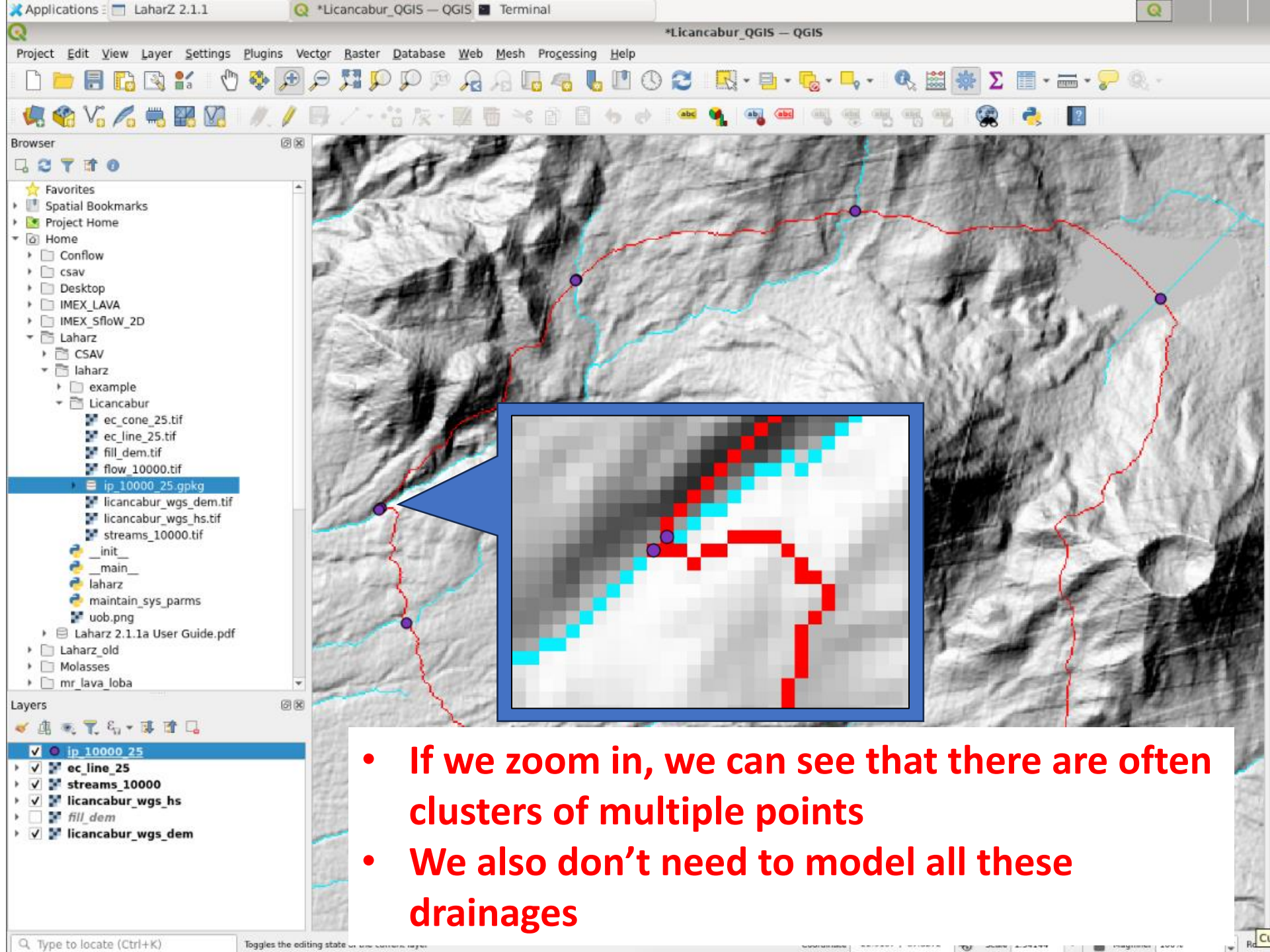
For the stream network and energy cone line, change the “render type” to “Paletted/unique values” and click “classify”. Then right click on the listed value and choose “change color”



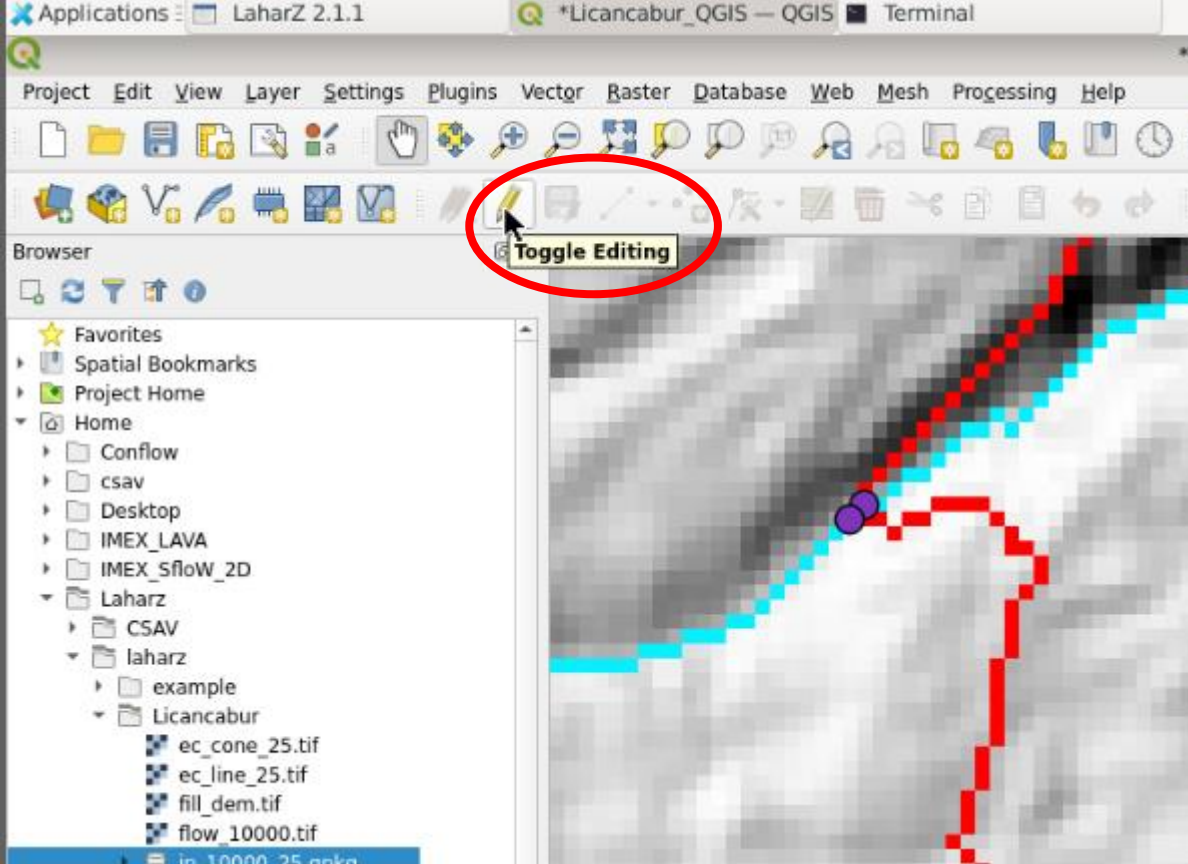
For the initiation points, change the color to something you like.

You can also change the size of the points if you want.



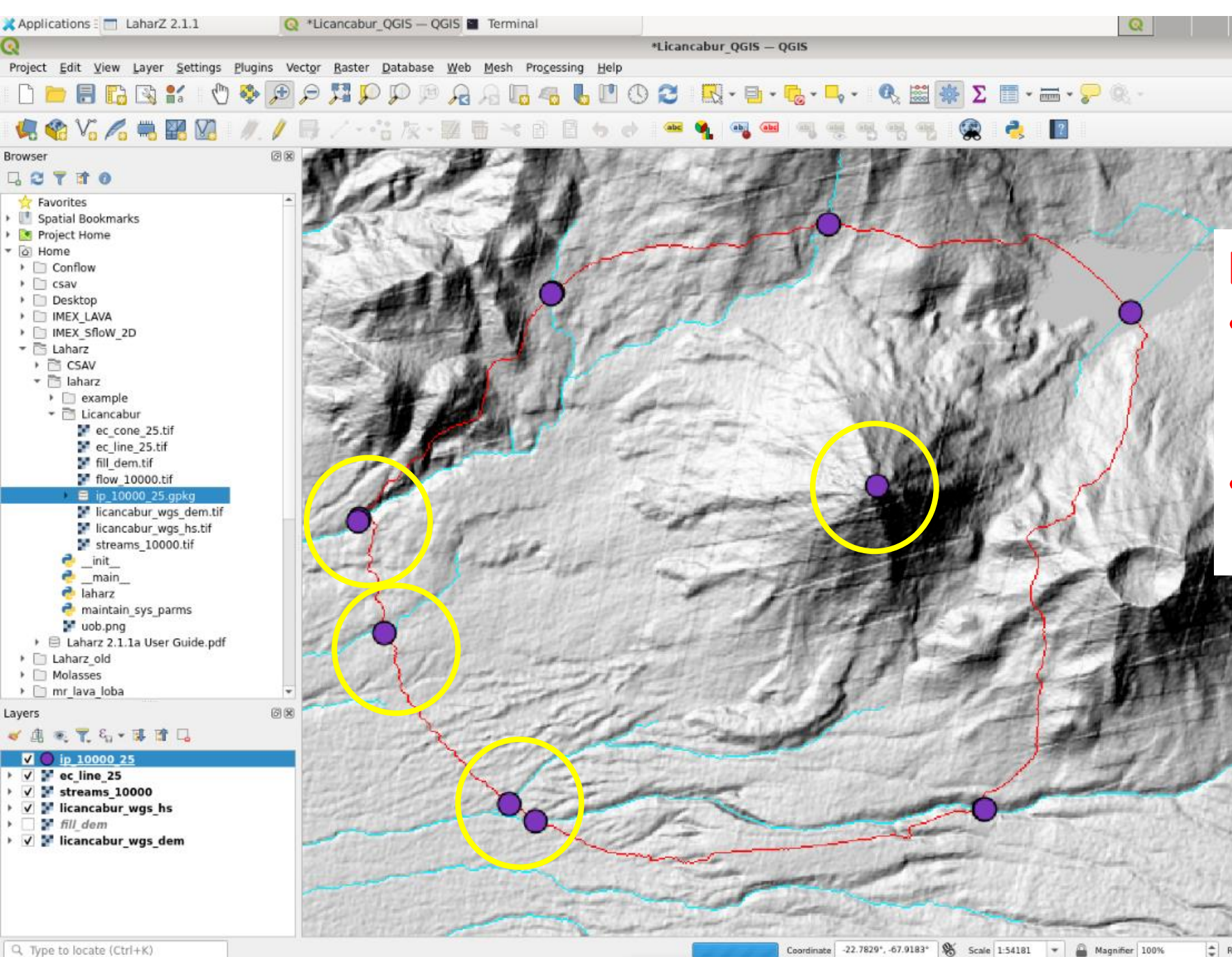


- If we zoom in, we can see that there are often clusters of multiple points
- We also don't need to model all these drainages



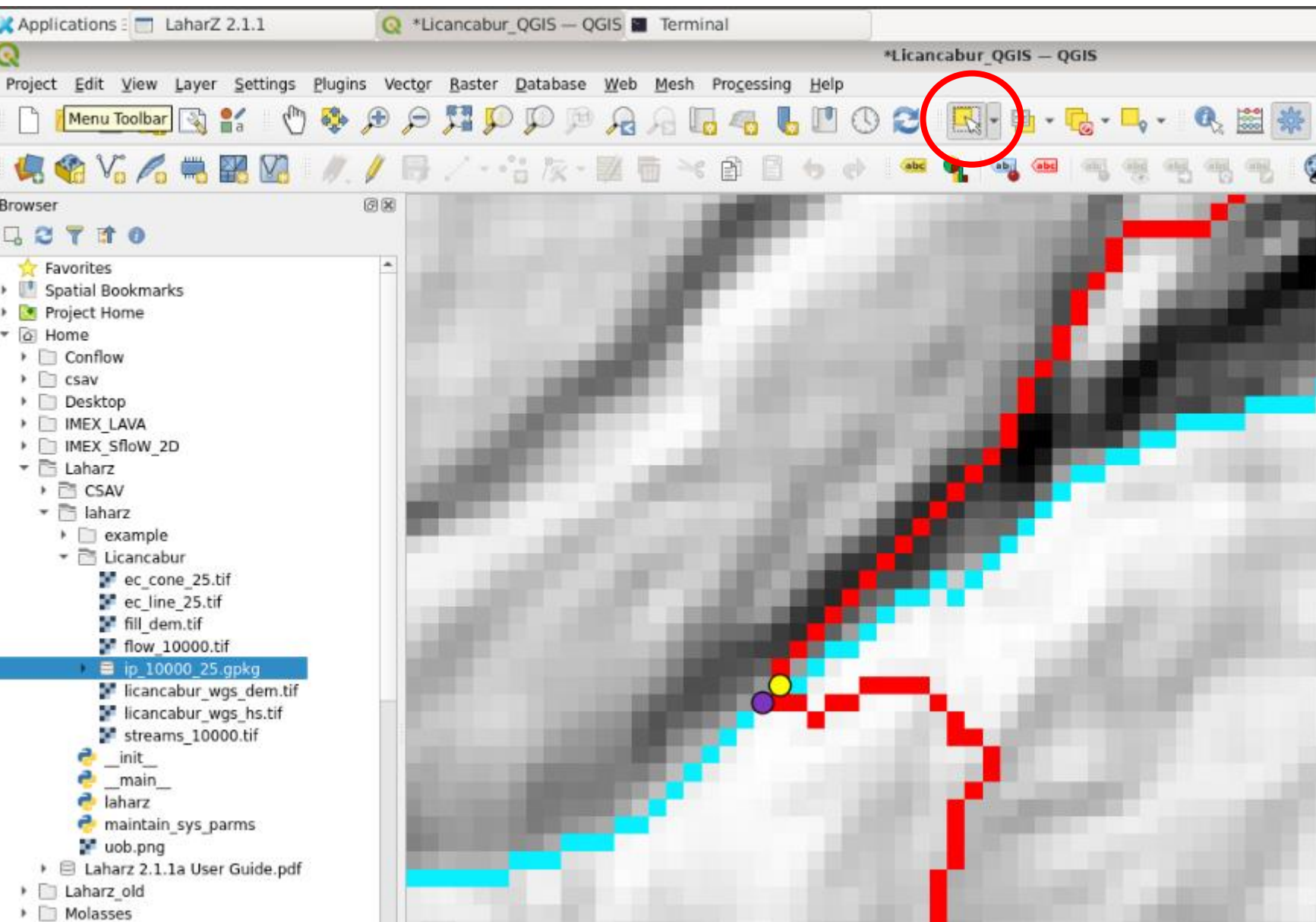
Let's delete some points

- Turn editing on by clicking the pencil “Toggle editing”



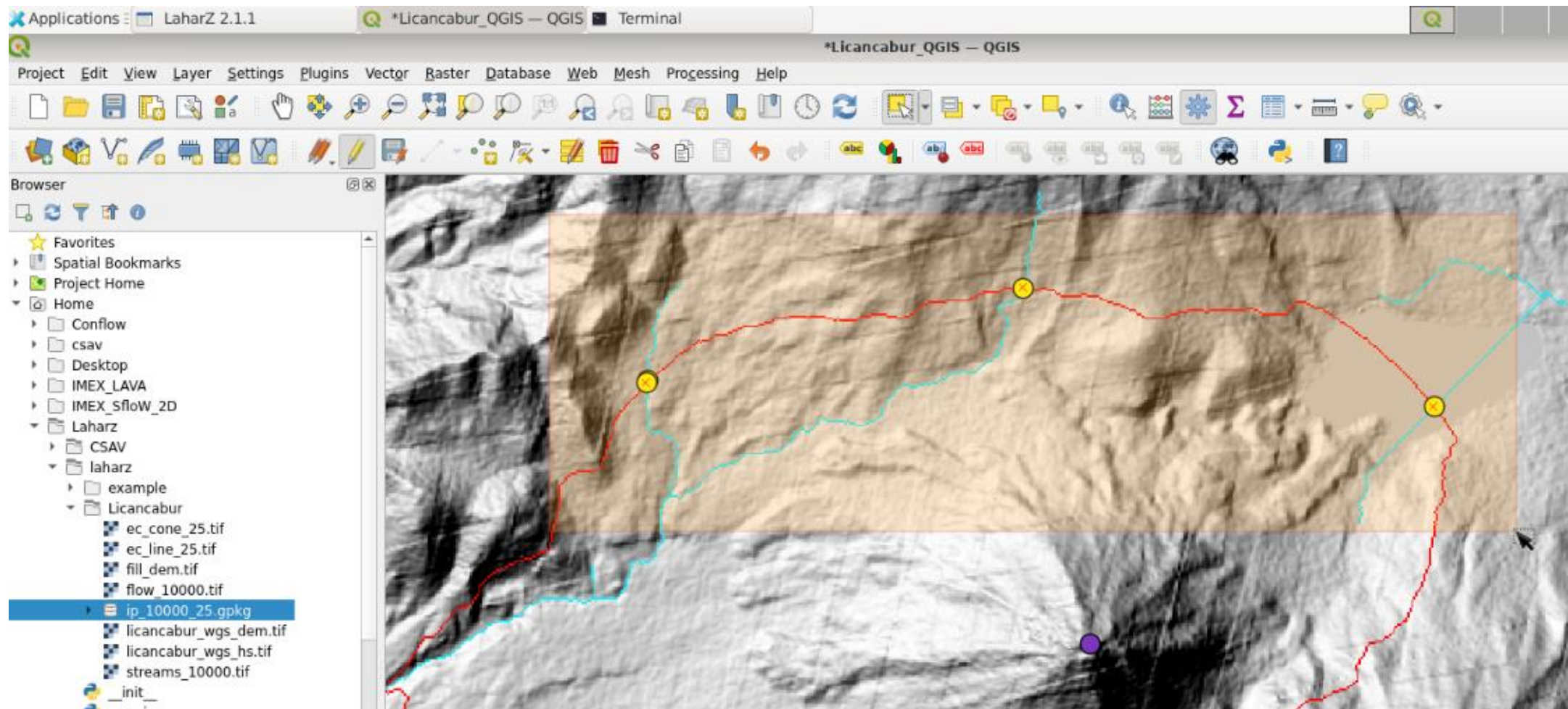
Let's delete some points

- **Let's keep just one point in each of the areas circled**
- **Also keep the summit point**



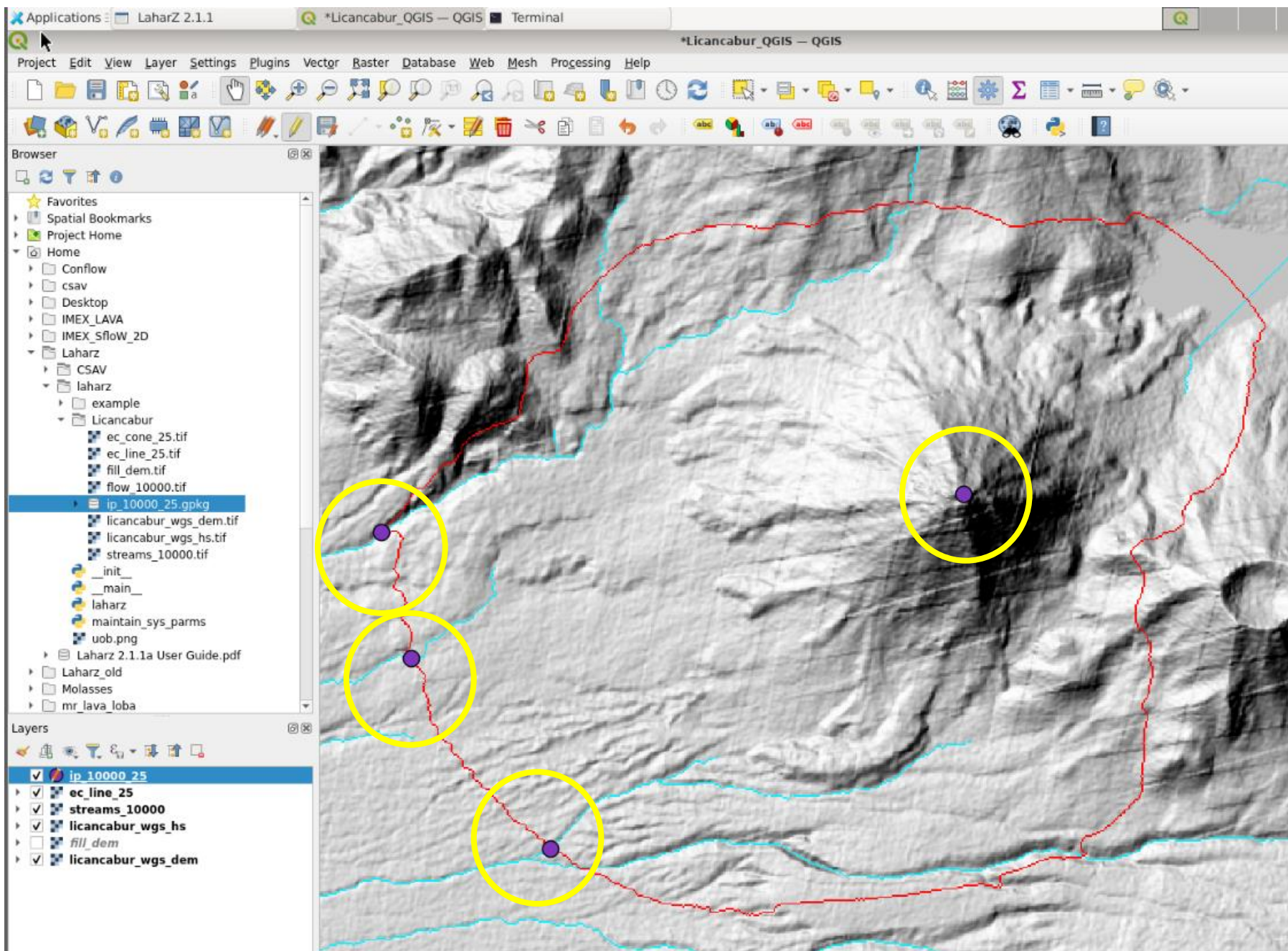
Let's delete some points

- Turn on the “Select features by area” tool and zoom in to each point cluster
- Click on the point you want to delete
- It will turn yellow
- The press the “delete” key on your keyboard



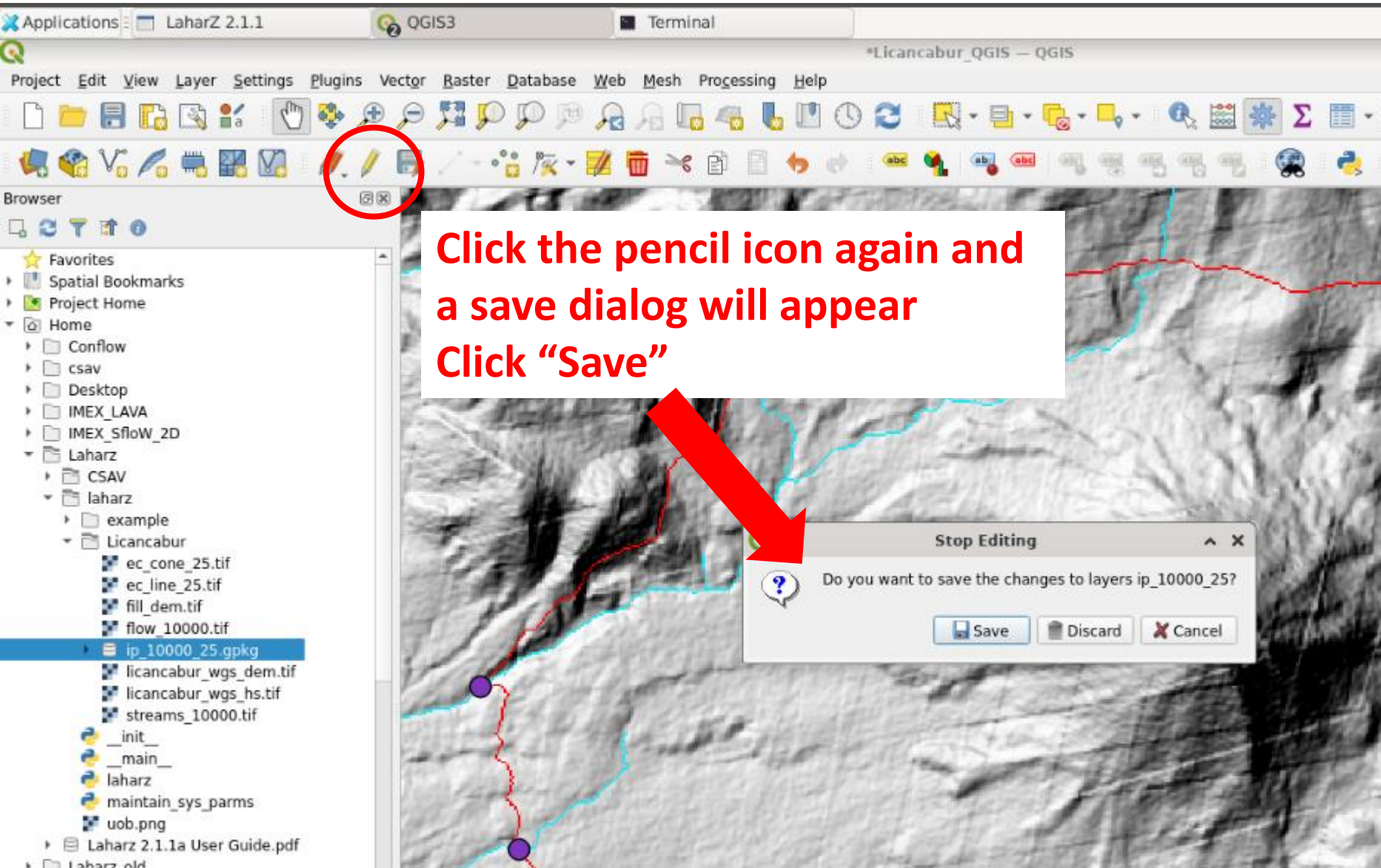
Let's delete some points

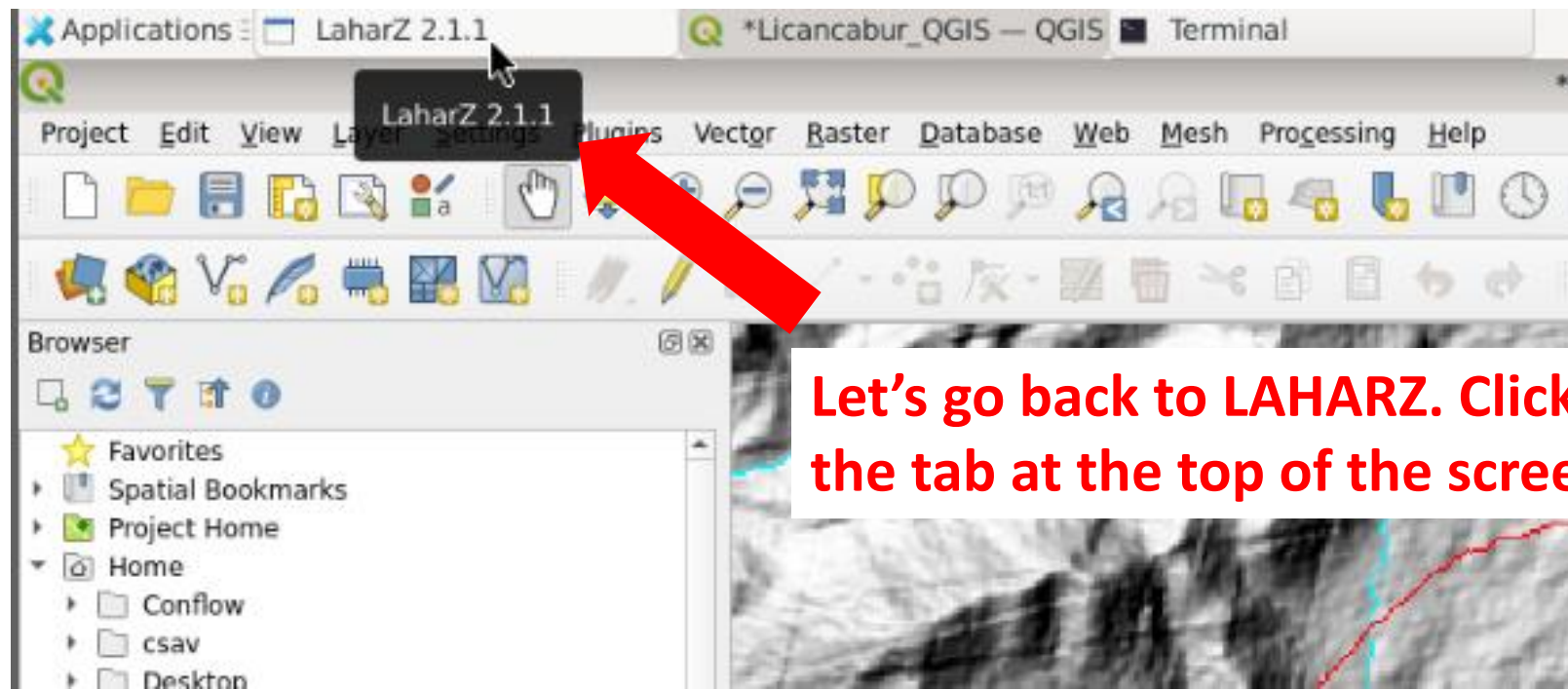
- You can also use the “Select features by area” tool to drag a selection box around multiple points
- They will turn yellow
- The press the “delete” key on your keyboard



Let's delete some points

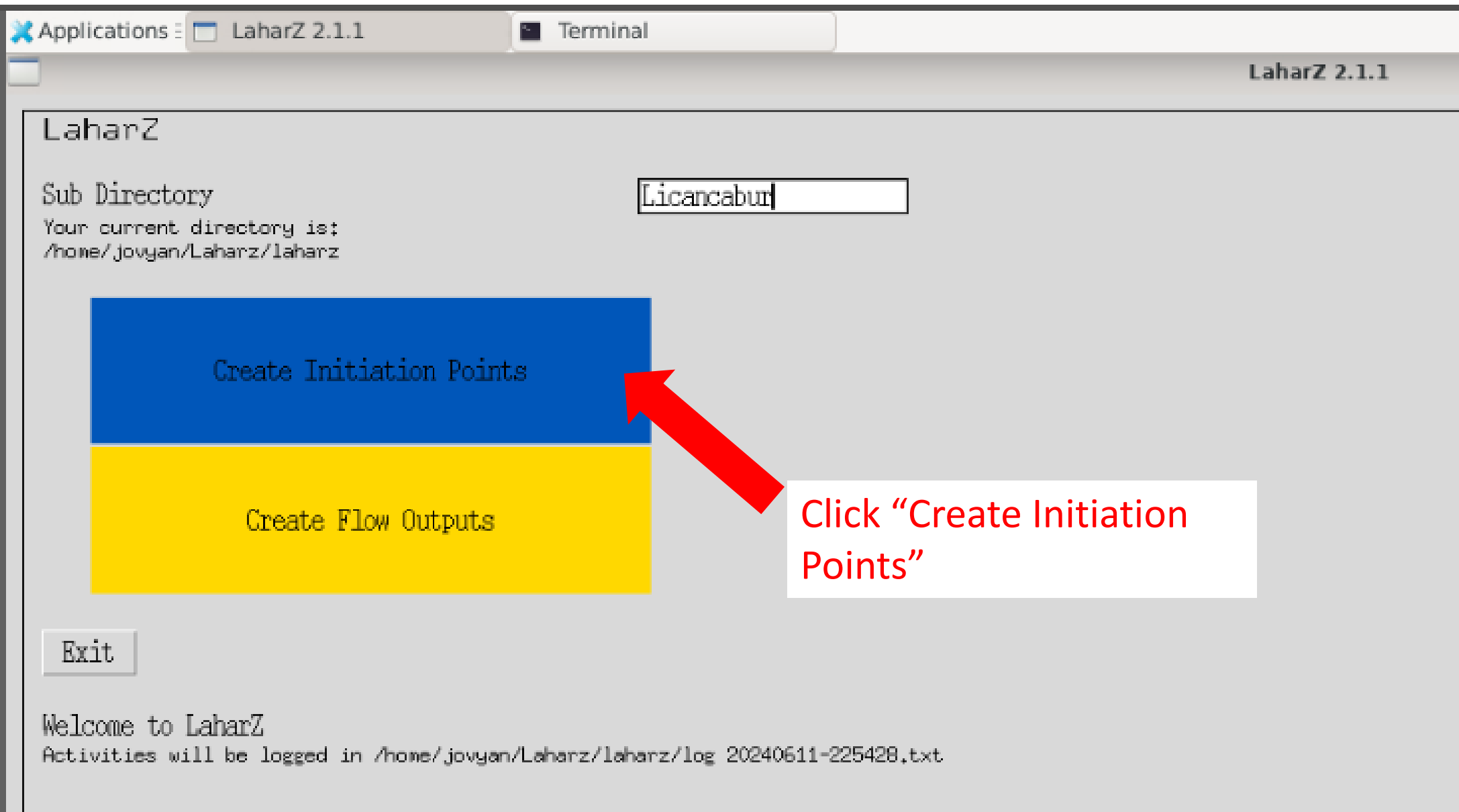
- We should now have 3 start points and 1 summit point left
- Click the pencil icon again and a save dialog will appear





Let's go back to LAHARZ. Click the tab at the top of the screen

Let's explore what a different stream threshold does.



Generate Initiation Points

DEM File	fill_dem.tif	Name of your DEM file in Licancabur
Stream and Flow Direction Files		
☒ Generate files ☐ Use your own files		
Uncheck box to disable overwrite check		
Streams file	☒ streams_20000.tif	Name of your streams file to be created in Licancabur
Stream Threshold	20000.0	Threshold value for accumulation to form stream
Flow direction file	☒ flow_20000.tif	Name of your flow direction file to be created in Lic
Determine Energy Cone Apex		
☐ Use Search File ☒ Use Longitude/Latitude		
Apex	-67.883,-22.833	Longitude, Latitude
Incremental Height	0.0	Incremental height in metres
H/L Ratio	0.25	H/L Ratio
Sea Level	0.0	No initiation points will be created at sea level or below
☒ Plot Energy Cone Graphics		
Uncheck box to disable overwrite check		
Energy Cone Graphics File	☒ ec_cone_25.tif	Name of the file for the energy cone graphics in Licancabur
Extent	1.3	Extent to plot the energy cone/surface (1.3 = 130% of L)
Energy Cone Line File	☒ ec_line_25.tif	Name of the file for the energy cone line in Licancabur
Initiation Points File	☒ ip_20000_25.gpkg	Name of the file for the initiations points in Licancabu
Back	Initiation points	

Most fields will already be filled in from the last time

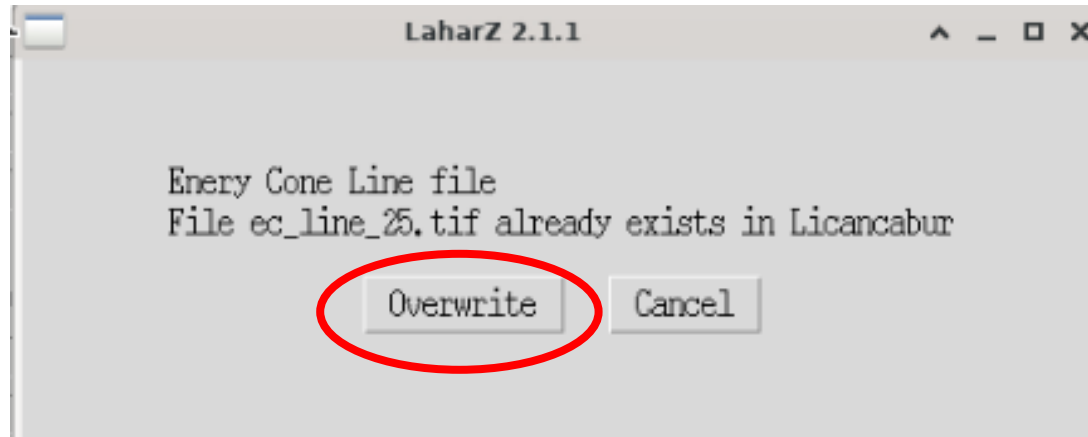
For this step, we're going to try a stream threshold of **20000**.

- Streams file: streams_20000.tif
- Stream threshold: 20000
- Flow direction file: flow_20000.tif

Name the initiation points with the stream threshold and H/L ratio:

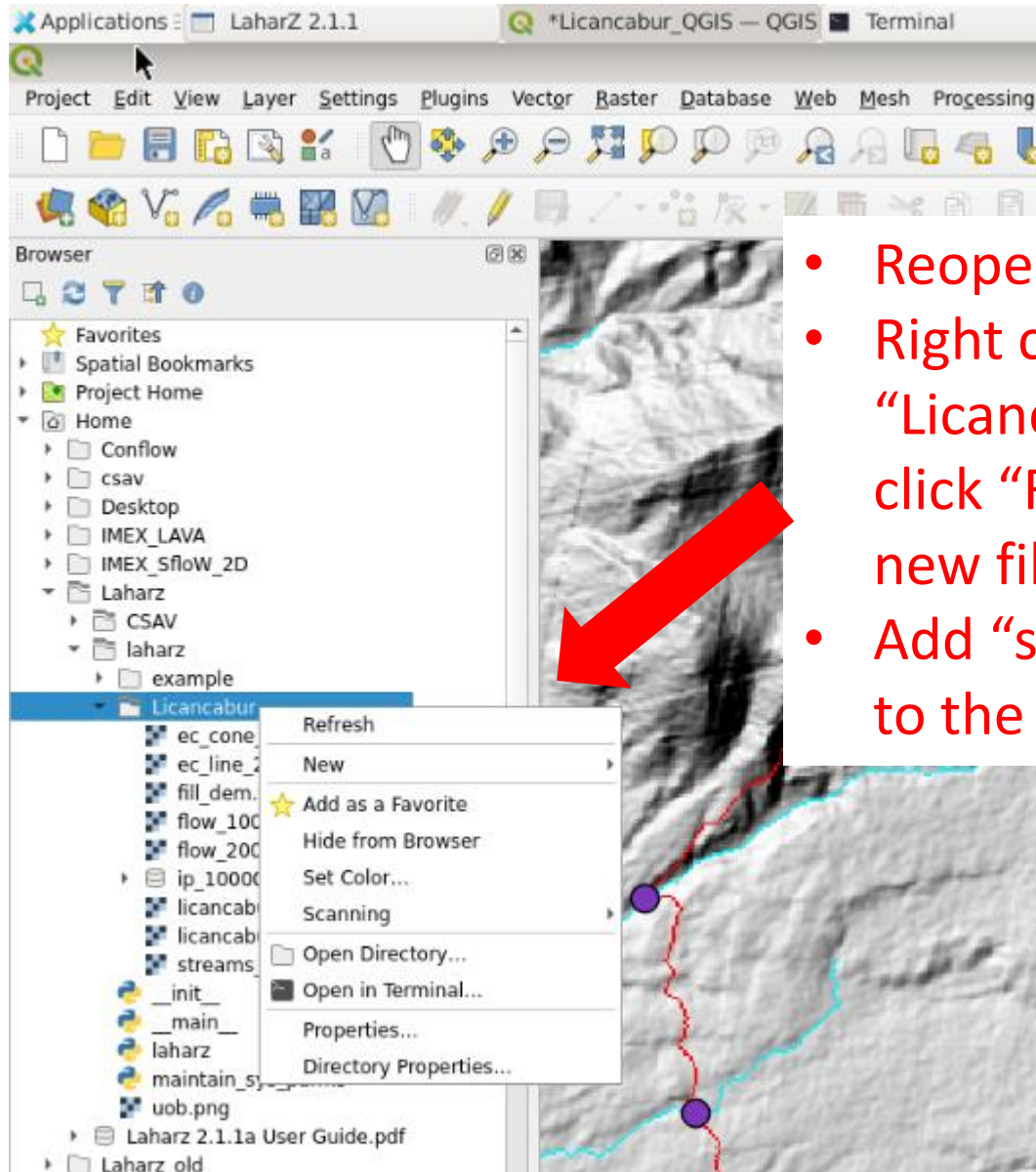
- ip_20000_25.gpkg

When you're done adding these choices, click "Initiation points" button

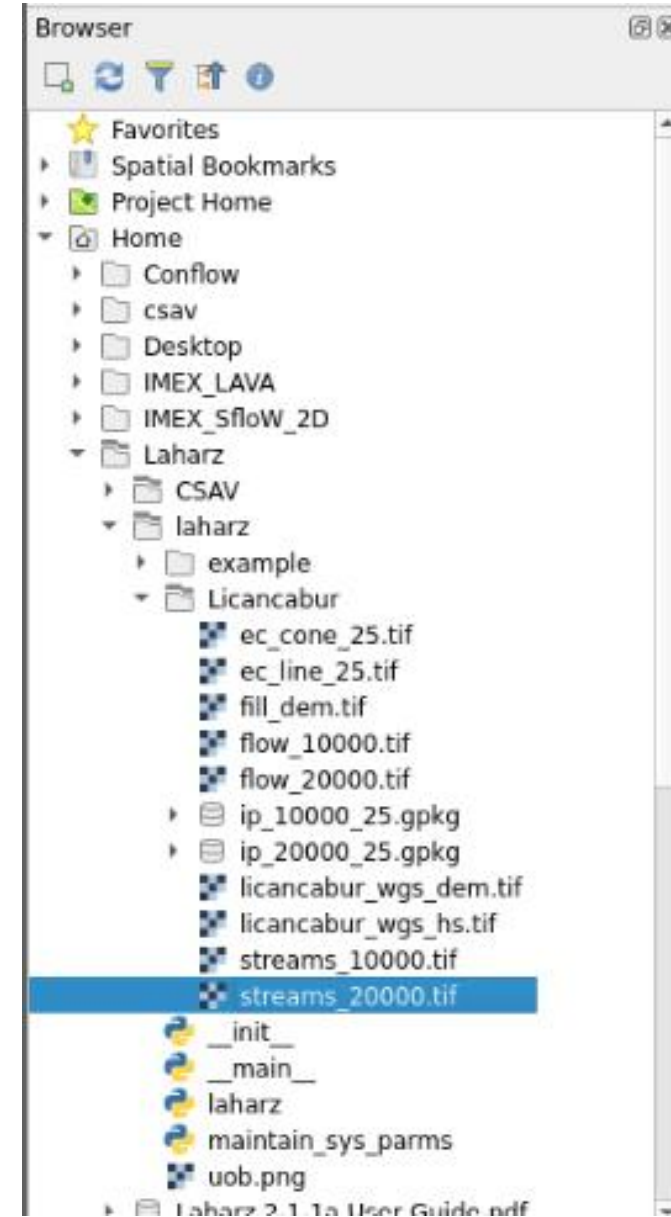


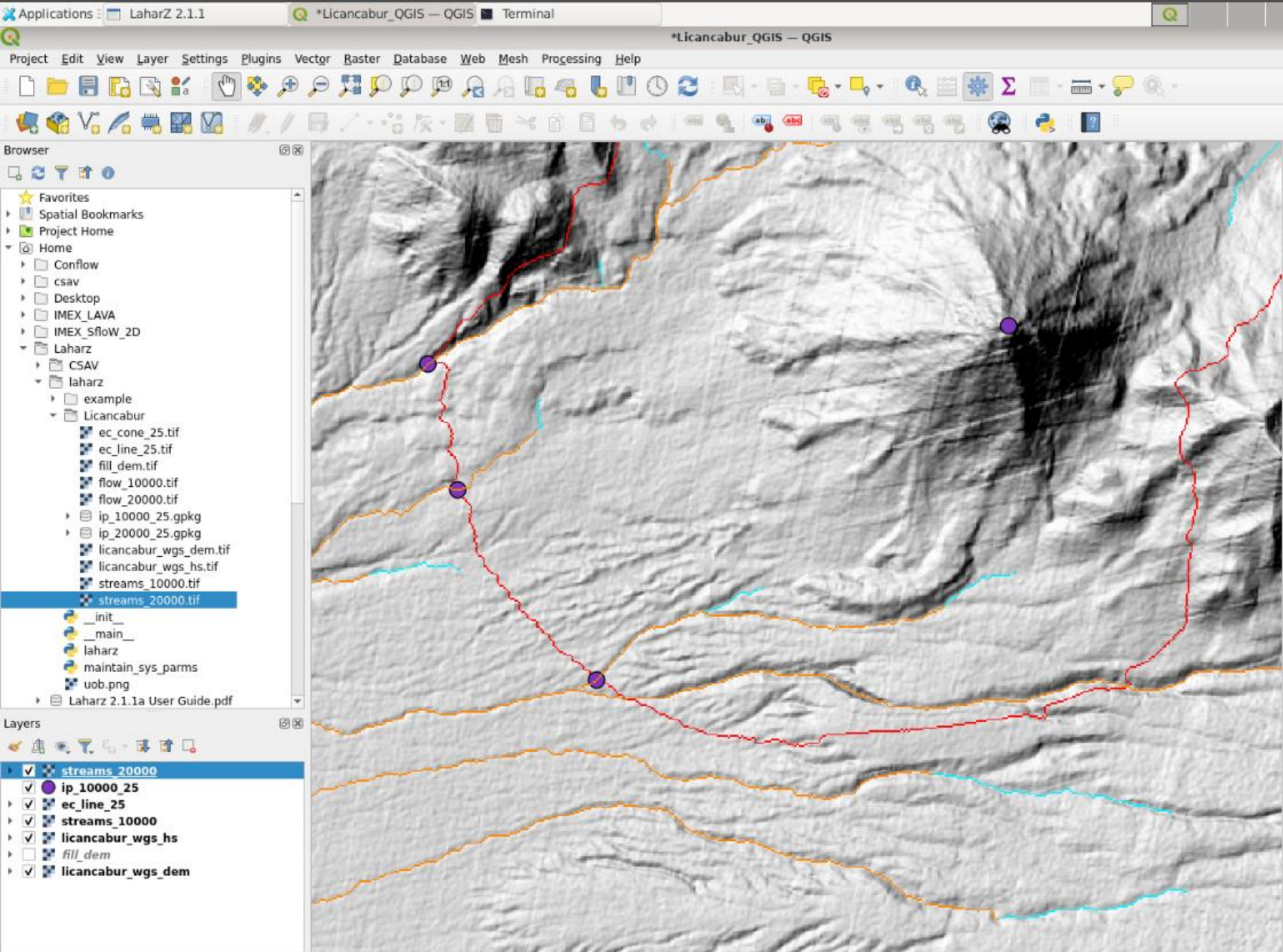
Choose "Overwrite" in the dialog boxes that pop up

Let's look at the new files.



- Reopen Licancabur_QGIS
- Right click on the "Licancabur" folder and click "Refresh" to see your new files
- Add "streams_20000.tif" to the map



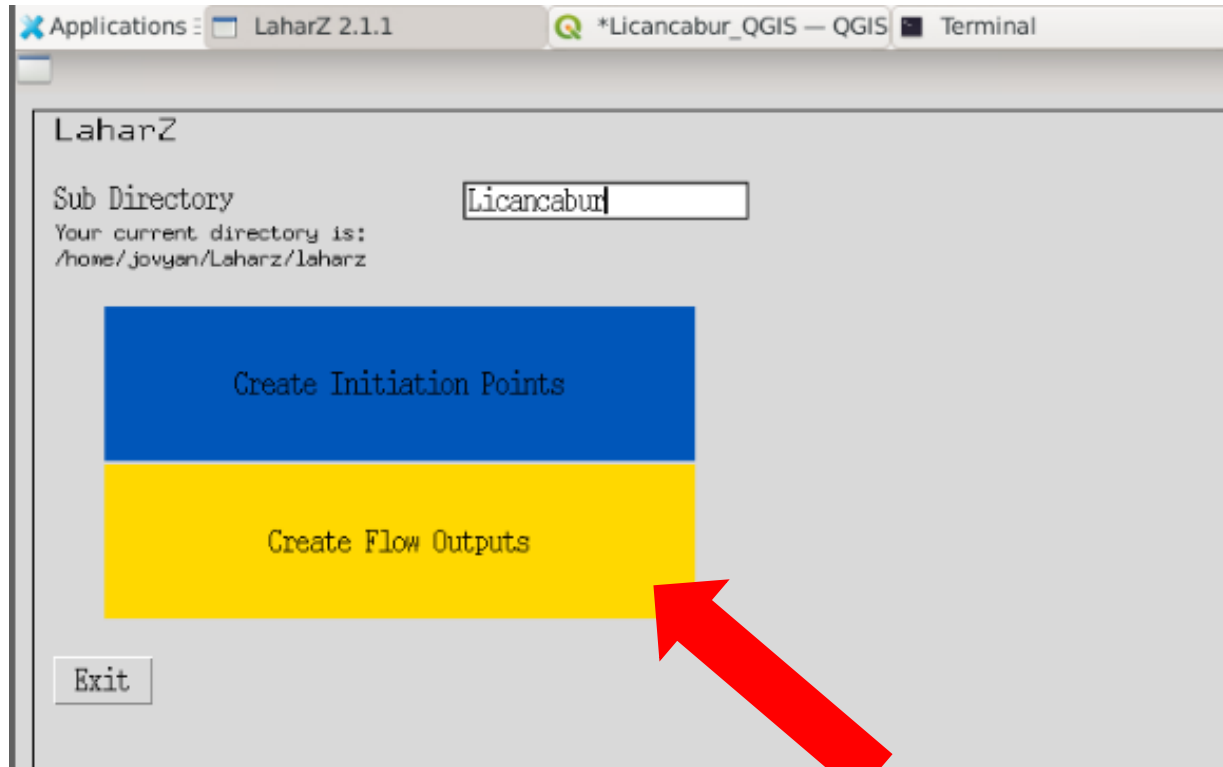


Lower stream threshold
(10000): longer stream
network with more branches

Higher stream threshold
(20000): shorter stream
network with fewer branches

For the rest of the exercise,
we're going to use the 10000
stream threshold files

Let's run LAHARZ!



Click "Create Flow Outputs"

Applications ▾ LaharZ 2.1.1 *Licancabur_QGIS — QGIS Terminal

LaharZ 2.1.1

Generate Flows

DEM File	fill_dem.tif	Name of your DEM file in Licancabur
Streams File	streams_20000.tif	Name of your streams file in Licancabur (not used to generate flows)
Flow Direction File	flow_10000.tif	Name of your flow direction file in Licancabur
Initiation Points File	ip_10000_25.gpkg	Name of the file with the initiations points in Licancabur
Volume(s)	1e7, 3e7, 1e8	Volumes (m ³) in a list separated by commas
Scenario	Lahar	Name of the scenario to use
Cross sectional area:	0.05	Value of the c1 parameter
Planar Area	200	Value of the c2 parameter
Cross Sectional Area =	c 1 V 20 3	
Planar Area =	c 2 V 20 3	
Sea Level	0.0	Flow will stop at sea level
Flow Output Directory	☒ lahars	Directory for flow output files
Uncheck box to disable overwrite check		

Most fields will already be filled in, but we need to change “20000” to “10000”

Flow direction file: flow_10000.tif

Initiation points file: ip_10000_25.gpkg

Now we add some volumes

We want the smallest volume to be:

- 10,000,000 (=10 million m³=0.01km³=1x10⁷=1e7)
- The easiest way to enter volumes is using scientific notation: 1e7

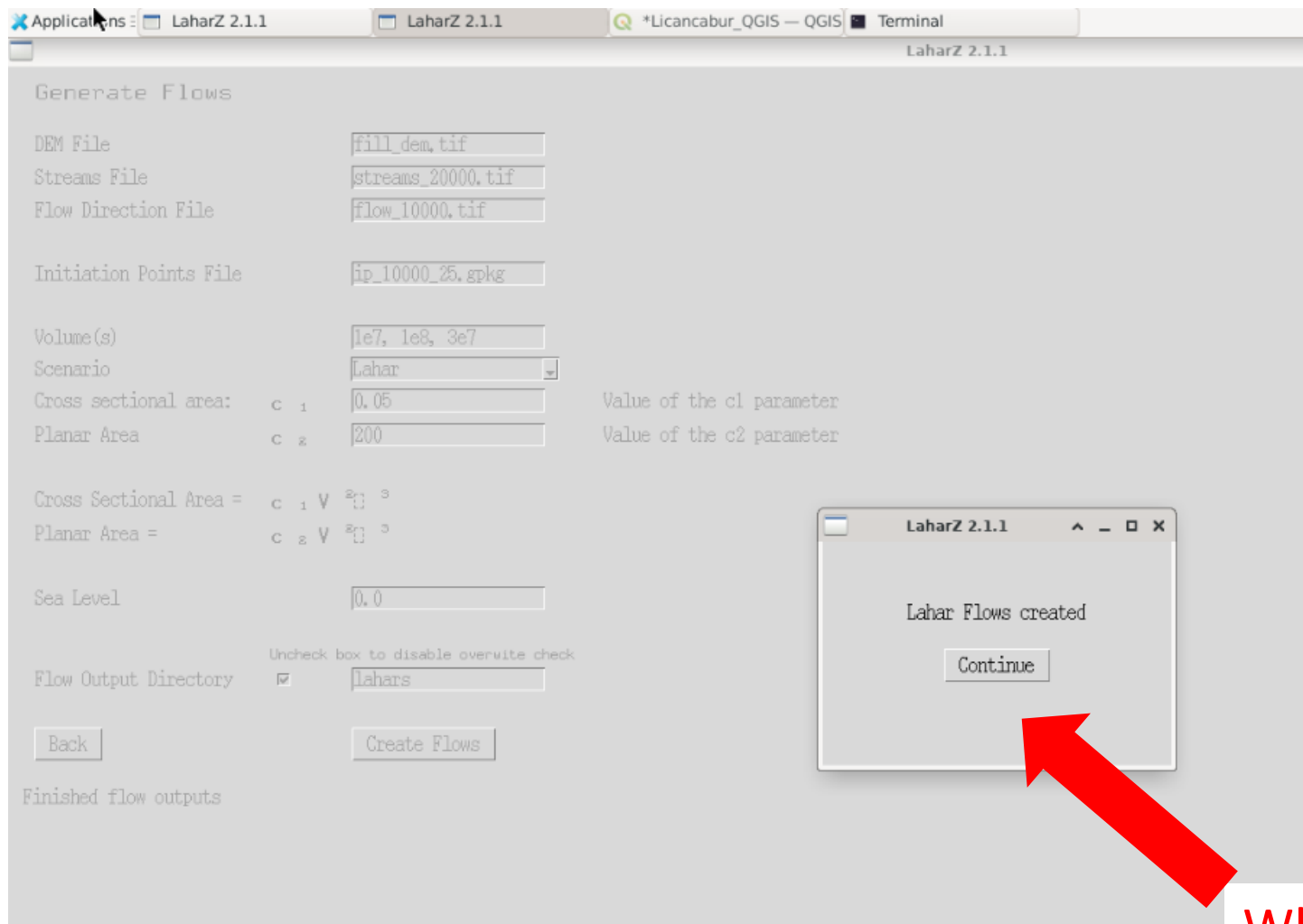
We also want to use:

- 30,000,000 (=30 million m³=0.03 km³=3x10⁷=3e7)
- 100,000,000 (=100 million m³=0.1km³=1x10⁸=1e8)

Enter “1e7, 3e7, 1e8” in the Volumes box

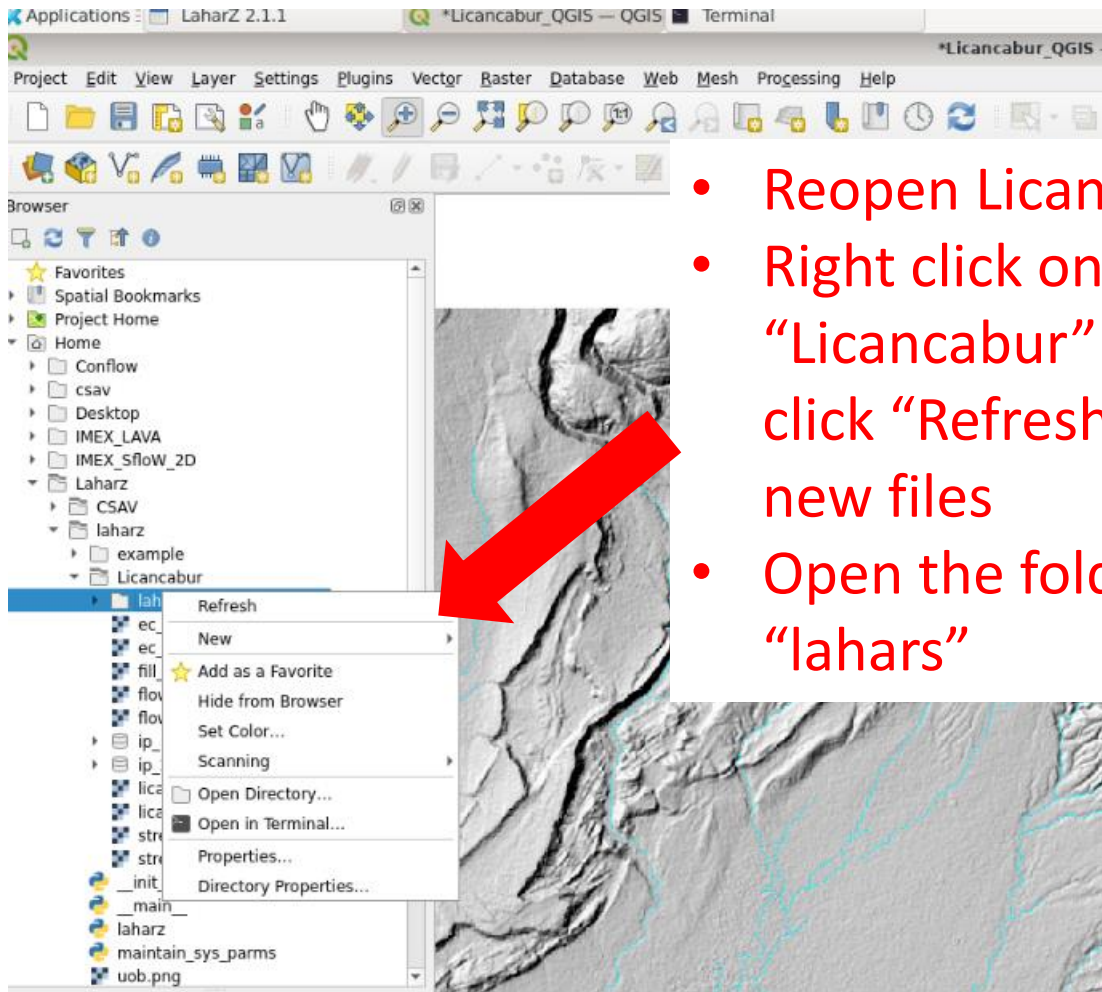
Everything else can stay as the default

When you’re done adding these choices, click
“Create flows” button

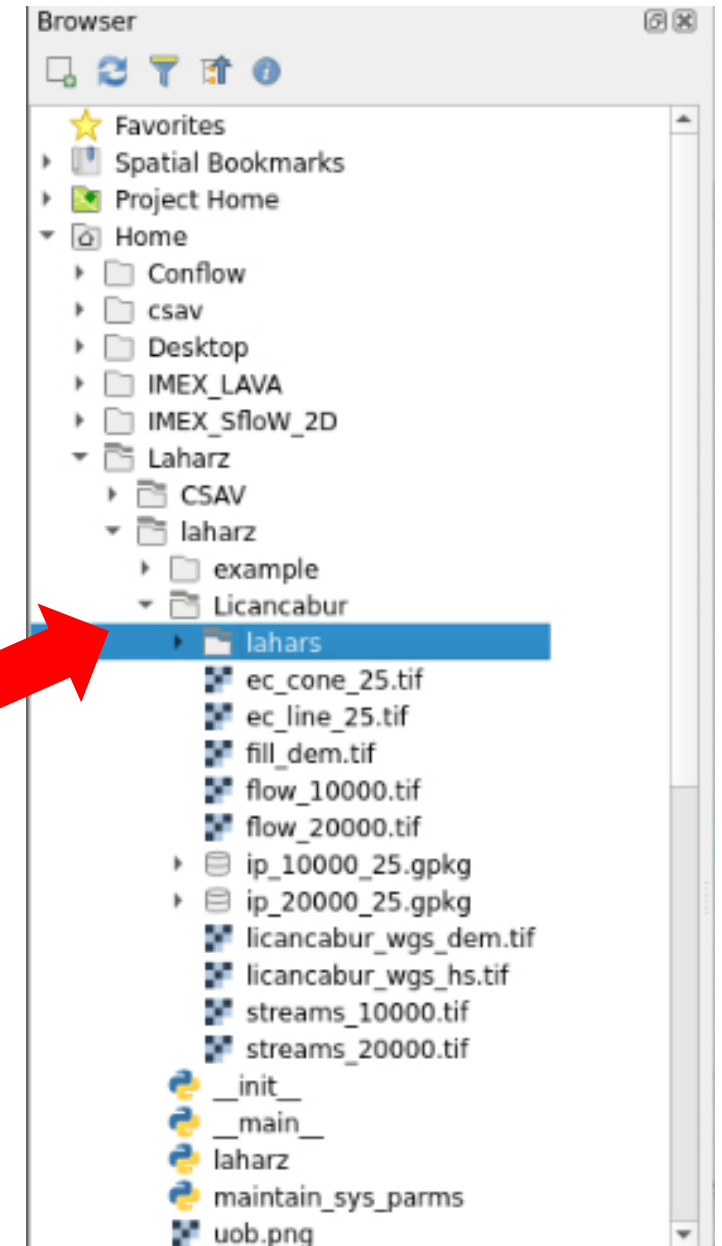


When it's done, click "Continue"

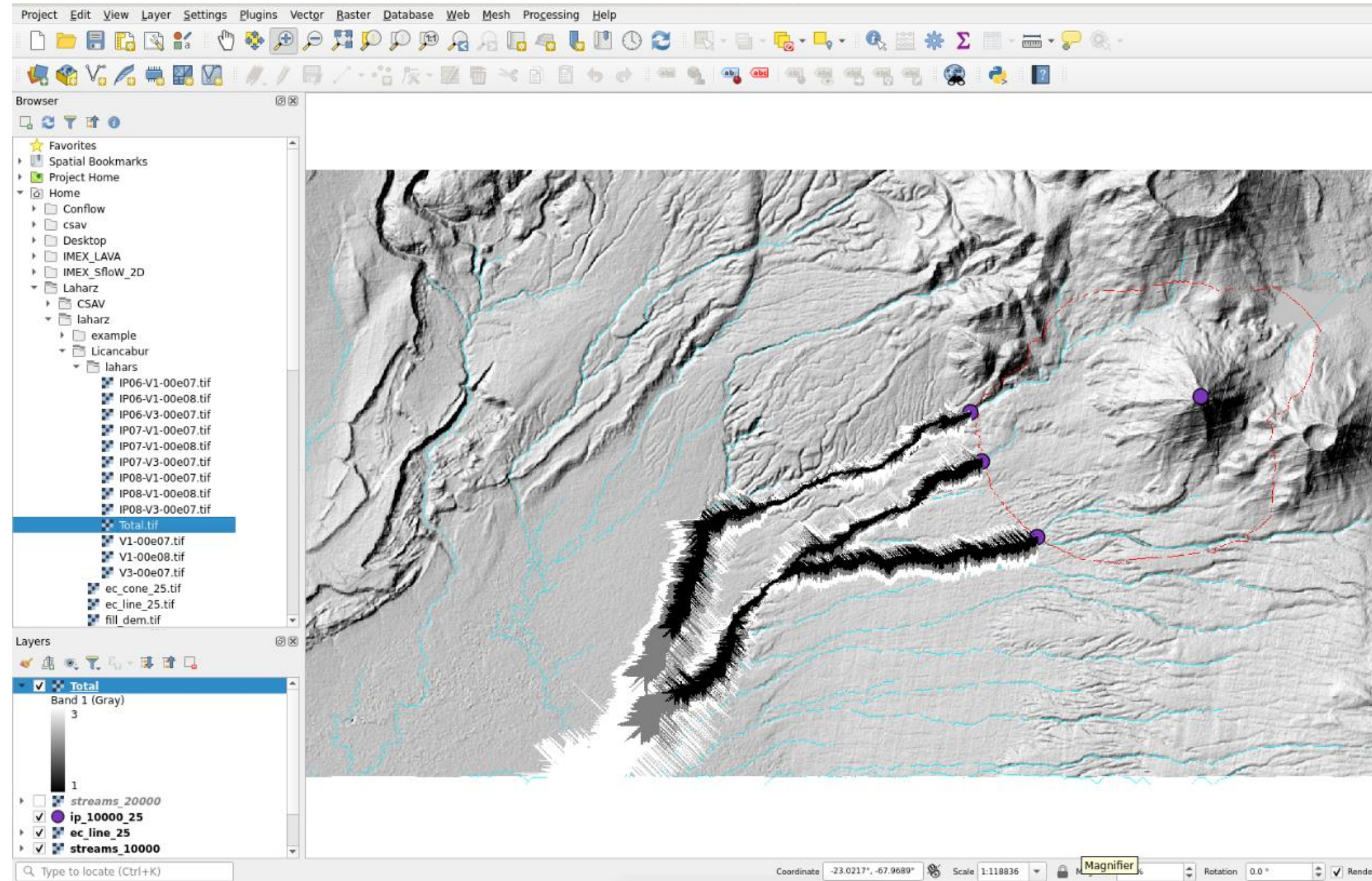
Let's look at the new files.

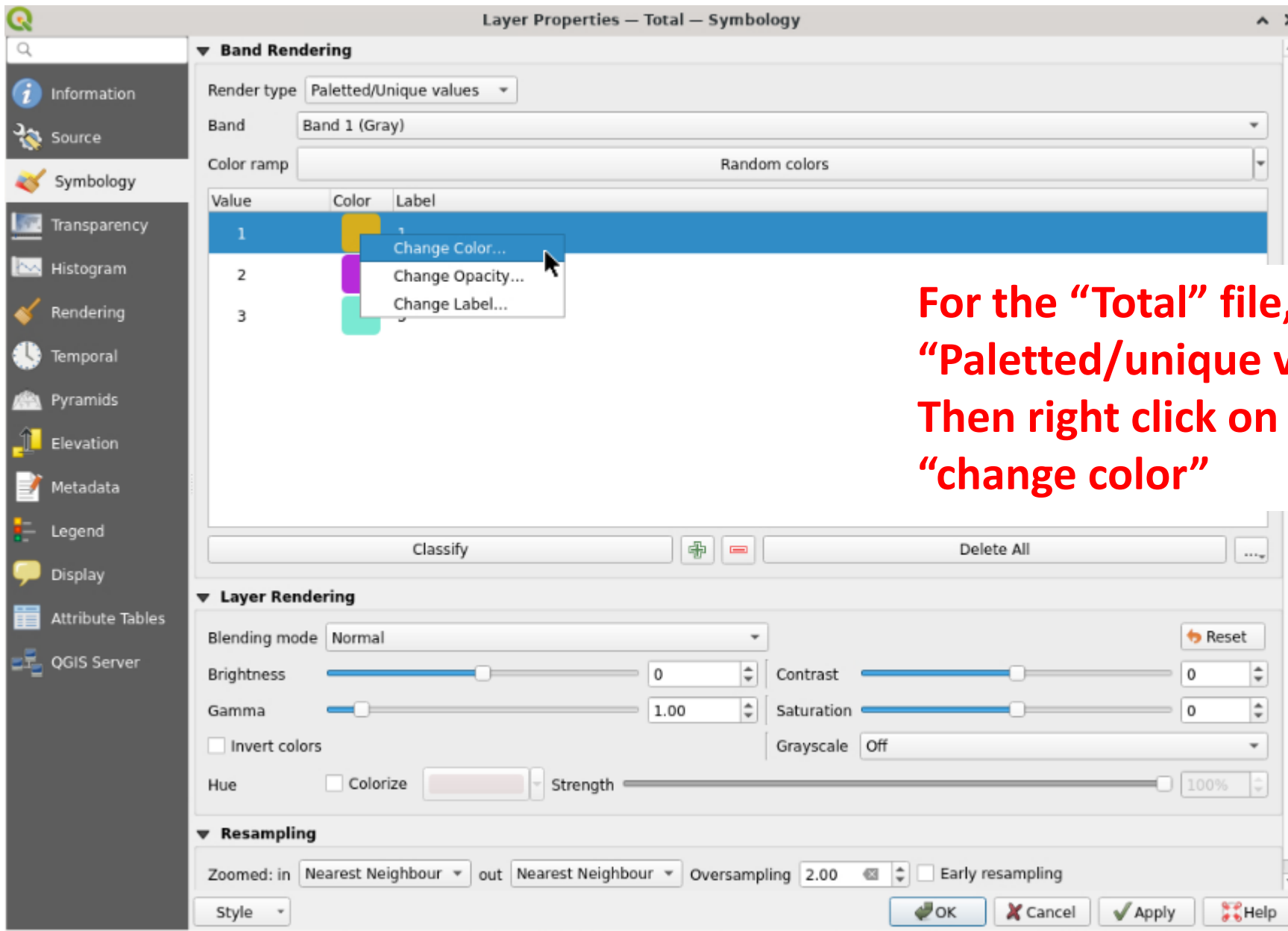


- Reopen Licancabur_QGIS
- Right click on the “Licancabur” folder and click “Refresh” to see your new files
- Open the folder called “lahars”



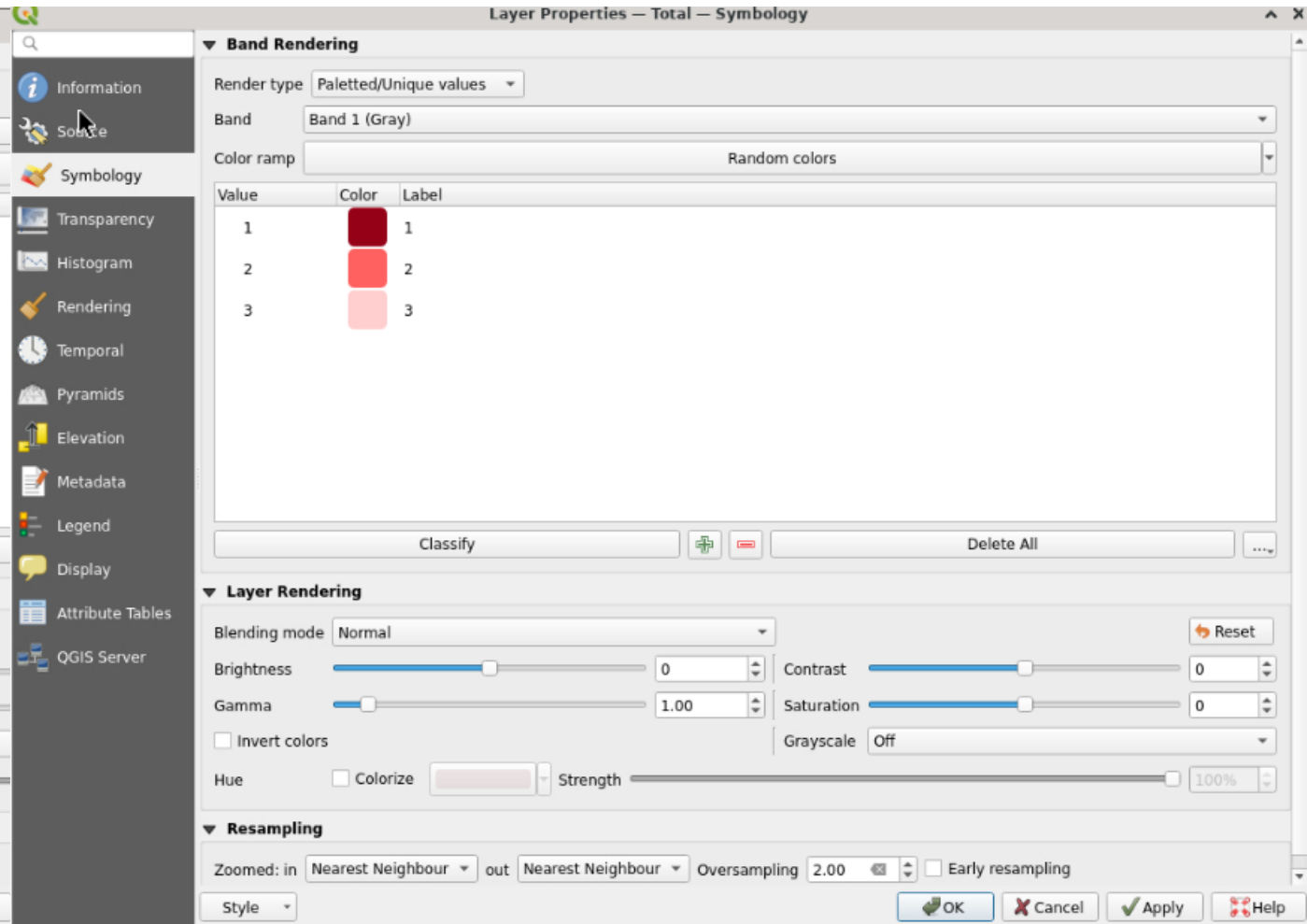
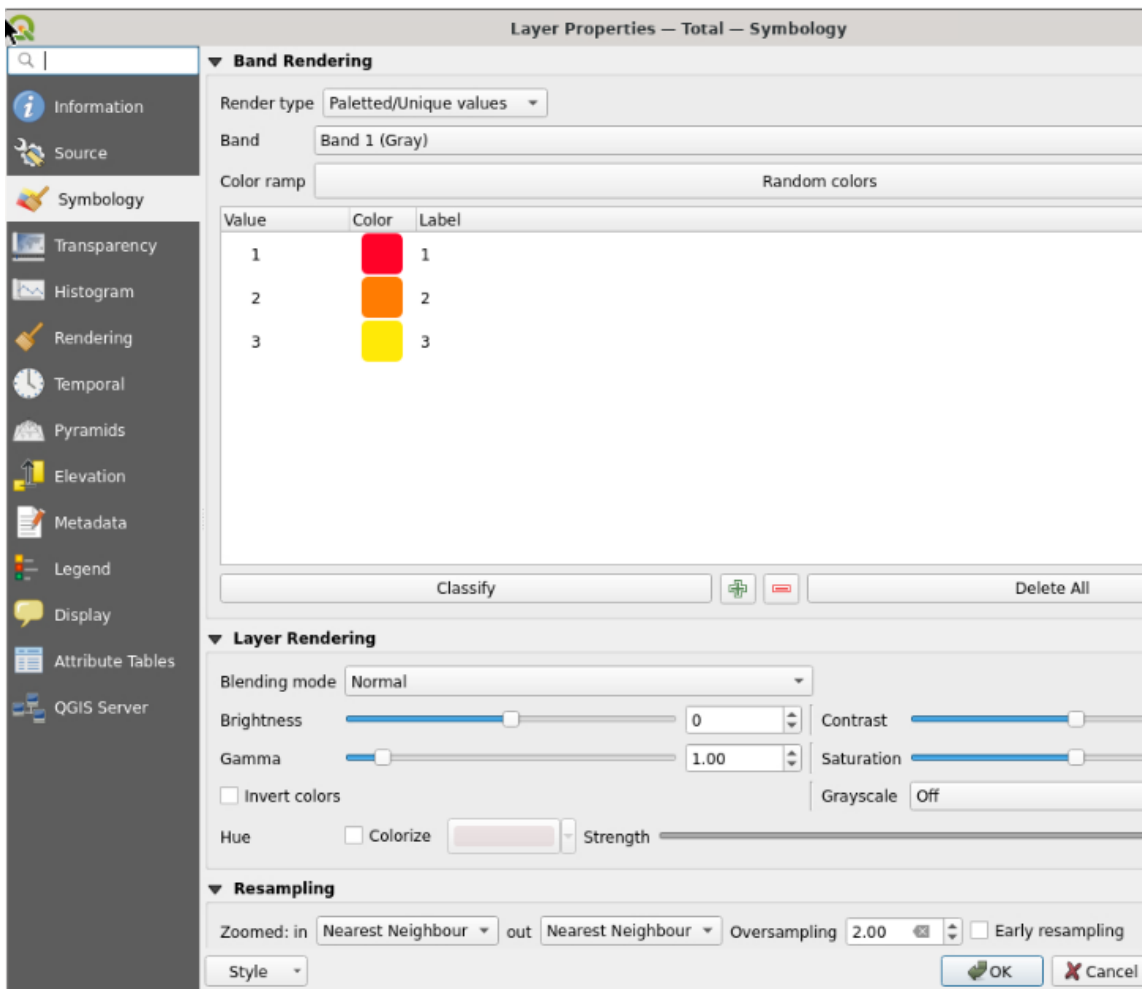
Let's change the colors and make the map prettier.

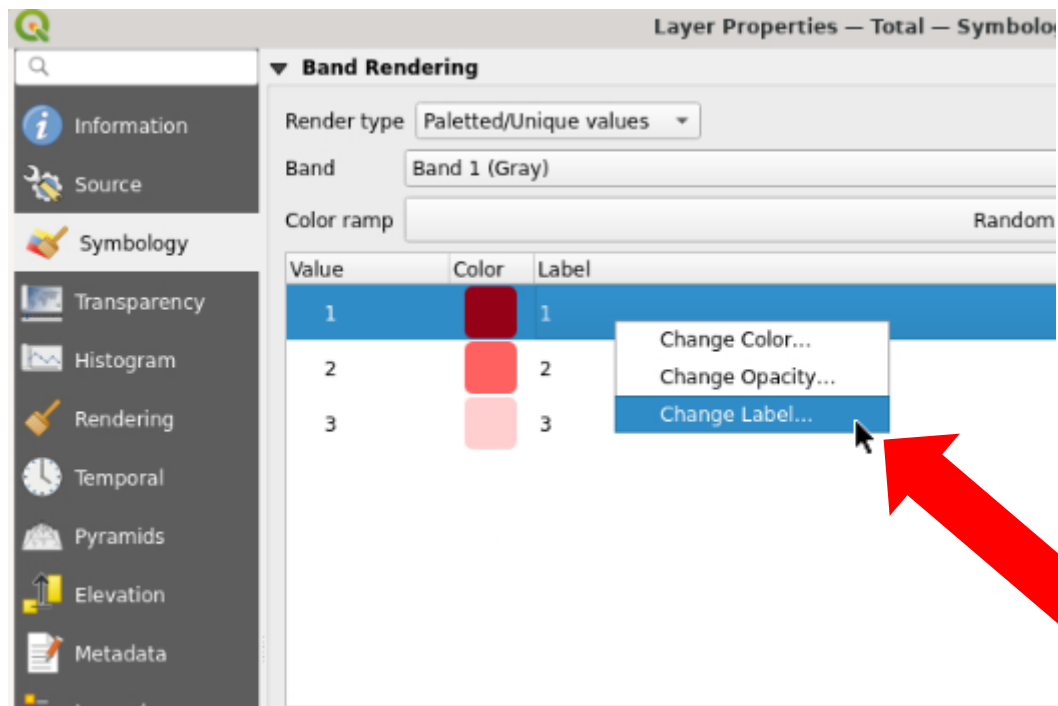




For the “Total” file, change the “render type” to “Paletted/unique values” and click “classify”. Then right click on the listed value and choose “change color”

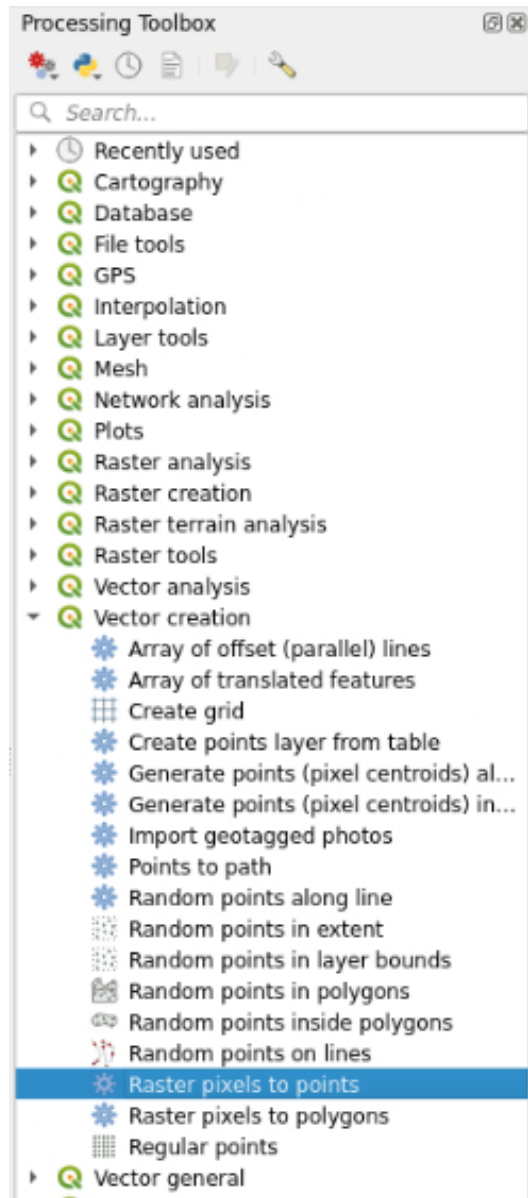
- The lowest value (1) is the lowest volume (1×10^7).
- This is the highest hazard zone because smaller lahars occur more often.
- Choose a color scheme that makes sense, like red-orange-yellow or dark red to light red.
- We'll talk more about color schemes later!





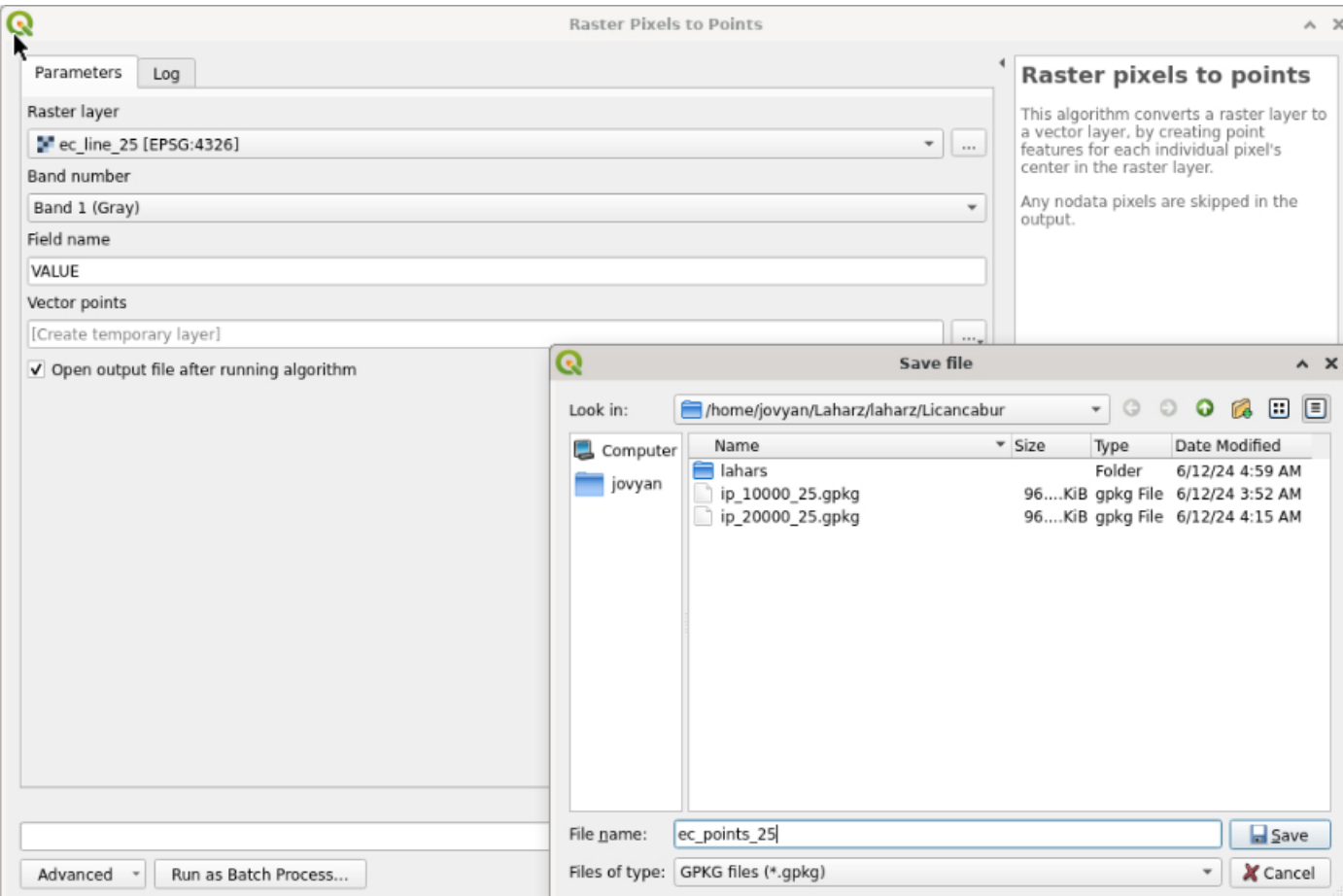
**You can also change the label for each layer to:
0.01 km³, 0.03 km³, 0.1 km³**

Let's make a proximal hazard zone layer



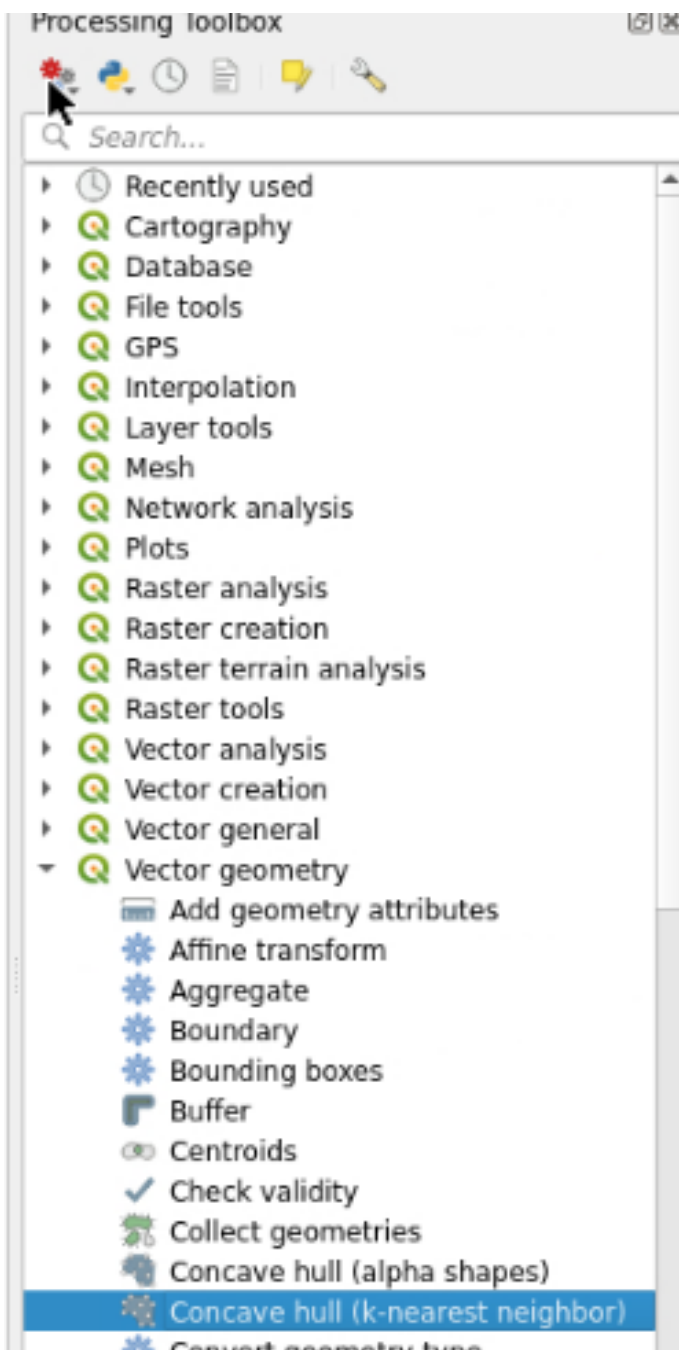
Open the “Raster pixels to points” tool

Open the “Raster pixels to points” tool



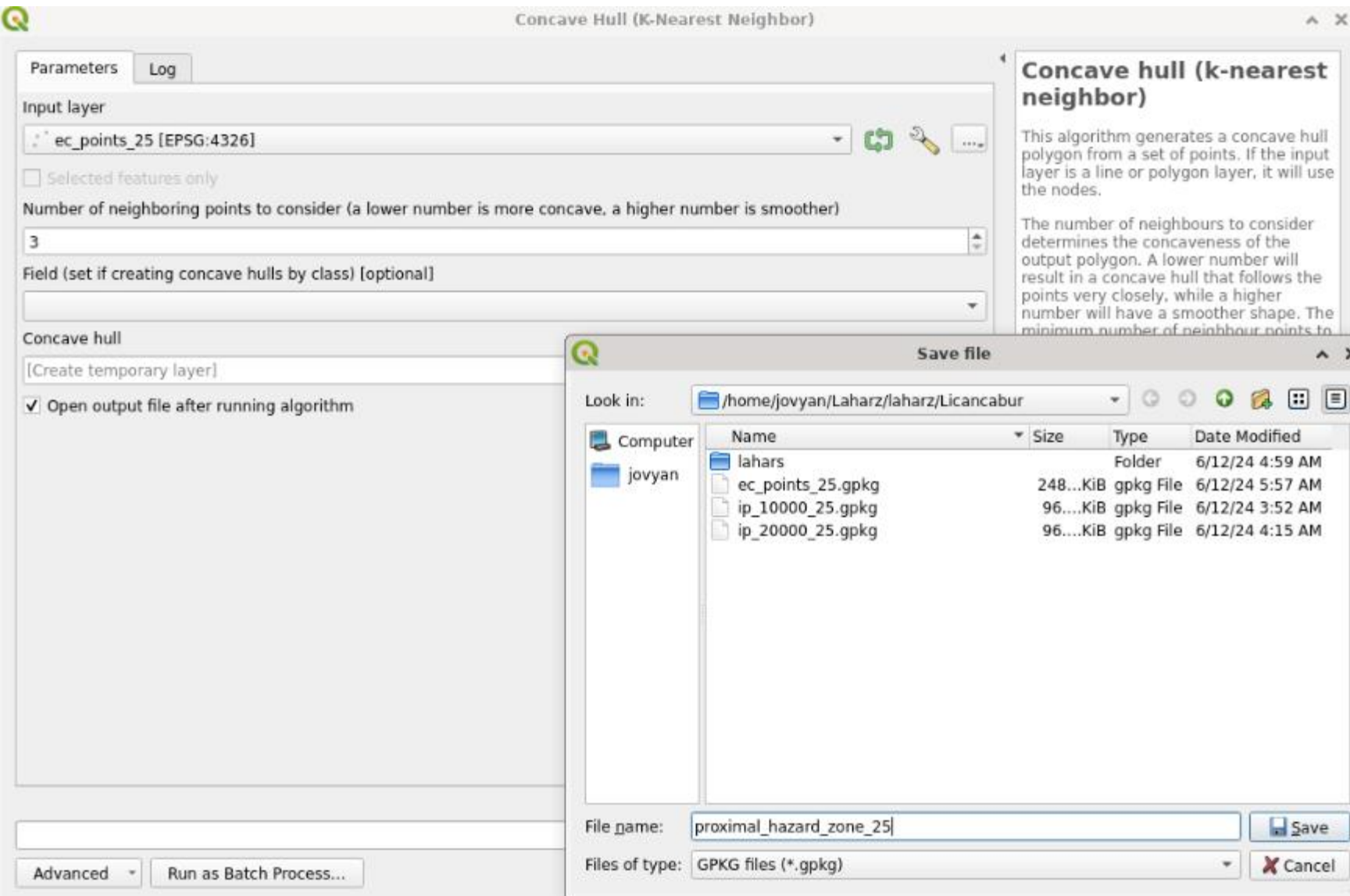
Raster layer: ec_line_25

- Vector points: Save to File
- File name: ec_points_25
- Click “save”
- Click “run”



Open the “Concave hull (k-nearest neighbor)” tool

Open the “Concave hull (k-nearest neighbor)” tool

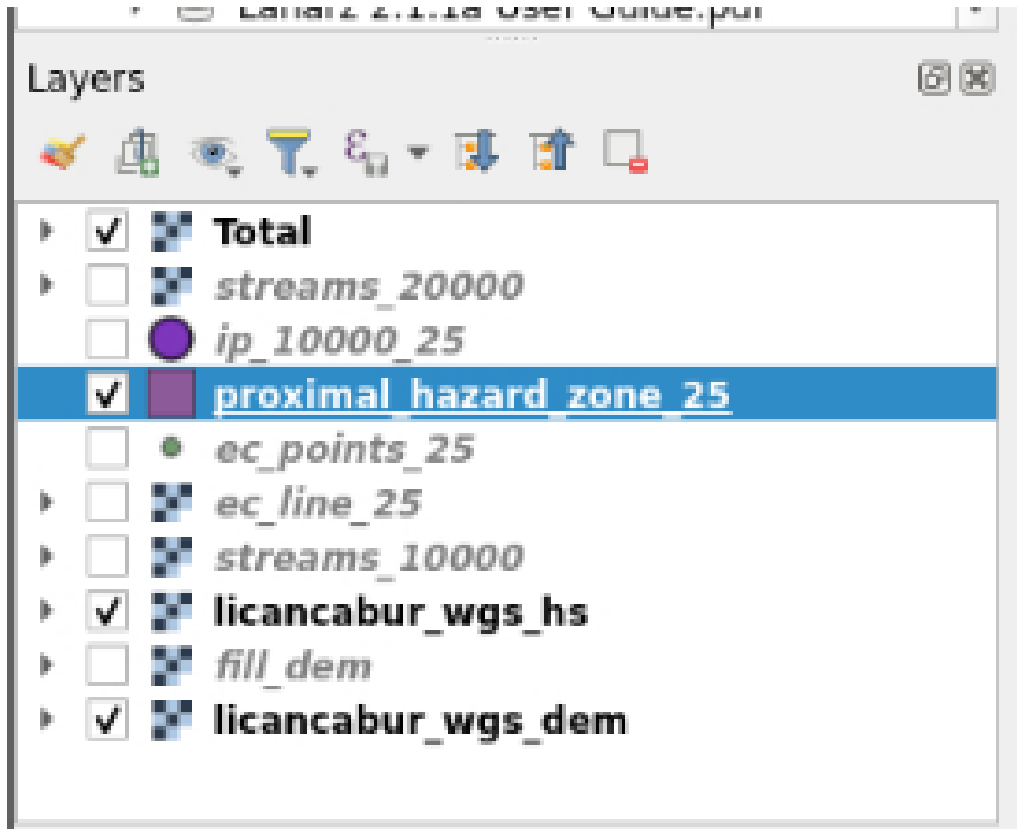


Input layer: ec_points_25

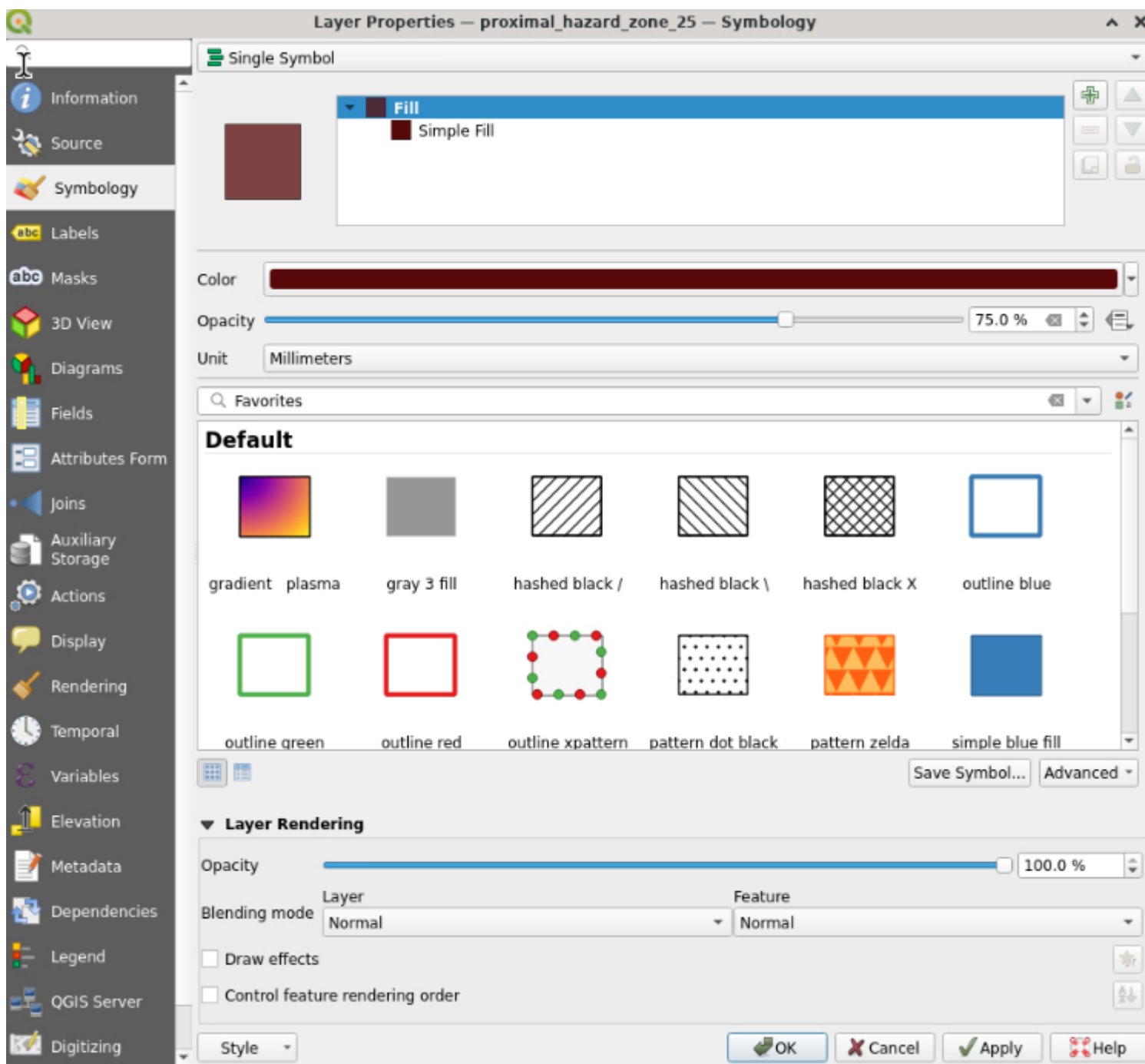
Number of points: 3

- Concave hull: Save to File
- File name: proximal_hazard_zone_25
- Click “save”
- Click “run”

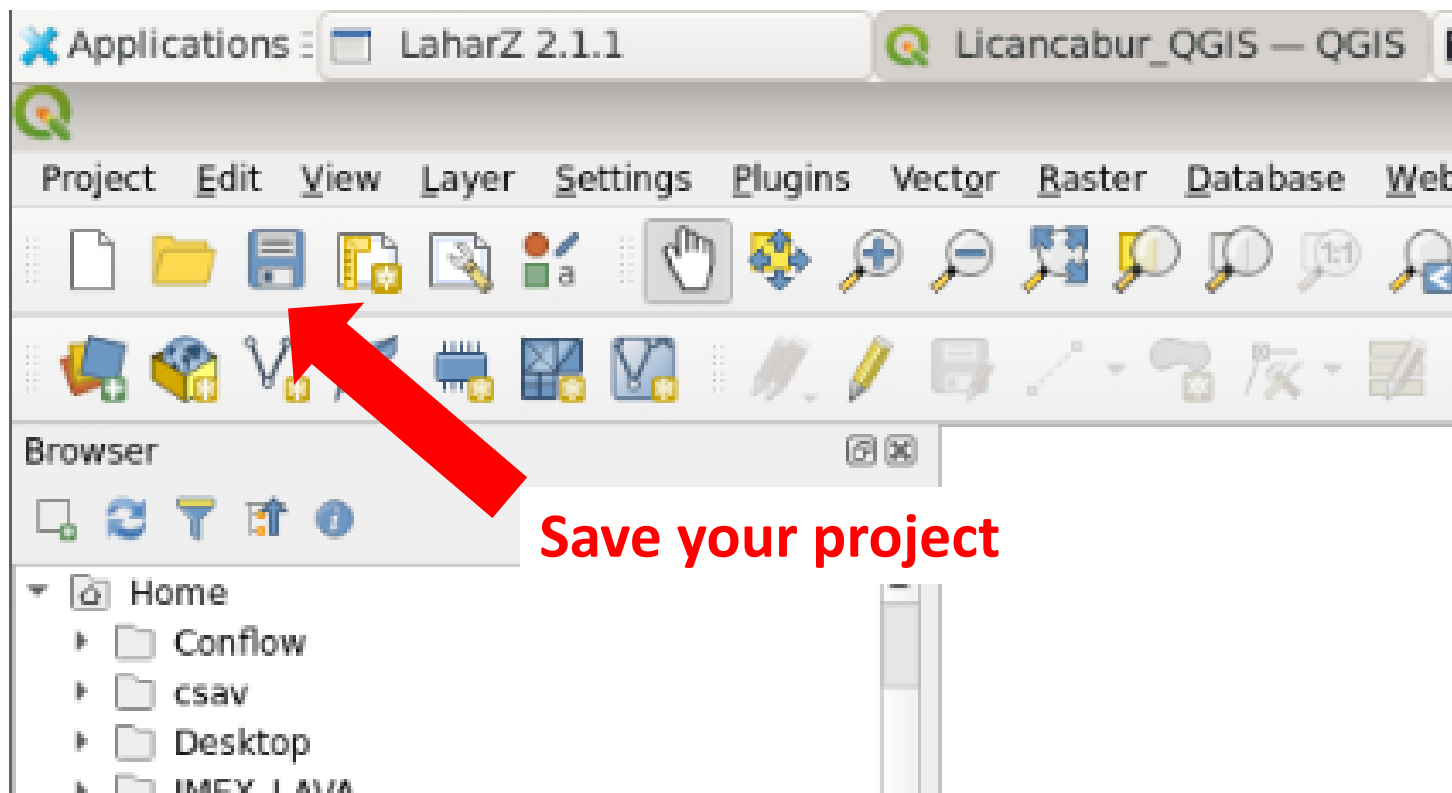
Let's make the map neater



- Turn off layers that aren't needed



Change the color and opacity of the proximal hazard zone layer



Save your project

Congratulations!

¡Felicidades!

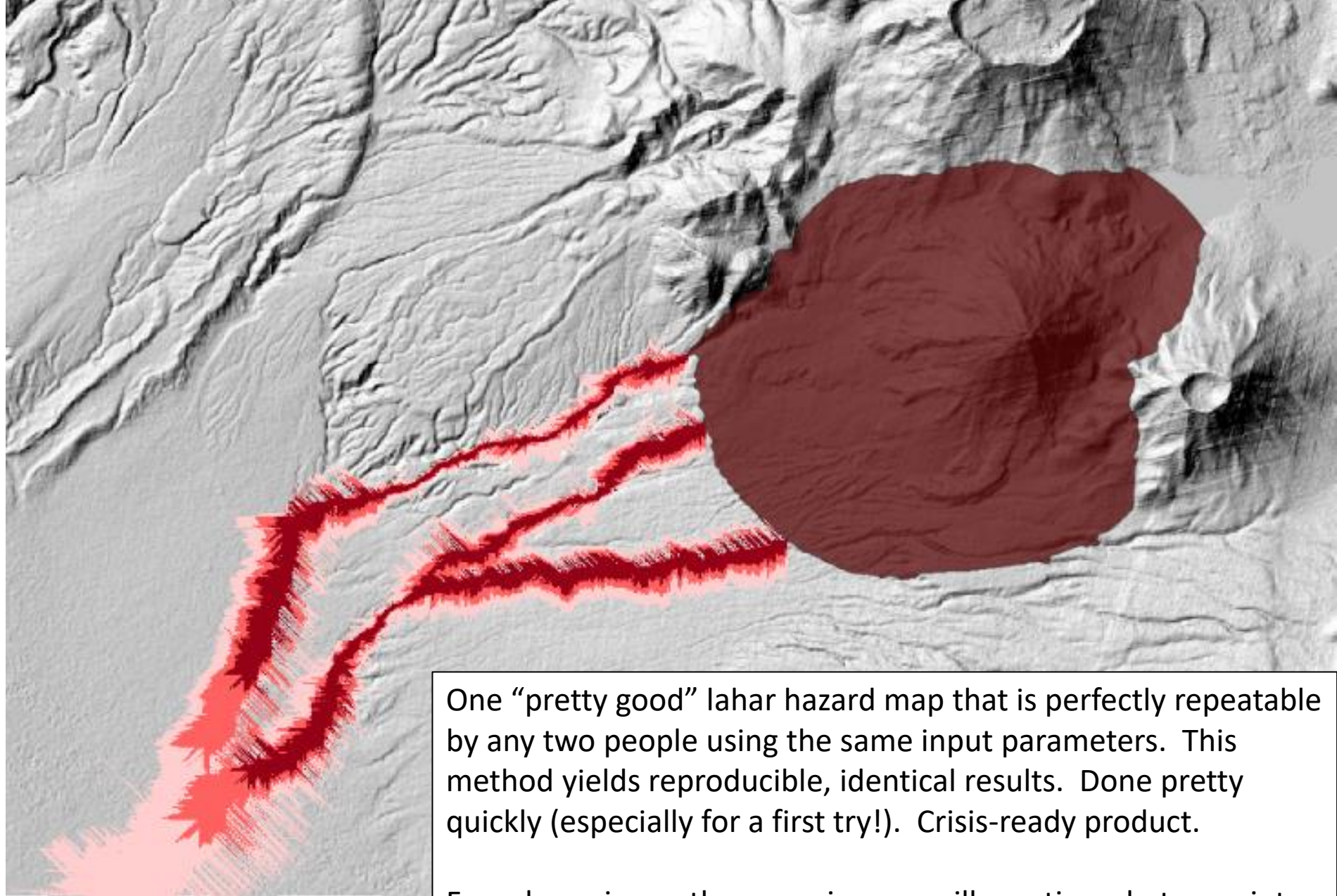
تهانينا

Selamat!

Binabati kita!

Baie geluk!

Félicitations!



One “pretty good” lahar hazard map that is perfectly repeatable by any two people using the same input parameters. This method yields reproducible, identical results. Done pretty quickly (especially for a first try!). Crisis-ready product.

From here, in another exercise, we will practice what goes into the decisions of the H/L cone, volumes, stream thresholds, and map design.